Anàlisi de dades òmiques (M0-157). PAC1.

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Index

1. Introducció

Aquest arxiu conté les sol · lucions analítiques als problemes plantejats en la PAC1 *Anàlisi de dades òmiques (M0-157)*. El projecte total del treball es pot consultar al: meu repositori github, cliqueu **aquí**

2. Carrega de dades

Abans que res, carreguem les dades al nostre entorn de treball. Treballarem amb un dels exemples de mostra, veure **aquí**. Aquest exemple porta dos arxius, DataInfo_S013 i DataValues_S013. Carreguem els dos fitxers:

```
DataInfo_S013 <- read.csv("DataInfo_S013.csv", sep = ",")
DataValues_S013 <- read.csv("DataValues_S013.csv", sep = ",")
```

Com és habitual amb dades òmiques, tenim moltes varibales i un nombre reduït de casos d'estudi. Tot i així, abans que res mirem l'estructura dels dos.

3. Eploració previa de les dades descarregades

```
dim(DataInfo_S013)
```

```
## [1] 695   4
```

```
dim(DataValues_S013)
```

```
## [1]  39 696
```

Aqui ja veiem que en l'arxiu Info hi ha 695 files que han de correspondre a les variables que hi ha a Values. En aquest darrer tenim 696 columnes i 39 files. Les files corresponen a cada cas d'estudi més les 695 columnes són les variables que es tenen de cada un d'ells.

Per veure què són les variables, mirem les 20 primeres files d'Info i podem fer-nos una idea.

```r
head(DataInfo_S013,20)
```

```
##                     X            VarName    varTpe Description
## 1            SUBJECTS           SUBJECTS   integer    dataDesc
## 2             SURGERY            SURGERY character    dataDesc
## 3                 AGE                AGE   integer    dataDesc
## 4              GENDER             GENDER character    dataDesc
## 5               Group              Group   integer    dataDesc
## 6            MEDDM_T0           MEDDM_T0   integer    dataDesc
## 7           MEDCOL_T0          MEDCOL_T0   integer    dataDesc
## 8           MEDINF_T0          MEDINF_T0   integer    dataDesc
## 9           MEDHTA_T0          MEDHTA_T0   integer    dataDesc
## 10             GLU_T0             GLU_T0   integer    dataDesc
## 11             INS_T0             INS_T0   numeric    dataDesc
## 12            HOMA_T0            HOMA_T0   numeric    dataDesc
## 13           HBA1C_T0           HBA1C_T0   numeric    dataDesc
## 14 HBA1C.mmol.mol_T0  HBA1C.mmol.mol_T0   numeric    dataDesc
## 15            PESO_T0            PESO_T0   integer    dataDesc
## 16             bmi_T0             bmi_T0   numeric    dataDesc
## 17              CC_T0              CC_T0   numeric    dataDesc
## 18            CINT_T0            CINT_T0   integer    dataDesc
## 19             CAD_T0             CAD_T0   integer    dataDesc
## 20             TAD_T0             TAD_T0   integer    dataDesc
```

Aqui ja veiem que hi ha 9 primers camps que semblen ser els que tenen la informació sobre el pacient de cada mostra. Mirem que contenen aquests camps a values:

```r
# En mirem 10 en realitat, per comprovar que la primera columna és un index
head(DataValues_S013[,1:10])
```

```
##   X.1 SUBJECTS SURGERY AGE GENDER Group MEDDM_T0 MEDCOL_T0 MEDINF_T0 MEDHTA_T0
## 1   1        1 by pass  27      F     1        0         0         0         1
## 2   2        2 by pass  19      F     2        0         0         0         0
## 3   3        3 by pass  42      F     1        0         0         0         0
## 4   4        4 by pass  37      F     2        0         0         0         0
## 5   5        5 tubular  42      F     1        0         0         0         0
## 6   6        6 by pass  24      F     2        0         0         0         0
```

Com suposavem, aquestes columnes corresponen a: 1. Un codi de pacient (que està repetit a les dues primeres columnes, d'aquí que hi hagi una fila més a Info que columnes a Values) 2. La cirugia a la que va ser sotmés el pacient 3. L'edat i el gènere 4. Un grup de tractament 5. 4 tractaments que venen amb 0 i 1

Per veure que la resta són variables metabòliques, mirem 10 columnes més

```r
head(DataValues_S013[,11:20],10)
```

```
##    GLU_T0 INS_T0 HOMA_T0 HBA1C_T0 HBA1C.mmol.mol_T0 PESO_T0 bmi_T0 CC_T0
## 1      85  11.40    2.40       NA                NA     151   62.9   0.7
## 2      78  12.10    2.32       NA                NA     139   47.0    NA
## 3      75   8.41    1.56      5.4             35.51      84   29.8   0.7
## 4      71  12.80    2.25      5.1             32.23     136   53.1   1.0
```

```
## 5      82   6.01    1.22      5.6              37.69      121   46.6   0.9
## 6      71   9.88    1.73      5.1              32.23      148   48.8   0.7
## 7      80   9.20    1.82      5.6              37.69      109   43.7   0.9
## 8      90   3.40    0.76      5.5              36.60      109   41.8   0.9
## 9      92   5.43    1.23      5.7              38.78      114   44.0   0.9
## 10     84   6.98    1.45      5.5              36.60      120   40.6    NA
##    CINT_T0 CAD_T0
## 1      116    167
## 2       NA     NA
## 3       90    126
## 4      157    162
## 5      123    132
## 6      110    148
## 7      122    141
## 8      124    136
## 9      136    148
## 10      NA     NA
```

Obviament ja ho veiem i també podem veure que hi ha més d'un valor que no es té per a alguna de les variables (NA). Anem a veure si n'hi ha molts

```r
sum(is.na(DataValues_S013))/(dim(DataValues_S013)[1]*dim(DataValues_S013)[2])
```

```
## [1] 0.1248895
```

Això apunta a que tenim un 12,5% de valors desconeguts. Aquest és un aspecte que caldrà tenir en compte en els anàlisi posteriors

4. Generació de l'obsject SummarizedExperiment

Ara que ja sabem que tenim a les dades, podem procedir a carregar-les en un objecte *SummarizedExperiment*. Per a fer-ho, convertirem però les dades a un data frame i li traurem la primera columna que ja hem vist que és redundant amb la segona. Aleshores ja ho podem carregar. En concret, carregarem:

1. Les dades de les 9 primeres columnes (obviant la primera que esta duplicada) de DataVAlues corresponen a dades dels pacients que els guardarem com a rowData ja que no formen part de l'anàlisi dels metabolits
2. La resta de dades de DataValues seràn els assays que és on es guarden les dades de les mostres que tenim
3. L'arxiu DataInfo com a colData que és on es guarda la informació de com és cada columna d'assays. D'aquí només hem d'agafar la part corresponent als metabolits. És a dir de la fila 11 endavant.

```r
# Primer carreguem la llibreria per poder tractar el SummarizedExperiment
library(SummarizedExperiment)
```

```
## Loading required package: MatrixGenerics

## Loading required package: matrixStats

##
## Attaching package: 'MatrixGenerics'

## The following objects are masked from 'package:matrixStats':
##
##     colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
##     colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
##     colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
##     colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
##     colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
```

```
##     colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
##     colWeightedMeans, colWeightedMedians, colWeightedSds,
##     colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
##     rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
##     rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
##     rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
##     rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
##     rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
##     rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
##     rowWeightedSds, rowWeightedVars

## Loading required package: GenomicRanges

## Loading required package: stats4

## Loading required package: BiocGenerics

##
## Attaching package: 'BiocGenerics'

## The following objects are masked from 'package:stats':
##
##     IQR, mad, sd, var, xtabs

## The following objects are masked from 'package:base':
##
##     anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##     colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##     get, grep, grepl, intersect, is.unsorted, lapply, Map, mapply,
##     match, mget, order, paste, pmax, pmax.int, pmin, pmin.int,
##     Position, rank, rbind, Reduce, rownames, sapply, saveRDS, setdiff,
##     table, tapply, union, unique, unsplit, which.max, which.min

## Loading required package: S4Vectors

##
## Attaching package: 'S4Vectors'

## The following object is masked from 'package:utils':
##
##     findMatches

## The following objects are masked from 'package:base':
##
##     expand.grid, I, unname

## Loading required package: IRanges

##
## Attaching package: 'IRanges'

## The following object is masked from 'package:grDevices':
##
##     windows

## Loading required package: GenomeInfoDb

## Loading required package: Biobase

## Welcome to Bioconductor
##
##     Vignettes contain introductory material; view with
```

```
##     'browseVignettes()'. To cite Bioconductor, see
##     'citation("Biobase")', and for packages 'citation("pkgname")'.

##
## Attaching package: 'Biobase'

## The following object is masked from 'package:MatrixGenerics':
##
##     rowMedians

## The following objects are masked from 'package:matrixStats':
##
##     anyMissing, rowMedians
```

```
# Després generem les dades de pacients i de metabolits i la llista de rownames que volem tal i com hem
pacients <- DataValues_S013[,2:10]
metabolits <- DataValues_S013[,11:696]
colNames <- DataInfo_S013[10:695,]
# Ara ja podem carregar el SummarizedExperiment
seMA <- SummarizedExperiment(assays = list(counts = as.matrix(metabolits)),
                             colData = colNames,
                             rowData = pacients)
seMA
```

```
## class: SummarizedExperiment
## dim: 39 686
## metadata(0):
## assays(1): counts
## rownames: NULL
## rowData names(9): SUBJECTS SURGERY ... MEDINF_T0 MEDHTA_T0
## colnames(686): GLU_T0 INS_T0 ... SM.C24.0_T5 SM.C24.1_T5
## colData names(4): X VarName varTpe Description
```

Sembla que l'objecte s'ha generat correctament

Mirem un parell de funcions que es poden fer.

```
# Comprovem com es diuen les columnes que tenim en cada una de les classes de dins de l'objecte
dimnames(seMA)
```

```
## [[1]]
## NULL
##
## [[2]]
##   [1] "GLU_T0"            "INS_T0"            "HOMA_T0"
##   [4] "HBA1C_T0"          "HBA1C.mmol.mol_T0" "PESO_T0"
##   [7] "bmi_T0"            "CC_T0"             "CINT_T0"
##  [10] "CAD_T0"            "TAD_T0"            "TAS_T0"
##  [13] "TG_T0"             "COL_T0"            "LDL_T0"
##  [16] "HDL_T0"            "VLDL_T0"           "PCR_T0"
##  [19] "LEP_T0"            "ADIPO_T0"          "GOT_T0"
##  [22] "GPT_T0"            "GGT_T0"            "URICO_T0"
##  [25] "CREAT_T0"          "UREA_T0"           "HIERRO_T0"
##  [28] "TRANSF_T0"         "FERR_T0"           "Ile_T0"
##  [31] "Leu_T0"            "Val_T0"            "Ala_T0"
##  [34] "Pro_T0"            "Gly_T0"            "Ser_T0"
##  [37] "Trp_T0"            "Phe_T0"            "Met_T0"
##  [40] "Orn_T0"            "Arg_T0"            "His_T0"
```

```
##  [43] "Asn_T0"             "Asp_T0"             "Glu_T0"
##  [46] "Gln_T0"             "Cit_T0"             "Tyr_T0"
##  [49] "Thr_T0"             "Lys_T0"             "Creatinine_T0"
##  [52] "Kynurenine_T0"      "Putrescine_T0"      "Sarcosine_T0"
##  [55] "Serotonin_T0"       "Taurine_T0"         "SDMA_T0"
##  [58] "C0_T0"              "C2_T0"              "C3.OH_T0"
##  [61] "C6..C4.1.DC._T0"    "C5.DC..C6.OH._T0"   "C7.DC_T0"
##  [64] "C8_T0"              "C10_T0"             "C10.1_T0"
##  [67] "C10.2_T0"           "C14.1_T0"           "C14.2_T0"
##  [70] "C16.1_T0"           "C16.2_T0"           "C16.2.OH_T0"
##  [73] "C18.1_T0"           "C18.1.OH_T0"        "C18.2_T0"
##  [76] "lysoPC.a.C16.0_T0" "lysoPC.a.C16.1_T0" "lysoPC.a.C17.0_T0"
##  [79] "lysoPC.a.C18.0_T0" "lysoPC.a.C18.1_T0" "lysoPC.a.C18.2_T0"
##  [82] "lysoPC.a.C20.3_T0" "lysoPC.a.C20.4_T0" "lysoPC.a.C24.0_T0"
##  [85] "lysoPC.a.C26.0_T0" "lysoPC.a.C26.1_T0" "lysoPC.a.C28.0_T0"
##  [88] "lysoPC.a.C28.1_T0" "PC.aa.C24.0_T0"    "PC.aa.C28.1_T0"
##  [91] "PC.aa.C30.0_T0"    "PC.aa.C32.0_T0"    "PC.aa.C32.1_T0"
##  [94] "PC.aa.C32.3_T0"    "PC.aa.C34.1_T0"    "PC.aa.C34.2_T0"
##  [97] "PC.aa.C34.3_T0"    "PC.aa.C34.4_T0"    "PC.aa.C36.0_T0"
## [100] "PC.aa.C36.1_T0"    "PC.aa.C36.2_T0"    "PC.aa.C36.3_T0"
## [103] "PC.aa.C36.4_T0"    "PC.aa.C36.5_T0"    "PC.aa.C38.0_T0"
## [106] "PC.aa.C38.1_T0"    "PC.aa.C38.3_T0"    "PC.aa.C38.4_T0"
## [109] "PC.aa.C38.5_T0"    "PC.aa.C38.6_T0"    "PC.aa.C40.1_T0"
## [112] "PC.aa.C40.2_T0"    "PC.aa.C40.3_T0"    "PC.aa.C40.4_T0"
## [115] "PC.aa.C40.5_T0"    "PC.aa.C40.6_T0"    "PC.aa.C42.0_T0"
## [118] "PC.aa.C42.1_T0"    "PC.aa.C42.2_T0"    "PC.aa.C42.4_T0"
## [121] "PC.aa.C42.5_T0"    "PC.aa.C42.6_T0"    "PC.ae.C30.0_T0"
## [124] "PC.ae.C32.1_T0"    "PC.ae.C32.2_T0"    "PC.ae.C34.0_T0"
## [127] "PC.ae.C34.1_T0"    "PC.ae.C34.2_T0"    "PC.ae.C34.3_T0"
## [130] "PC.ae.C36.0_T0"    "PC.ae.C36.1_T0"    "PC.ae.C36.2_T0"
## [133] "PC.ae.C36.3_T0"    "PC.ae.C36.4_T0"    "PC.ae.C36.5_T0"
## [136] "PC.ae.C38.0_T0"    "PC.ae.C38.2_T0"    "PC.ae.C38.3_T0"
## [139] "PC.ae.C38.4_T0"    "PC.ae.C38.5_T0"    "PC.ae.C38.6_T0"
## [142] "PC.ae.C40.1_T0"    "PC.ae.C40.2_T0"    "PC.ae.C40.3_T0"
## [145] "PC.ae.C40.4_T0"    "PC.ae.C40.5_T0"    "PC.ae.C40.6_T0"
## [148] "PC.ae.C42.1_T0"    "PC.ae.C42.2_T0"    "PC.ae.C42.3_T0"
## [151] "PC.ae.C42.4_T0"    "PC.ae.C42.5_T0"    "PC.ae.C44.3_T0"
## [154] "PC.ae.C44.4_T0"    "PC.ae.C44.5_T0"    "PC.ae.C44.6_T0"
## [157] "SM..OH..C14.1_T0"  "SM..OH..C16.1_T0"  "SM..OH..C22.1_T0"
## [160] "SM..OH..C22.2_T0"  "SM..OH..C24.1_T0"  "SM.C16.0_T0"
## [163] "SM.C16.1_T0"       "SM.C18.0_T0"       "SM.C18.1_T0"
## [166] "SM.C20.2_T0"       "SM.C24.0_T0"       "SM.C24.1_T0"
## [169] "MEDDM_T2"          "MEDCOL_T2"         "MEDINF_T2"
## [172] "MEDHTA_T2"         "GLU_T2"            "INS_T2"
## [175] "HOMA_T2"           "HBA1C_T2"          "HBA1C.mmol.mol_T2"
## [178] "PESO_T2"           "bmi_T2"            "CC_T2"
## [181] "CINT_T2"           "CAD_T2"            "TAD_T2"
## [184] "TAS_T2"            "TG_T2"             "COL_T2"
## [187] "LDL_T2"            "HDL_T2"            "VLDL_T2"
## [190] "PCR_T2"            "LEP_T2"            "ADIPO_T2"
## [193] "GOT_T2"            "GPT_T2"            "GGT_T2"
## [196] "URICO_T2"          "CREAT_T2"          "UREA_T2"
## [199] "HIERRO_T2"         "TRANSF_T2"         "FERR_T2"
## [202] "Ile_T2"            "Leu_T2"            "Val_T2"
```

```
## [205] "Ala_T2"            "Pro_T2"            "Gly_T2"
## [208] "Ser_T2"            "Trp_T2"            "Phe_T2"
## [211] "Met_T2"            "Orn_T2"            "Arg_T2"
## [214] "His_T2"            "Asn_T2"            "Asp_T2"
## [217] "Glu_T2"            "Gln_T2"            "Cit_T2"
## [220] "Tyr_T2"            "Thr_T2"            "Lys_T2"
## [223] "Creatinine_T2"     "Kynurenine_T2"     "Putrescine_T2"
## [226] "Sarcosine_T2"      "Serotonin_T2"      "Taurine_T2"
## [229] "SDMA_T2"           "C0_T2"             "C2_T2"
## [232] "C3.OH_T2"          "C6..C4.1.DC._T2"   "C5.DC..C6.OH._T2"
## [235] "C7.DC_T2"          "C8_T2"             "C10_T2"
## [238] "C10.1_T2"          "C10.2_T2"          "C14.1_T2"
## [241] "C14.2_T2"          "C16.1_T2"          "C16.2_T2"
## [244] "C16.2.OH_T2"       "C18.1_T2"          "C18.1.OH_T2"
## [247] "C18.2_T2"          "X"                 "lysoPC.a.C14.0_T2"
## [250] "lysoPC.a.C16.0_T2" "lysoPC.a.C16.1_T2" "lysoPC.a.C17.0_T2"
## [253] "lysoPC.a.C18.0_T2" "lysoPC.a.C18.1_T2" "lysoPC.a.C18.2_T2"
## [256] "lysoPC.a.C20.3_T2" "lysoPC.a.C20.4_T2" "lysoPC.a.C24.0_T2"
## [259] "lysoPC.a.C26.0_T2" "lysoPC.a.C26.1_T2" "lysoPC.a.C28.0_T2"
## [262] "lysoPC.a.C28.1_T2" "PC.aa.C24.0_T2"    "PC.aa.C28.1_T2"
## [265] "PC.aa.C30.0_T2"    "PC.aa.C32.0_T2"    "PC.aa.C32.1_T2"
## [268] "PC.aa.C32.3_T2"    "PC.aa.C34.1_T2"    "PC.aa.C34.2_T2"
## [271] "PC.aa.C34.3_T2"    "PC.aa.C34.4_T2"    "PC.aa.C36.0_T2"
## [274] "PC.aa.C36.1_T2"    "PC.aa.C36.2_T2"    "PC.aa.C36.3_T2"
## [277] "PC.aa.C36.4_T2"    "PC.aa.C36.5_T2"    "PC.aa.C38.0_T2"
## [280] "PC.aa.C38.1_T2"    "PC.aa.C38.3_T2"    "PC.aa.C38.4_T2"
## [283] "PC.aa.C38.5_T2"    "PC.aa.C38.6_T2"    "PC.aa.C40.1_T2"
## [286] "PC.aa.C40.2_T2"    "PC.aa.C40.3_T2"    "PC.aa.C40.4_T2"
## [289] "PC.aa.C40.5_T2"    "PC.aa.C40.6_T2"    "PC.aa.C42.0_T2"
## [292] "PC.aa.C42.1_T2"    "PC.aa.C42.2_T2"    "PC.aa.C42.4_T2"
## [295] "PC.aa.C42.5_T2"    "PC.aa.C42.6_T2"    "PC.ae.C30.0_T2"
## [298] "PC.ae.C32.1_T2"    "PC.ae.C32.2_T2"    "PC.ae.C34.0_T2"
## [301] "PC.ae.C34.1_T2"    "PC.ae.C34.2_T2"    "PC.ae.C34.3_T2"
## [304] "PC.ae.C36.0_T2"    "PC.ae.C36.1_T2"    "PC.ae.C36.2_T2"
## [307] "PC.ae.C36.3_T2"    "PC.ae.C36.4_T2"    "PC.ae.C36.5_T2"
## [310] "PC.ae.C38.0_T2"    "PC.ae.C38.2_T2"    "PC.ae.C38.3_T2"
## [313] "PC.ae.C38.4_T2"    "PC.ae.C38.5_T2"    "PC.ae.C38.6_T2"
## [316] "PC.ae.C40.1_T2"    "PC.ae.C40.2_T2"    "PC.ae.C40.3_T2"
## [319] "PC.ae.C40.4_T2"    "PC.ae.C40.5_T2"    "PC.ae.C40.6_T2"
## [322] "PC.ae.C42.1_T2"    "PC.ae.C42.2_T2"    "PC.ae.C42.3_T2"
## [325] "PC.ae.C42.4_T2"    "PC.ae.C42.5_T2"    "PC.ae.C44.3_T2"
## [328] "PC.ae.C44.4_T2"    "PC.ae.C44.5_T2"    "PC.ae.C44.6_T2"
## [331] "SM..OH..C14.1_T2"  "SM..OH..C16.1_T2"  "SM..OH..C22.1_T2"
## [334] "SM..OH..C22.2_T2"  "SM..OH..C24.1_T2"  "SM.C16.0_T2"
## [337] "SM.C16.1_T2"       "SM.C18.0_T2"       "SM.C18.1_T2"
## [340] "SM.C20.2_T2"       "SM.C24.0_T2"       "SM.C24.1_T2"
## [343] "MEDDM_T4"          "MEDCOL_T4"         "MEDINF_T4"
## [346] "MEDHTA_T4"         "GLU_T4"            "INS_T4"
## [349] "HOMA_T4"           "HBA1C_T4"          "HBA1C.mmol.mol_T4"
## [352] "PESO_T4"           "bmi_T4"            "CC_T4"
## [355] "CINT_T4"           "CAD_T4"            "TAD_T4"
## [358] "TAS_T4"            "TG_T4"             "COL_T4"
## [361] "LDL_T4"            "HDL_T4"            "VLDL_T4"
## [364] "PCR_T4"            "LEP_T4"            "ADIPO_T4"
```

```
## [367] "GOT_T4"             "GPT_T4"             "GGT_T4"
## [370] "URICO_T4"           "CREAT_T4"           "UREA_T4"
## [373] "HIERRO_T4"          "TRANSF_T4"          "FERR_T4"
## [376] "Ile_T4"             "Leu_T4"             "Val_T4"
## [379] "Ala_T4"             "Pro_T4"             "Gly_T4"
## [382] "Ser_T4"             "Trp_T4"             "Phe_T4"
## [385] "Met_T4"             "Orn_T4"             "Arg_T4"
## [388] "His_T4"             "Asn_T4"             "Asp_T4"
## [391] "Glu_T4"             "Gln_T4"             "Cit_T4"
## [394] "Tyr_T4"             "Thr_T4"             "Lys_T4"
## [397] "Creatinine_T4"      "Kynurenine_T4"      "Putrescine_T4"
## [400] "Sarcosine_T4"       "Serotonin_T4"       "Taurine_T4"
## [403] "SDMA_T4"            "C0_T4"              "C2_T4"
## [406] "C3.OH_T4"           "C6..C4.1.DC._T4"    "C5.DC..C6.OH._T4"
## [409] "C7.DC_T4"           "C8_T4"              "C10_T4"
## [412] "C10.1_T4"           "C10.2_T4"           "C14.1_T4"
## [415] "C14.2_T4"           "C16.1_T4"           "C16.2_T4"
## [418] "C16.2.OH_T4"        "C18.1_T4"           "C18.1.OH_T4"
## [421] "C18.2_T4"           "lysoPC.a.C16.0_T4" "lysoPC.a.C16.1_T4"
## [424] "lysoPC.a.C17.0_T4" "lysoPC.a.C18.0_T4" "lysoPC.a.C18.1_T4"
## [427] "lysoPC.a.C18.2_T4" "lysoPC.a.C20.3_T4" "lysoPC.a.C20.4_T4"
## [430] "lysoPC.a.C24.0_T4" "lysoPC.a.C26.0_T4" "lysoPC.a.C26.1_T4"
## [433] "lysoPC.a.C28.0_T4" "lysoPC.a.C28.1_T4" "PC.aa.C24.0_T4"
## [436] "PC.aa.C28.1_T4"    "PC.aa.C30.0_T4"    "PC.aa.C32.0_T4"
## [439] "PC.aa.C32.1_T4"    "PC.aa.C32.3_T4"    "PC.aa.C34.1_T4"
## [442] "PC.aa.C34.2_T4"    "PC.aa.C34.3_T4"    "PC.aa.C34.4_T4"
## [445] "PC.aa.C36.0_T4"    "PC.aa.C36.1_T4"    "PC.aa.C36.2_T4"
## [448] "PC.aa.C36.3_T4"    "PC.aa.C36.4_T4"    "PC.aa.C36.5_T4"
## [451] "PC.aa.C38.0_T4"    "PC.aa.C38.1_T4"    "PC.aa.C38.3_T4"
## [454] "PC.aa.C38.4_T4"    "PC.aa.C38.5_T4"    "PC.aa.C38.6_T4"
## [457] "PC.aa.C40.1_T4"    "PC.aa.C40.2_T4"    "PC.aa.C40.3_T4"
## [460] "PC.aa.C40.4_T4"    "PC.aa.C40.5_T4"    "PC.aa.C40.6_T4"
## [463] "PC.aa.C42.0_T4"    "PC.aa.C42.1_T4"    "PC.aa.C42.2_T4"
## [466] "PC.aa.C42.4_T4"    "PC.aa.C42.5_T4"    "PC.aa.C42.6_T4"
## [469] "PC.ae.C30.0_T4"    "PC.ae.C32.1_T4"    "PC.ae.C32.2_T4"
## [472] "PC.ae.C34.0_T4"    "PC.ae.C34.1_T4"    "PC.ae.C34.2_T4"
## [475] "PC.ae.C34.3_T4"    "PC.ae.C36.0_T4"    "PC.ae.C36.1_T4"
## [478] "PC.ae.C36.2_T4"    "PC.ae.C36.3_T4"    "PC.ae.C36.4_T4"
## [481] "PC.ae.C36.5_T4"    "PC.ae.C38.0_T4"    "PC.ae.C38.2_T4"
## [484] "PC.ae.C38.3_T4"    "PC.ae.C38.4_T4"    "PC.ae.C38.5_T4"
## [487] "PC.ae.C38.6_T4"    "PC.ae.C40.1_T4"    "PC.ae.C40.2_T4"
## [490] "PC.ae.C40.3_T4"    "PC.ae.C40.4_T4"    "PC.ae.C40.5_T4"
## [493] "PC.ae.C40.6_T4"    "PC.ae.C42.1_T4"    "PC.ae.C42.2_T4"
## [496] "PC.ae.C42.3_T4"    "PC.ae.C42.4_T4"    "PC.ae.C42.5_T4"
## [499] "PC.ae.C44.3_T4"    "PC.ae.C44.4_T4"    "PC.ae.C44.5_T4"
## [502] "PC.ae.C44.6_T4"    "SM..OH..C14.1_T4"  "SM..OH..C16.1_T4"
## [505] "SM..OH..C22.1_T4"  "SM..OH..C22.2_T4"  "SM..OH..C24.1_T4"
## [508] "SM.C16.0_T4"       "SM.C16.1_T4"       "SM.C18.0_T4"
## [511] "SM.C18.1_T4"       "SM.C20.2_T4"       "SM.C24.0_T4"
## [514] "SM.C24.1_T4"       "MEDDM_T5"          "MEDCOL_T5"
## [517] "MEDINF_T5"         "MEDHTA_T5"         "GLU_T5"
## [520] "INS_T5"            "HOMA_T5"           "HBA1C_T5"
## [523] "HBA1C.mmol.mol_T5" "PESO_T5"           "bmi_T5"
## [526] "CC_T5"             "CINT_T5"           "CAD_T5"
```

```
## [529] "TAD_T5"            "TAS_T5"            "TG_T5"
## [532] "COL_T5"            "LDL_T5"            "HDL_T5"
## [535] "VLDL_T5"           "PCR_T5"            "LEP_T5"
## [538] "ADIPO_T5"          "GOT_T5"            "GPT_T5"
## [541] "GGT_T5"            "URICO_T5"          "CREAT_T5"
## [544] "UREA_T5"           "HIERRO_T5"         "TRANSF_T5"
## [547] "FERR_T5"           "Ile_T5"            "Leu_T5"
## [550] "Val_T5"            "Ala_T5"            "Pro_T5"
## [553] "Gly_T5"            "Ser_T5"            "Trp_T5"
## [556] "Phe_T5"            "Met_T5"            "Orn_T5"
## [559] "Arg_T5"            "His_T5"            "Asn_T5"
## [562] "Asp_T5"            "Glu_T5"            "Gln_T5"
## [565] "Cit_T5"            "Tyr_T5"            "Thr_T5"
## [568] "Lys_T5"            "Creatinine_T5"     "Kynurenine_T5"
## [571] "Putrescine_T5"     "Sarcosine_T5"      "Serotonin_T5"
## [574] "Taurine_T5"        "SDMA_T5"           "C0_T5"
## [577] "C2_T5"             "C3.OH_T5"          "C6..C4.1.DC._T5"
## [580] "C5.DC..C6.OH._T5"  "C7.DC_T5"          "C8_T5"
## [583] "C10_T5"            "C10.1_T5"          "C10.2_T5"
## [586] "C14.1_T5"          "C14.2_T5"          "C16.1_T5"
## [589] "C16.2_T5"          "C16.2.OH_T5"       "C18.1_T5"
## [592] "C18.1.OH_T5"       "C18.2_T5"          "lysoPC.a.C16.0_T5"
## [595] "lysoPC.a.C16.1_T5" "lysoPC.a.C17.0_T5" "lysoPC.a.C18.0_T5"
## [598] "lysoPC.a.C18.1_T5" "lysoPC.a.C18.2_T5" "lysoPC.a.C20.3_T5"
## [601] "lysoPC.a.C20.4_T5" "lysoPC.a.C24.0_T5" "lysoPC.a.C26.0_T5"
## [604] "lysoPC.a.C26.1_T5" "lysoPC.a.C28.0_T5" "lysoPC.a.C28.1_T5"
## [607] "PC.aa.C24.0_T5"    "PC.aa.C28.1_T5"    "PC.aa.C30.0_T5"
## [610] "PC.aa.C32.0_T5"    "PC.aa.C32.1_T5"    "PC.aa.C32.3_T5"
## [613] "PC.aa.C34.1_T5"    "PC.aa.C34.2_T5"    "PC.aa.C34.3_T5"
## [616] "PC.aa.C34.4_T5"    "PC.aa.C36.0_T5"    "PC.aa.C36.1_T5"
## [619] "PC.aa.C36.2_T5"    "PC.aa.C36.3_T5"    "PC.aa.C36.4_T5"
## [622] "PC.aa.C36.5_T5"    "PC.aa.C38.0_T5"    "PC.aa.C38.1_T5"
## [625] "PC.aa.C38.3_T5"    "PC.aa.C38.4_T5"    "PC.aa.C38.5_T5"
## [628] "PC.aa.C38.6_T5"    "PC.aa.C40.1_T5"    "PC.aa.C40.2_T5"
## [631] "PC.aa.C40.3_T5"    "PC.aa.C40.4_T5"    "PC.aa.C40.5_T5"
## [634] "PC.aa.C40.6_T5"    "PC.aa.C42.0_T5"    "PC.aa.C42.1_T5"
## [637] "PC.aa.C42.2_T5"    "PC.aa.C42.4_T5"    "PC.aa.C42.5_T5"
## [640] "PC.aa.C42.6_T5"    "PC.ae.C30.0_T5"    "PC.ae.C32.1_T5"
## [643] "PC.ae.C32.2_T5"    "PC.ae.C34.0_T5"    "PC.ae.C34.1_T5"
## [646] "PC.ae.C34.2_T5"    "PC.ae.C34.3_T5"    "PC.ae.C36.0_T5"
## [649] "PC.ae.C36.1_T5"    "PC.ae.C36.2_T5"    "PC.ae.C36.3_T5"
## [652] "PC.ae.C36.4_T5"    "PC.ae.C36.5_T5"    "PC.ae.C38.0_T5"
## [655] "PC.ae.C38.2_T5"    "PC.ae.C38.3_T5"    "PC.ae.C38.4_T5"
## [658] "PC.ae.C38.5_T5"    "PC.ae.C38.6_T5"    "PC.ae.C40.1_T5"
## [661] "PC.ae.C40.2_T5"    "PC.ae.C40.3_T5"    "PC.ae.C40.4_T5"
## [664] "PC.ae.C40.5_T5"    "PC.ae.C40.6_T5"    "PC.ae.C42.1_T5"
## [667] "PC.ae.C42.2_T5"    "PC.ae.C42.3_T5"    "PC.ae.C42.4_T5"
## [670] "PC.ae.C42.5_T5"    "PC.ae.C44.3_T5"    "PC.ae.C44.4_T5"
## [673] "PC.ae.C44.5_T5"    "PC.ae.C44.6_T5"    "SM..OH..C14.1_T5"
## [676] "SM..OH..C16.1_T5"  "SM..OH..C22.1_T5"  "SM..OH..C22.2_T5"
## [679] "SM..OH..C24.1_T5"  "SM.C16.0_T5"       "SM.C16.1_T5"
## [682] "SM.C18.0_T5"       "SM.C18.1_T5"       "SM.C20.2_T5"
## [685] "SM.C24.0_T5"       "SM.C24.1_T5"
```

```r
# I ara mirem les 10 primeres columnes i les 5 primeres files d'assays
assay(seMA)[1:5, 1:10]
```

```
##      GLU_T0 INS_T0 HOMA_T0 HBA1C_T0 HBA1C.mmol.mol_T0 PESO_T0 bmi_T0 CC_T0
## [1,]     85  11.40    2.40       NA                NA     151   62.9   0.7
## [2,]     78  12.10    2.32       NA                NA     139   47.0    NA
## [3,]     75   8.41    1.56      5.4             35.51      84   29.8   0.7
## [4,]     71  12.80    2.25      5.1             32.23     136   53.1   1.0
## [5,]     82   6.01    1.22      5.6             37.69     121   46.6   0.9
##      CINT_T0 CAD_T0
## [1,]     116    167
## [2,]      NA     NA
## [3,]      90    126
## [4,]     157    162
## [5,]     123    132
```

```r
# I també mirem com ha quedat colData
colData(seMA)
```

```
## DataFrame with 686 rows and 4 columns
##                                   X           VarName      varTpe Description
##                         <character>       <character> <character> <character>
## GLU_T0                       GLU_T0            GLU_T0     integer    dataDesc
## INS_T0                       INS_T0            INS_T0     numeric    dataDesc
## HOMA_T0                     HOMA_T0           HOMA_T0     numeric    dataDesc
## HBA1C_T0                   HBA1C_T0          HBA1C_T0     numeric    dataDesc
## HBA1C.mmol.mol_T0 HBA1C.mmol.mol_T0 HBA1C.mmol.mol_T0     numeric    dataDesc
## ...                             ...               ...         ...         ...
## SM.C18.0_T5              SM.C18.0_T5       SM.C18.0_T5     numeric    dataDesc
## SM.C18.1_T5              SM.C18.1_T5       SM.C18.1_T5     numeric    dataDesc
## SM.C20.2_T5              SM.C20.2_T5       SM.C20.2_T5     numeric    dataDesc
## SM.C24.0_T5              SM.C24.0_T5       SM.C24.0_T5     numeric    dataDesc
## SM.C24.1_T5              SM.C24.1_T5       SM.C24.1_T5     numeric    dataDesc
```

```r
# I les dades bàsiques dels pacients
rowData(seMA)
```

```
## DataFrame with 39 rows and 9 columns
##      SUBJECTS     SURGERY       AGE      GENDER     Group   MEDDM_T0 MEDCOL_T0
##     <integer> <character> <integer> <character> <integer> <integer> <integer>
## 1           1     by pass        27           F         1         0         0
## 2           2     by pass        19           F         2         0         0
## 3           3     by pass        42           F         1         0         0
## 4           4     by pass        37           F         2         0         0
## 5           5     tubular        42           F         1         0         0
## ...       ...         ...       ...         ...       ...       ...       ...
## 35         35     tubular        39           M         2         0         0
## 36         36     tubular        35           M         1         0         0
## 37         37     by pass        46           M         2         0         0
## 38         38     tubular        41           M         1         0         0
## 39         39     by pass        26           M         1         0         0
##     MEDINF_T0 MEDHTA_T0
##     <integer> <integer>
## 1           0         1
## 2           0         0
## 3           0         0
```

```
## 4           0          0
## 5           0          0
## ...       ...        ...
## 35          0          1
## 36          0          0
## 37          1          0
## 38          0          0
## 39          0          0
```

5. Anàlisi exploratoria inicial de les dades

5.1. Descripció de les dades

Fins ara hem vist que les nostres dades tenen NA, el que pot suposar un problema. Anem a veure també si la variabilitat de rangs és molt gran´i ens cal normalitzar les dades. Fem un summary a més per veure com varien els estadístics bàsics. Si ho fem amb tot l'arxiu no s'arriba a apreciar la sortida, així que en fem uns quants

```
summary(assay(seMA)[,1:150])
```

```
##      GLU_T0          INS_T0         HOMA_T0          HBA1C_T0    
##  Min.   : 71.0   Min.   : 3.40   Min.   : 0.760   Min.   :5.100  
##  1st Qu.: 91.0   1st Qu.:11.85   1st Qu.: 2.435   1st Qu.:5.400  
##  Median :103.0   Median :16.00   Median : 4.210   Median :5.600  
##  Mean   :106.5   Mean   :17.60   Mean   : 4.890   Mean   :5.592  
##  3rd Qu.:109.5   3rd Qu.:21.15   3rd Qu.: 5.720   3rd Qu.:5.800  
##  Max.   :263.0   Max.   :43.00   Max.   :13.600   Max.   :6.400  
##                                                   NA's   :15     
##  HBA1C.mmol.mol_T0    PESO_T0          bmi_T0          CC_T0       
##  Min.   :32.23     Min.   : 84.0   Min.   :29.80   Min.   :0.7000  
##  1st Qu.:35.51     1st Qu.:119.5   1st Qu.:44.40   1st Qu.:0.9000  
##  Median :37.69     Median :135.0   Median :48.80   Median :0.9000  
##  Mean   :37.60     Mean   :140.0   Mean   :50.52   Mean   :0.9343  
##  3rd Qu.:39.88     3rd Qu.:155.0   3rd Qu.:55.35   3rd Qu.:1.0000  
##  Max.   :46.44     Max.   :200.0   Max.   :68.60   Max.   :1.7000  
##  NA's   :15                                        NA's   :4       
##     CINT_T0          CAD_T0          TAD_T0           TAS_T0     
##  Min.   : 90.0   Min.   : 85.0   Min.   : 60.00   Min.   :111.0  
##  1st Qu.:122.5   1st Qu.:136.0   1st Qu.: 74.50   1st Qu.:121.0  
##  Median :133.0   Median :148.0   Median : 83.00   Median :128.0  
##  Mean   :135.7   Mean   :147.1   Mean   : 83.58   Mean   :133.5  
##  3rd Qu.:147.5   3rd Qu.:158.5   3rd Qu.: 90.00   3rd Qu.:144.0  
##  Max.   :186.0   Max.   :182.0   Max.   :125.00   Max.   :180.0  
##  NA's   :4       NA's   :4       NA's   :8        NA's   :8      
##      TG_T0           COL_T0          LDL_T0          HDL_T0      
##  Min.   : 30.0   Min.   :102.0   Min.   : 24.0   Min.   : 11.00  
##  1st Qu.: 84.0   1st Qu.:180.5   1st Qu.:100.3   1st Qu.: 39.25  
##  Median :118.0   Median :207.0   Median :130.0   Median : 47.00  
##  Mean   :129.1   Mean   :206.6   Mean   :128.2   Mean   : 49.97  
##  3rd Qu.:151.0   3rd Qu.:236.0   3rd Qu.:158.8   3rd Qu.: 59.50  
##  Max.   :268.0   Max.   :295.0   Max.   :191.0   Max.   :107.00  
##                                  NA's   :1       NA's   :1       
##     VLDL_T0          PCR_T0          LEP_T0           ADIPO_T0    
##  Min.   : 6.00   Min.   : 0.00   Min.   : 27.00   Min.   : 1.720  
```

```
##  1st Qu.:16.80   1st Qu.: 3.51   1st Qu.: 49.25   1st Qu.: 4.700
##  Median :23.60   Median : 8.25   Median : 67.20   Median : 7.460
##  Mean   :25.81   Mean   :10.13   Mean   : 78.33   Mean   : 7.702
##  3rd Qu.:30.20   3rd Qu.:13.90   3rd Qu.: 98.25   3rd Qu.: 8.590
##  Max.   :53.60   Max.   :29.80   Max.   :155.00   Max.   :17.100
##                  NA's   :14      NA's   :13       NA's   :14
##     GOT_T0          GPT_T0           GGT_T0         URICO_T0
##  Min.   :11.0   Min.   : 11.00   Min.   :12.00   Min.   :2.700
##  1st Qu.:15.0   1st Qu.: 31.00   1st Qu.:20.00   1st Qu.:4.700
##  Median :20.0   Median : 40.00   Median :26.00   Median :5.500
##  Mean   :23.9   Mean   : 44.54   Mean   :31.41   Mean   :5.501
##  3rd Qu.:25.0   3rd Qu.: 48.50   3rd Qu.:37.00   3rd Qu.:6.200
##  Max.   :83.0   Max.   :157.00   Max.   :86.00   Max.   :8.600
##                                  NA's   :2       NA's   :2
##    CREAT_T0         UREA_T0        HIERRO_T0        TRANSF_T0
##  Min.   :0.5000   Min.   :14.00   Min.   : 13.00   Min.   :159.0
##  1st Qu.:0.7000   1st Qu.:24.00   1st Qu.: 50.00   1st Qu.:236.5
##  Median :0.8000   Median :29.00   Median : 69.00   Median :268.5
##  Mean   :0.7923   Mean   :31.38   Mean   : 75.61   Mean   :262.9
##  3rd Qu.:0.9000   3rd Qu.:38.00   3rd Qu.: 92.25   3rd Qu.:289.8
##  Max.   :1.2000   Max.   :56.00   Max.   :166.00   Max.   :374.0
##                                   NA's   :1        NA's   :9
##    FERR_T0           Ile_T0           Leu_T0          Val_T0
##  Min.   : 11.10   Min.   : 53.50   Min.   : 90.1   Min.   :183.0
##  1st Qu.: 25.00   1st Qu.: 83.55   1st Qu.:150.5   1st Qu.:256.0
##  Median : 49.00   Median : 91.70   Median :183.0   Median :311.0
##  Mean   : 77.06   Mean   : 98.98   Mean   :185.2   Mean   :318.4
##  3rd Qu.: 97.00   3rd Qu.:116.00   3rd Qu.:217.0   3rd Qu.:369.5
##  Max.   :372.00   Max.   :150.00   Max.   :281.0   Max.   :508.0
##  NA's   :2
##      Ala_T0          Pro_T0          Gly_T0          Ser_T0
##  Min.   :194.0   Min.   :123.0   Min.   :153.0   Min.   : 94.5
##  1st Qu.:414.0   1st Qu.:164.5   1st Qu.:227.5   1st Qu.:133.5
##  Median :459.0   Median :222.0   Median :270.0   Median :159.0
##  Mean   :471.9   Mean   :234.6   Mean   :291.1   Mean   :161.4
##  3rd Qu.:543.0   3rd Qu.:257.5   3rd Qu.:311.5   3rd Qu.:180.0
##  Max.   :907.0   Max.   :592.0   Max.   :586.0   Max.   :310.0
##
##      Trp_T0           Phe_T0           Met_T0           Orn_T0
##  Min.   : 41.00   Min.   : 48.40   Min.   :-99.00   Min.   : 39.6
##  1st Qu.: 52.15   1st Qu.: 70.60   1st Qu.: 23.25   1st Qu.: 74.6
##  Median : 62.30   Median : 87.30   Median : 28.30   Median :103.0
##  Mean   : 67.22   Mean   : 87.48   Mean   : 23.08   Mean   :102.4
##  3rd Qu.: 76.95   3rd Qu.:102.00   3rd Qu.: 33.30   3rd Qu.:128.5
##  Max.   :123.00   Max.   :139.00   Max.   : 53.00   Max.   :195.0
##
##      Arg_T0          His_T0           Asn_T0          Asp_T0
##  Min.   : 90.7   Min.   : 70.50   Min.   :27.30   Min.   : 7.10
##  1st Qu.:120.5   1st Qu.: 80.65   1st Qu.:39.55   1st Qu.:15.65
##  Median :143.0   Median : 94.40   Median :47.40   Median :20.50
##  Mean   :148.5   Mean   : 97.68   Mean   :48.81   Mean   :23.12
##  3rd Qu.:167.5   3rd Qu.:108.00   3rd Qu.:56.00   3rd Qu.:27.85
##  Max.   :254.0   Max.   :167.00   Max.   :82.10   Max.   :59.30
##
```

```
##      Glu_T0           Gln_T0          Cit_T0         Tyr_T0
##  Min.   : 16.90   Min.   : 241.0   Min.   :12.00   Min.   : 49.30
##  1st Qu.: 51.70   1st Qu.: 628.0   1st Qu.:26.25   1st Qu.: 71.65
##  Median : 66.40   Median : 713.0   Median :30.30   Median : 89.00
##  Mean   : 84.68   Mean   : 749.9   Mean   :32.76   Mean   : 92.46
##  3rd Qu.: 86.25   3rd Qu.: 872.5   3rd Qu.:38.35   3rd Qu.:108.00
##  Max.   :351.00   Max.   :1320.0   Max.   :59.50   Max.   :162.00
##
##      Thr_T0          Lys_T0      Creatinine_T0    Kynurenine_T0
##  Min.   : 67.5   Min.   :266.0   Min.   : 39.10   Min.   :1.610
##  1st Qu.:112.0   1st Qu.:316.5   1st Qu.: 65.85   1st Qu.:2.540
##  Median :144.0   Median :378.0   Median : 78.40   Median :3.160
##  Mean   :151.5   Mean   :381.8   Mean   : 80.43   Mean   :3.303
##  3rd Qu.:177.5   3rd Qu.:434.5   3rd Qu.: 99.20   3rd Qu.:3.775
##  Max.   :286.0   Max.   :562.0   Max.   :115.00   Max.   :6.510
##
##  Putrescine_T0       Sarcosine_T0     Serotonin_T0       Taurine_T0
##  Min.   :-99.0000   Min.   : 2.050   Min.   :-99.0000   Min.   : 41.00
##  1st Qu.:  0.1295   1st Qu.: 5.895   1st Qu.:  0.3565   1st Qu.: 78.55
##  Median :  0.2120   Median : 7.700   Median :  0.6500   Median :107.00
##  Mean   :-12.7355   Mean   : 8.648   Mean   : -9.5029   Mean   :108.77
##  3rd Qu.:  0.2670   3rd Qu.:12.200   3rd Qu.:  0.9180   3rd Qu.:132.50
##  Max.   :  0.3660   Max.   :15.700   Max.   :  1.8100   Max.   :181.00
##
##     SDMA_T0          C0_T0           C2_T0          C3.OH_T0
##  Min.   :0.820   Min.   :21.60   Min.   : 0.360   Min.   :0.1500
##  1st Qu.:1.115   1st Qu.:36.00   1st Qu.: 5.215   1st Qu.:0.2100
##  Median :1.290   Median :43.30   Median : 7.270   Median :0.2400
##  Mean   :1.317   Mean   :45.60   Mean   : 7.762   Mean   :0.2474
##  3rd Qu.:1.535   3rd Qu.:52.05   3rd Qu.: 9.200   3rd Qu.:0.2800
##  Max.   :1.940   Max.   :79.30   Max.   :19.500   Max.   :0.3700
##
##  C6..C4.1.DC._T0   C5.DC..C6.OH._T0     C7.DC_T0           C8_T0
##  Min.   :-99.000   Min.   :-9.0000   Min.   :-9.0000   Min.   :0.3230
##  1st Qu.: -9.000   1st Qu.: 0.0410   1st Qu.: 0.0395   1st Qu.:0.4385
##  Median :  0.094   Median : 0.0490   Median : 0.0470   Median :0.5190
##  Mean   :-14.232   Mean   :-1.1096   Mean   :-1.1098   Mean   :0.5419
##  3rd Qu.:  0.118   3rd Qu.: 0.0565   3rd Qu.: 0.0555   3rd Qu.:0.6235
##  Max.   :  0.201   Max.   : 0.0780   Max.   : 0.0760   Max.   :0.8710
##
##      C10_T0          C10.1_T0         C10.2_T0         C14.1_T0
##  Min.   :0.2500   Min.   :0.1300   Min.   :0.1000   Min.   :0.0900
##  1st Qu.:0.4200   1st Qu.:0.1900   1st Qu.:0.1300   1st Qu.:0.1450
##  Median :0.5200   Median :0.2300   Median :0.1500   Median :0.1900
##  Mean   :0.5274   Mean   :0.2482   Mean   :0.1621   Mean   :0.1944
##  3rd Qu.:0.6350   3rd Qu.:0.3050   3rd Qu.:0.1850   3rd Qu.:0.2450
##  Max.   :0.9300   Max.   :0.4200   Max.   :0.2600   Max.   :0.3000
##
##     C14.2_T0         C16.1_T0         C16.2_T0         C16.2.OH_T0
##  Min.   :0.0600   Min.   :0.0600   Min.   :0.03000   Min.   :0.00000
##  1st Qu.:0.1000   1st Qu.:0.0900   1st Qu.:0.03000   1st Qu.:0.04000
##  Median :0.1200   Median :0.1000   Median :0.04000   Median :0.04000
##  Mean   :0.1244   Mean   :0.1126   Mean   :0.04487   Mean   :0.04385
##  3rd Qu.:0.1400   3rd Qu.:0.1300   3rd Qu.:0.05000   3rd Qu.:0.05500
```

```
##  Max.   :0.2100   Max.   :0.1900   Max.   :0.08000   Max.   :0.07000
##
##     C18.1_T0       C18.1.OH_T0        C18.2_T0      lysoPC.a.C16.0_T0
##  Min.   :0.0700   Min.   :0.03000   Min.   :0.0600   Min.   : 72.3
##  1st Qu.:0.1200   1st Qu.:0.04000   1st Qu.:0.0900   1st Qu.:118.5
##  Median :0.1600   Median :0.04000   Median :0.1100   Median :144.0
##  Mean   :0.1703   Mean   :0.04615   Mean   :0.1074   Mean   :151.7
##  3rd Qu.:0.2150   3rd Qu.:0.05000   3rd Qu.:0.1300   3rd Qu.:174.0
##  Max.   :0.2900   Max.   :0.07000   Max.   :0.1700   Max.   :308.0
##
##  lysoPC.a.C16.1_T0 lysoPC.a.C17.0_T0 lysoPC.a.C18.0_T0 lysoPC.a.C18.1_T0
##  Min.   :1.940     Min.   :0.790     Min.   : 14.30    Min.   :12.70
##  1st Qu.:2.870     1st Qu.:1.445     1st Qu.: 34.05    1st Qu.:22.25
##  Median :3.810     Median :2.110     Median : 45.20    Median :28.80
##  Mean   :4.283     Mean   :2.101     Mean   : 50.51    Mean   :32.32
##  3rd Qu.:5.500     3rd Qu.:2.425     3rd Qu.: 60.70    3rd Qu.:40.55
##  Max.   :9.890     Max.   :4.340     Max.   :120.00    Max.   :58.20
##
##  lysoPC.a.C18.2_T0 lysoPC.a.C20.3_T0 lysoPC.a.C20.4_T0 lysoPC.a.C24.0_T0
##  Min.   : 19.10    Min.   :1.190     Min.   : 4.21     Min.   :0.2400
##  1st Qu.: 34.75    1st Qu.:2.940     1st Qu.: 7.50     1st Qu.:0.3500
##  Median : 39.90    Median :4.090     Median :10.50     Median :0.4600
##  Mean   : 43.82    Mean   :4.373     Mean   :10.77     Mean   :0.4974
##  3rd Qu.: 50.75    3rd Qu.:5.335     3rd Qu.:13.55     3rd Qu.:0.5550
##  Max.   :109.00    Max.   :9.410     Max.   :19.60     Max.   :0.9200
##
##  lysoPC.a.C26.0_T0 lysoPC.a.C26.1_T0 lysoPC.a.C28.0_T0 lysoPC.a.C28.1_T0
##  Min.   :-9.0000   Min.   :0.1200    Min.   :-9.0000   Min.   :0.2400
##  1st Qu.: 0.3100   1st Qu.:0.2600    1st Qu.: 0.2650   1st Qu.:0.3800
##  Median : 0.4200   Median :0.3300    Median : 0.3400   Median :0.4600
##  Mean   : 0.2841   Mean   :0.3408    Mean   :-0.7536   Mean   :0.5238
##  3rd Qu.: 0.5050   3rd Qu.:0.4150    3rd Qu.: 0.4100   3rd Qu.:0.5900
##  Max.   : 2.1300   Max.   :0.6600    Max.   : 1.7200   Max.   :1.5000
##
##  PC.aa.C24.0_T0   PC.aa.C28.1_T0  PC.aa.C30.0_T0  PC.aa.C32.0_T0
##  Min.   :-9.000   Min.   :1.970   Min.   :0.630   Min.   : 5.360
##  1st Qu.: 0.145   1st Qu.:2.380   1st Qu.:1.420   1st Qu.: 8.525
##  Median : 0.180   Median :2.910   Median :1.710   Median :10.200
##  Mean   :-1.650   Mean   :3.014   Mean   :1.895   Mean   :10.695
##  3rd Qu.: 0.250   3rd Qu.:3.640   3rd Qu.:2.305   3rd Qu.:12.100
##  Max.   : 0.780   Max.   :4.770   Max.   :3.520   Max.   :17.300
##
##  PC.aa.C32.1_T0  PC.aa.C32.3_T0   PC.aa.C34.1_T0   PC.aa.C34.2_T0
##  Min.   : 2.64   Min.   :0.1200   Min.   : 50.60   Min.   :154.0
##  1st Qu.: 7.11   1st Qu.:0.1750   1st Qu.: 99.75   1st Qu.:245.0
##  Median :10.50   Median :0.2400   Median :136.00   Median :302.0
##  Mean   :11.77   Mean   :0.2362   Mean   :139.15   Mean   :338.7
##  3rd Qu.:12.80   3rd Qu.:0.2800   3rd Qu.:161.00   3rd Qu.:389.5
##  Max.   :29.60   Max.   :0.4400   Max.   :310.00   Max.   :904.0
##
##  PC.aa.C34.3_T0  PC.aa.C34.4_T0   PC.aa.C36.0_T0  PC.aa.C36.1_T0
##  Min.   : 3.94   Min.   :0.2000   Min.   :0.610   Min.   : 8.40
##  1st Qu.: 6.34   1st Qu.:0.3700   1st Qu.:1.030   1st Qu.:13.30
##  Median : 9.73   Median :0.6300   Median :1.300   Median :15.70
```

```
##  Mean   :10.29   Mean   :0.7015   Mean   :1.568   Mean   :18.32
##  3rd Qu.:11.00   3rd Qu.:0.9050   3rd Qu.:1.880   3rd Qu.:23.00
##  Max.   :24.40   Max.   :1.8600   Max.   :4.560   Max.   :43.30
##
##  PC.aa.C36.2_T0  PC.aa.C36.3_T0   PC.aa.C36.4_T0   PC.aa.C36.5_T0
##  Min.   : 60.6   Min.   : 36.60   Min.   : 40.10   Min.   : 2.770
##  1st Qu.: 89.8   1st Qu.: 58.90   1st Qu.: 95.15   1st Qu.: 6.595
##  Median :112.0   Median : 69.80   Median :120.00   Median : 9.200
##  Mean   :125.9   Mean   : 78.45   Mean   :120.26   Mean   : 9.591
##  3rd Qu.:151.0   3rd Qu.: 91.20   3rd Qu.:149.50   3rd Qu.:11.750
##  Max.   :282.0   Max.   :187.00   Max.   :214.00   Max.   :23.700
##
##  PC.aa.C38.0_T0  PC.aa.C38.1_T0   PC.aa.C38.3_T0   PC.aa.C38.4_T0
##  Min.   :1.340   Min.   :-9.000   Min.   : 16.90   Min.   : 34.20
##  1st Qu.:2.385   1st Qu.: 0.910   1st Qu.: 29.15   1st Qu.: 66.85
##  Median :3.000   Median : 1.100   Median : 37.40   Median : 92.20
##  Mean   :3.258   Mean   : 1.219   Mean   : 41.54   Mean   : 92.36
##  3rd Qu.:3.960   3rd Qu.: 1.625   3rd Qu.: 51.25   3rd Qu.:111.50
##  Max.   :6.180   Max.   : 6.230   Max.   :110.00   Max.   :164.00
##
##  PC.aa.C38.5_T0  PC.aa.C38.6_T0   PC.aa.C40.1_T0   PC.aa.C40.2_T0
##  Min.   :11.20   Min.   : 13.90   Min.   :-9.000   Min.   :0.1700
##  1st Qu.:20.55   1st Qu.: 43.90   1st Qu.: 0.335   1st Qu.:0.2300
##  Median :29.50   Median : 56.70   Median : 0.460   Median :0.3400
##  Mean   :28.74   Mean   : 62.95   Mean   :-1.550   Mean   :0.6377
##  3rd Qu.:36.75   3rd Qu.: 77.45   3rd Qu.: 0.565   3rd Qu.:0.4300
##  Max.   :47.70   Max.   :128.00   Max.   : 3.330   Max.   :5.7700
##
##  PC.aa.C40.3_T0  PC.aa.C40.4_T0  PC.aa.C40.5_T0   PC.aa.C40.6_T0
##  Min.   :0.370   Min.   :1.080   Min.   : 1.750   Min.   : 5.39
##  1st Qu.:0.490   1st Qu.:1.860   1st Qu.: 3.680   1st Qu.:16.15
##  Median :0.580   Median :2.500   Median : 4.910   Median :19.50
##  Mean   :0.871   Mean   :2.777   Mean   : 5.178   Mean   :22.21
##  3rd Qu.:0.780   3rd Qu.:3.385   3rd Qu.: 6.185   3rd Qu.:29.65
##  Max.   :4.820   Max.   :5.970   Max.   :11.100   Max.   :41.10
##
##  PC.aa.C42.0_T0   PC.aa.C42.1_T0   PC.aa.C42.2_T0   PC.aa.C42.4_T0
##  Min.   :0.3700   Min.   :0.2000   Min.   :0.1600   Min.   :0.1100
##  1st Qu.:0.5600   1st Qu.:0.2750   1st Qu.:0.2150   1st Qu.:0.1800
##  Median :0.6800   Median :0.3500   Median :0.2800   Median :0.2200
##  Mean   :0.7562   Mean   :0.4087   Mean   :0.4064   Mean   :0.3379
##  3rd Qu.:0.8700   3rd Qu.:0.4250   3rd Qu.:0.3300   3rd Qu.:0.2450
##  Max.   :1.7400   Max.   :1.3800   Max.   :2.3100   Max.   :2.6100
##
##  PC.aa.C42.5_T0   PC.aa.C42.6_T0     PC.ae.C30.0_T0   PC.ae.C32.1_T0
##  Min.   :0.1300   Min.   :-9.00000   Min.   :0.2100   Min.   :1.220
##  1st Qu.:0.2600   1st Qu.: 0.27500   1st Qu.:0.2700   1st Qu.:1.675
##  Median :0.3200   Median : 0.34000   Median :0.3300   Median :2.160
##  Mean   :0.3818   Mean   :-0.08128   Mean   :0.3262   Mean   :2.219
##  3rd Qu.:0.4000   3rd Qu.: 0.42500   3rd Qu.:0.3600   3rd Qu.:2.625
##  Max.   :1.4400   Max.   : 1.09000   Max.   :0.5000   Max.   :4.120
##
##  PC.ae.C32.2_T0   PC.ae.C34.0_T0  PC.ae.C34.1_T0  PC.ae.C34.2_T0
##  Min.   :0.3500   Min.   :0.250   Min.   :2.760   Min.   : 3.030
```

```
##  1st Qu.:0.5450   1st Qu.:0.390   1st Qu.:4.135   1st Qu.: 5.310
##  Median :0.6700   Median :0.470   Median :4.850   Median : 6.170
##  Mean   :0.6997   Mean   :0.511   Mean   :5.147   Mean   : 7.382
##  3rd Qu.:0.8300   3rd Qu.:0.630   3rd Qu.:6.395   3rd Qu.: 8.590
##  Max.   :1.2000   Max.   :0.840   Max.   :9.350   Max.   :19.000
##
##  PC.ae.C34.3_T0   PC.ae.C36.0_T0  PC.ae.C36.1_T0   PC.ae.C36.2_T0
##  Min.   : 1.450   Min.   :0.470   Min.   : 1.510   Min.   : 3.230
##  1st Qu.: 3.285   1st Qu.:0.760   1st Qu.: 2.310   1st Qu.: 4.390
##  Median : 4.140   Median :0.940   Median : 3.120   Median : 5.210
##  Mean   : 4.581   Mean   :1.008   Mean   : 3.922   Mean   : 5.852
##  3rd Qu.: 5.030   3rd Qu.:1.135   3rd Qu.: 3.695   3rd Qu.: 6.965
##  Max.   :10.100   Max.   :1.980   Max.   :18.200   Max.   :10.800
##
##  PC.ae.C36.3_T0   PC.ae.C36.4_T0   PC.ae.C36.5_T0   PC.ae.C38.0_T0
##  Min.   : 1.660   Min.   : 2.180   Min.   : 1.760   Min.   :0.490
##  1st Qu.: 2.710   1st Qu.: 6.490   1st Qu.: 4.950   1st Qu.:1.265
##  Median : 3.120   Median : 7.940   Median : 6.240   Median :1.630
##  Mean   : 3.812   Mean   : 8.826   Mean   : 6.642   Mean   :1.623
##  3rd Qu.: 4.610   3rd Qu.:10.850   3rd Qu.: 8.215   3rd Qu.:1.945
##  Max.   :11.000   Max.   :20.100   Max.   :13.100   Max.   :2.960
##
##  PC.ae.C38.2_T0   PC.ae.C38.3_T0   PC.ae.C38.4_T0   PC.ae.C38.5_T0
##  Min.   : 0.350   Min.   : 0.830   Min.   : 3.100   Min.   : 3.370
##  1st Qu.: 1.195   1st Qu.: 1.250   1st Qu.: 4.440   1st Qu.: 7.470
##  Median : 1.580   Median : 1.530   Median : 5.480   Median : 8.610
##  Mean   : 2.752   Mean   : 2.235   Mean   : 5.729   Mean   : 9.477
##  3rd Qu.: 2.115   3rd Qu.: 1.990   3rd Qu.: 6.425   3rd Qu.:10.500
##  Max.   :25.500   Max.   :11.200   Max.   :11.100   Max.   :19.200
##
##  PC.ae.C38.6_T0  PC.ae.C40.1_T0  PC.ae.C40.2_T0  PC.ae.C40.3_T0
##  Min.   :1.380   Min.   :0.450   Min.   :0.620   Min.   : 0.560
##  1st Qu.:2.715   1st Qu.:0.895   1st Qu.:0.965   1st Qu.: 0.740
##  Median :3.740   Median :1.160   Median :1.170   Median : 0.880
##  Mean   :3.995   Mean   :1.373   Mean   :1.557   Mean   : 1.574
##  3rd Qu.:4.915   3rd Qu.:1.570   3rd Qu.:1.390   3rd Qu.: 1.160
##  Max.   :7.690   Max.   :4.540   Max.   :8.440   Max.   :11.400
##
##  PC.ae.C40.4_T0  PC.ae.C40.5_T0  PC.ae.C40.6_T0  PC.ae.C42.1_T0
##  Min.   :1.020   Min.   :1.050   Min.   :1.340   Min.   :0.2100
##  1st Qu.:1.645   1st Qu.:1.570   1st Qu.:1.700   1st Qu.:0.3600
##  Median :1.940   Median :1.790   Median :2.200   Median :0.4300
##  Mean   :2.380   Mean   :2.317   Mean   :2.309   Mean   :0.6195
##  3rd Qu.:2.610   3rd Qu.:2.385   3rd Qu.:3.035   3rd Qu.:0.5650
##  Max.   :7.970   Max.   :8.690   Max.   :3.540   Max.   :3.4000
##
##  PC.ae.C42.2_T0   PC.ae.C42.3_T0
##  Min.   :0.2900   Min.   :0.4600
##  1st Qu.:0.3800   1st Qu.:0.6050
##  Median :0.4700   Median :0.7900
##  Mean   :0.6236   Mean   :0.9569
##  3rd Qu.:0.6100   3rd Qu.:0.9450
##  Max.   :2.8200   Max.   :4.1900
##
```

Veiem que la variabilitat en els rangs és molt elevada i, per tant, caldrà normalitzar les dades. Recordem també que tenim un nombre de NAs força elevat el que implica fer alguna mena d'imputació per eliminar problemes computacionals.

Tot i això, abans de continuar, generaré el fitxer rdata amb seMA

```
save(seMA, file = "AntonRecasensMarcSe.rdata")
```

5.2. Preprocessat

Amb el que hem pogut veure fins aquó de les dades que tenim és que ens calen dos procesos per poder començar els anàlisi multivariants:

1. Imputar els missing values
2. Normalitzar les dades

Això ho podem fer amb POMA

```
# primer instal·lem el paquet i les llibreries
# BiocManager::install("POMA") # Això només el primer cop per instal·lar el paquet
library(POMA)
```

```
## Welcome to POMA!
## Version 1.16.0
## POMAShiny app: https://github.com/pcastellanoescuder/POMAShiny
```

```
library(ggtext)
```

```
## Warning: package 'ggtext' was built under R version 4.4.3
```

```
library(magrittr)
```

```
##
## Attaching package: 'magrittr'
```

```
## The following object is masked from 'package:GenomicRanges':
##
##     subtract
```

Ataquem primer la imputació.

Abans de fer-la cal dir que no podem imputar NA en columnes on no tenim cap valor. Per tant, mirem si hi ha alguna columna que no tingui cap valor i l'eliminem.

```
# Busquem columnes amb tot NAs
eliminats = c()
for (i in 1:685)
  if (sum(is.na(assay(seMA)[,i])) == 39)
  {print((paste(i, sum(is.na(assay(seMA)[,i])))))
    eliminats = c(eliminats, i)}
```

```
## [1] "248 39"
```

Eliminem per tant, la columna 248

```
seMAclean <- seMA[,-eliminats]
```

I fem la imputació dels NA utilitzant el knn que és possiblement el mètode més robust per fer-ho

```
seMAimputed <- seMAclean %>%
  PomaImpute(method = "knn", zeros_as_na = FALSE, remove_na = FALSE, cutoff = 20)
```

```
## -39 features removed.
```

Després comprovem que no tenim NAs

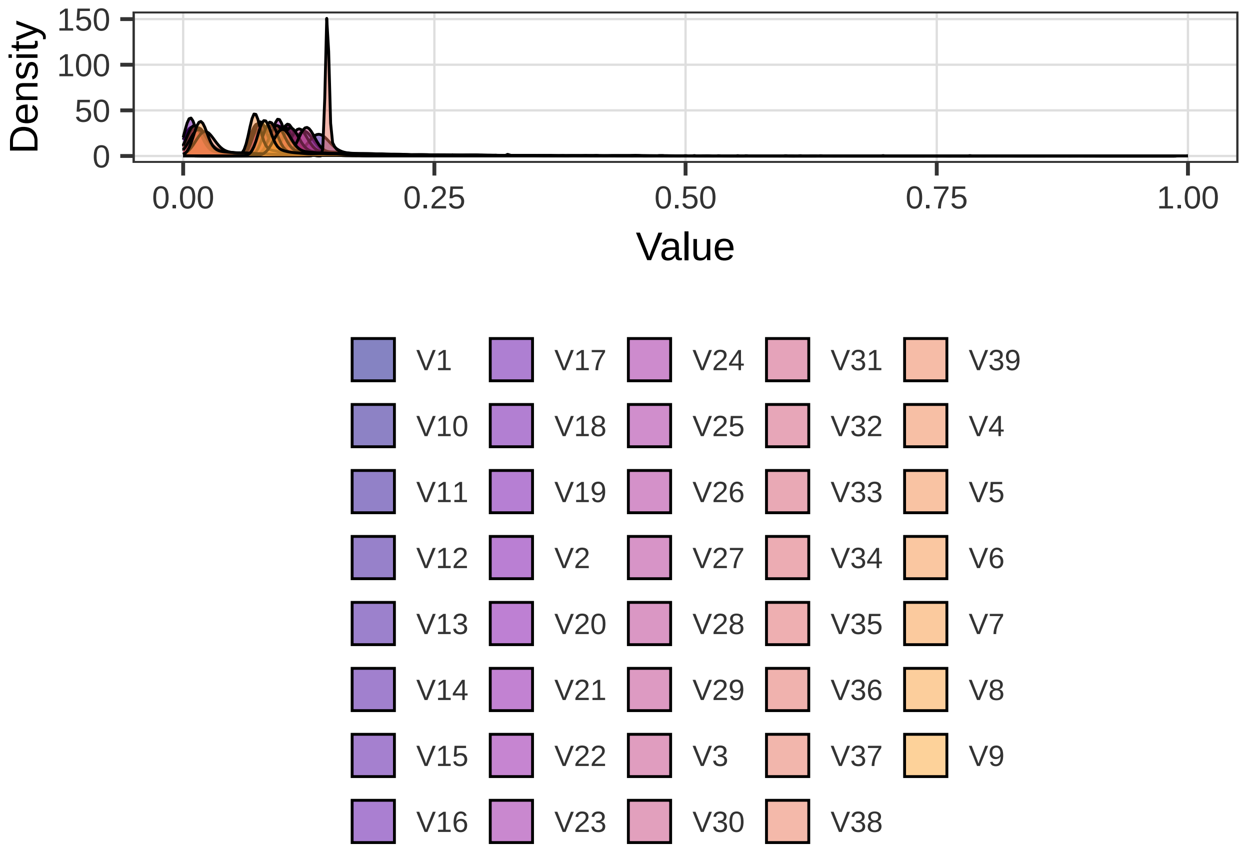
```
sum(is.na(assay(seMAimputed)))
```

```
## [1] 0
```

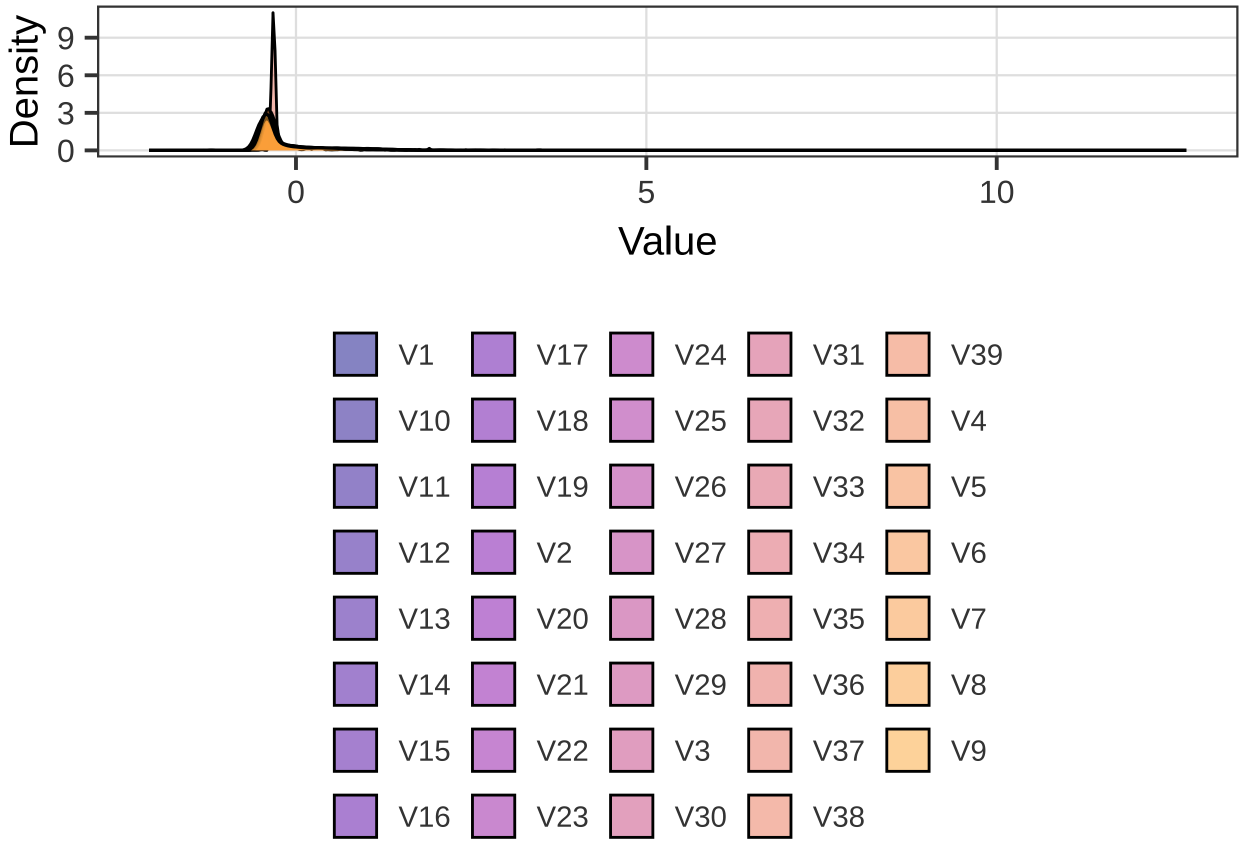
Doncs ara, ha podem normalitzar les dades. D'entre els mètodes que es poden utilitzar hem rebutjat els logaritmes perquè amb les dades que tenim ens generen NAs, ek mix_max perquè genera molta variabilitat en les dades en el nostre cas i el box_cox perquè no funciona amb la matriu que tenim.

Ens queden els dos mètodes d'escalat i el auto_scaling és el que dona el gràfic més homogeni. Així que ens quedem amb aquest

```
# Ensenyem el resultat de min_max per veure que les funcions de densitat són molt variables
seMAnormalized <- seMAimputed %>%
  PomaNorm(method = "min_max")
PomaDensity(seMAnormalized, x = "features")
```




```
# Ensenyem el resultat de min_max per veure que les funcions de densitat són molt variables
seMAnormalized <- seMAimputed %>%
  PomaNorm(method = "auto_scaling")
PomaDensity(seMAnormalized, x = "features")
```

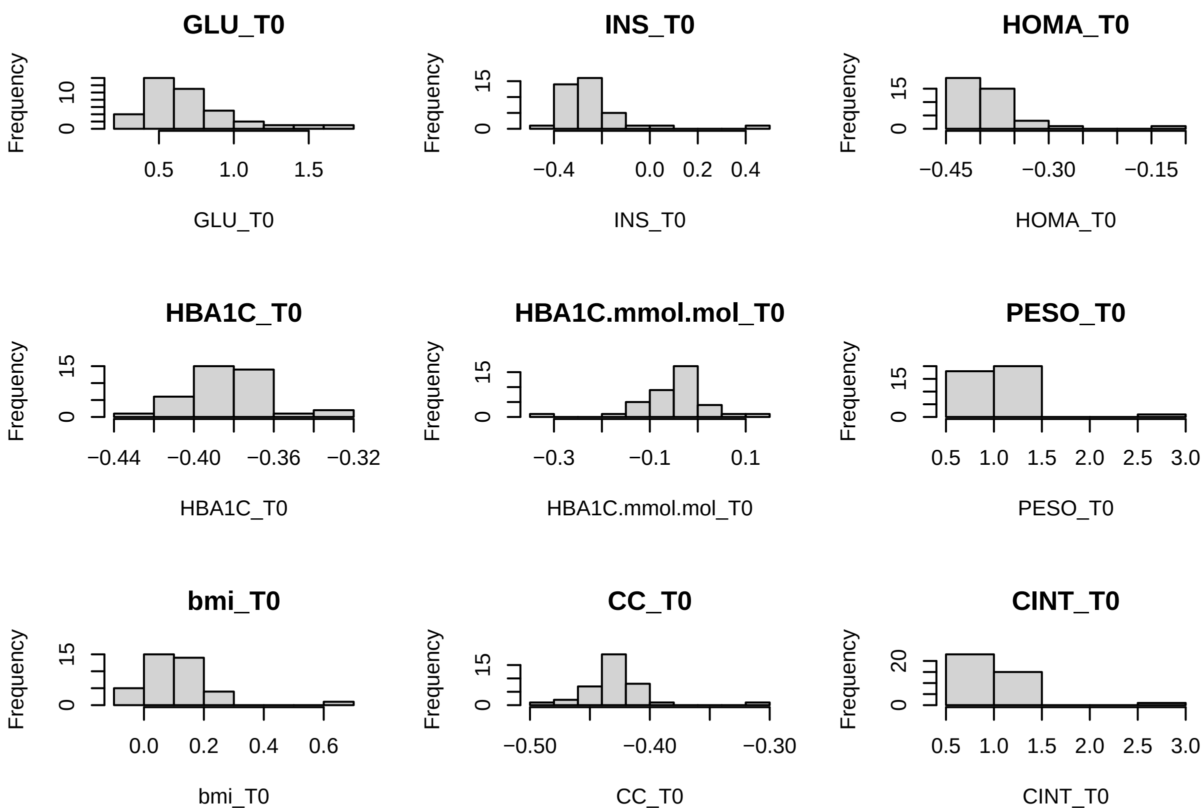


Sembla que hi ha algun outlier, però crec que no és preocupant

5.4. Analisi univariant

L'anàlisi univariant no té massa sentit amb una matriu tant gran de dades, però per provar alguna cosa, podem dir que fer alguns histogrames no està de mas. Farem 9 de les variables per veure una mica com es distribueixen després del preprocessament

```
opt <- par(mfrow=c(3,3))
for (i in 1:9)
  hist(assay(seMAnormalized)[,i],
       xlab = colData(seMAnormalized)[i,2],
       main = colData(seMAnormalized)[i,2])
```




```r
par(opt)
```

Podem apreciar que les distribucions no són molt homogenies, el que potser implica poques correlacions. Veurem més endavant

6. Reducció de les dimensions

Un pas important ara que hem vist com són les variables pot ser fer un Anàlisi de Components Principals (PCA)). Com que tenim els NAs eliminats i les dades normalitzades, podem fer-lo sense problemes addicionals. Reduim la PCA a 5 components ja que sinó en fa 39 i el resultat s'allarga molt

```r
PCAseMA <- prcomp(assay(seMAnormalized), scale. = TRUE, rank. = 5)
PCAseMA
```

```
## Standard deviations (1, .., p=39):
##  [1] 1.893256e+01 7.377935e+00 5.806778e+00 5.366514e+00 5.212884e+00
##  [6] 4.421286e+00 4.032098e+00 3.610802e+00 3.464758e+00 2.993985e+00
## [11] 2.905096e+00 2.840228e+00 2.671005e+00 2.633563e+00 2.591891e+00
## [16] 2.550440e+00 2.438817e+00 2.282599e+00 2.222872e+00 2.118927e+00
## [21] 2.067725e+00 2.042961e+00 1.985076e+00 1.889037e+00 1.853185e+00
## [26] 1.802430e+00 1.759200e+00 1.735222e+00 1.711374e+00 1.624513e+00
## [31] 1.531975e+00 1.494274e+00 1.382320e+00 1.374211e+00 1.328333e+00
## [36] 1.224837e+00 1.109147e+00 1.021286e+00 1.161219e-14
##
## Rotation (n x k) = (685 x 5):
##                              PC1           PC2           PC3           PC4
## GLU_T0             -1.199624e-02  4.941264e-02 -2.319322e-02 -1.078391e-02
```

```
## INS_T0            -3.040036e-02  5.672071e-02 -2.721737e-02  2.233584e-02
## HOMA_T0           -4.080096e-02  3.104342e-02 -2.125132e-02  7.222602e-03
## HBA1C_T0          -4.651415e-02 -1.834343e-02  2.008314e-02 -5.565033e-04
## HBA1C.mmol.mol_T0  3.552732e-02 -9.188496e-03  2.104319e-02  4.572940e-04
## PESO_T0           -2.326499e-02  8.125680e-02 -3.630520e-02 -3.611425e-03
## bmi_T0            -2.471270e-02  8.087699e-02 -3.633006e-02  1.303281e-02
## CC_T0             -5.244727e-02 -1.569056e-03  6.832376e-03  6.712868e-03
## CINT_T0           -2.108588e-02  1.005267e-01 -1.260106e-02  2.898376e-02
## CAD_T0            -1.094508e-02  1.031612e-01 -1.546234e-02 -6.292826e-03
## TAD_T0             3.368663e-02 -5.542617e-03  1.880689e-02 -1.076498e-02
## TAS_T0             4.155077e-02 -5.256121e-04  3.309474e-03 -4.469568e-03
## TG_T0              4.626877e-03  2.775474e-02 -5.204236e-02  5.830628e-02
## COL_T0            -3.209452e-03  4.890035e-02 -1.392578e-02  2.888548e-02
## LDL_T0            -9.495424e-03  6.748697e-02 -1.252003e-02  2.204182e-02
## HDL_T0             6.530946e-04 -3.647308e-02  2.226752e-02 -1.236797e-02
## VLDL_T0           -4.486175e-03  2.630072e-02 -5.128724e-02  5.894473e-02
## PCR_T0            -1.594345e-02 -2.534249e-02  2.709684e-02 -1.312719e-02
## LEP_T0            -2.977204e-02  5.974349e-02 -1.459339e-02  5.927173e-03
## ADIPO_T0          -2.157399e-02  5.445112e-03 -1.229830e-02 -2.448064e-02
## GOT_T0            -1.097243e-02  3.930316e-03  3.068829e-02  3.609539e-03
## GPT_T0            -1.042910e-02  6.848081e-03 -1.324231e-02 -8.294370e-03
## GGT_T0            -2.106519e-02  5.718606e-02 -4.429951e-02  4.709703e-02
## URICO_T0          -4.690040e-02  3.257911e-02 -1.050633e-02  2.145387e-02
## CREAT_T0          -5.238177e-02 -4.462782e-03  7.125217e-03  3.583363e-04
## UREA_T0           -1.102555e-02  4.186594e-02 -8.738753e-03  3.110859e-02
## HIERRO_T0         -3.876584e-03  4.665753e-02 -5.411954e-03  2.833490e-02
## TRANSF_T0          3.990148e-02 -6.735974e-03  1.411681e-02  2.756888e-02
## FERR_T0            1.100355e-02  2.202379e-02  1.604286e-02  1.859058e-03
## Ile_T0            -9.979157e-03  4.347839e-02 -8.025407e-02  3.102797e-02
## Leu_T0            -1.901120e-03  3.243955e-02 -8.925402e-02  2.524131e-02
## Val_T0             1.370998e-03  2.254413e-02 -9.141408e-02  3.520201e-02
## Ala_T0            -5.657694e-03  5.095690e-02 -1.196257e-01  2.891007e-02
## Pro_T0            -2.375758e-02  4.844247e-02 -8.208939e-02 -1.149389e-02
## Gly_T0            -1.004746e-02  3.877661e-02 -6.471853e-02 -2.918055e-02
## Ser_T0            -1.665486e-02  5.123520e-02 -7.174336e-02 -1.949394e-02
## Trp_T0            -1.706001e-02  1.060458e-02 -9.699474e-02  2.718742e-02
## Phe_T0            -8.719772e-03  4.965718e-02 -1.062346e-01  1.202598e-04
## Met_T0             2.364641e-02 -8.029138e-02  4.323907e-02 -2.661267e-02
## Orn_T0            -9.794162e-03  4.300912e-03 -8.253731e-02  2.358071e-02
## Arg_T0            -8.207693e-03  7.242484e-02 -9.691286e-02  5.283762e-02
## His_T0            -1.961687e-02  3.543715e-02 -1.080403e-01 -6.512983e-03
## Asn_T0            -2.278992e-02 -1.581569e-02 -2.588730e-02 -2.529066e-02
## Asp_T0            -2.454699e-02  7.611141e-02 -6.003059e-02  3.112978e-02
## Glu_T0            -1.891976e-02  9.132817e-02 -6.796917e-02  5.490545e-02
## Gln_T0            -6.384554e-03 -5.760449e-02 -2.247765e-02 -7.017922e-02
## Cit_T0            -3.067717e-02  4.560629e-02 -3.427953e-02  3.033582e-02
## Tyr_T0            -1.926940e-02  2.538624e-02 -1.016609e-01  1.033410e-02
## Thr_T0            -3.264978e-03 -2.056393e-02 -4.111030e-02  1.093298e-02
## Lys_T0            -6.434112e-03  1.624795e-02 -1.015303e-01  4.084040e-02
## Creatinine_T0     -1.030747e-02  5.069871e-02 -3.989496e-02 -1.454879e-03
## Kynurenine_T0     -4.960725e-02 -3.669416e-03 -2.642139e-02 -9.883789e-04
## Putrescine_T0     -7.308058e-03 -6.104151e-04 -1.309582e-02  5.102346e-03
## Sarcosine_T0      -3.160741e-02  4.123728e-02 -1.020420e-02  2.945476e-02
## Serotonin_T0      -7.413211e-03 -7.621286e-03  4.666980e-02  5.402820e-03
```

```
## Taurine_T0        -1.772370e-02  3.613755e-02 -3.522009e-02  1.739916e-02
## SDMA_T0           -5.241650e-02 -2.826580e-03  3.299265e-03  6.957672e-04
## C0_T0             -2.576613e-02  4.587907e-02 -8.069356e-02  4.560331e-02
## C2_T0             -2.218579e-02 -2.907379e-02  3.813234e-02 -2.346379e-02
## C3.OH_T0          -5.247407e-02 -8.697286e-03  6.063526e-03  1.223400e-03
## C6..C4.1.DC._T0    3.267974e-03  3.041575e-02  4.521626e-02  1.201054e-02
## C5.DC..C6.OH._T0  -3.565798e-02 -1.456302e-02  3.969868e-03  1.830535e-02
## C7.DC_T0          -3.495627e-02 -5.186282e-03  6.317214e-03 -5.121927e-03
## C8_T0             -5.250656e-02 -7.142245e-03  5.207529e-03  2.284466e-03
## C10_T0            -5.251524e-02 -5.433485e-03  3.792803e-03  2.205449e-03
## C10.1_T0          -5.249195e-02 -8.578609e-03  4.615177e-03  1.155361e-03
## C10.2_T0          -5.245744e-02 -8.645811e-03  6.548846e-03  1.100308e-03
## C14.1_T0          -5.249986e-02 -8.257121e-03  5.865208e-03  1.783103e-03
## C14.2_T0          -5.247850e-02 -8.609041e-03  6.262901e-03  1.412700e-03
## C16.1_T0          -5.248726e-02 -8.523857e-03  6.458853e-03  1.373352e-03
## C16.2_T0          -5.246910e-02 -9.095494e-03  6.594646e-03  1.459950e-03
## C16.2.OH_T0       -5.247213e-02 -9.103851e-03  6.569942e-03  1.380622e-03
## C18.1_T0          -5.253721e-02 -7.752416e-03  5.969628e-03  1.142369e-03
## C18.1.OH_T0       -5.246717e-02 -9.145540e-03  6.585084e-03  1.396145e-03
## C18.2_T0          -5.248329e-02 -8.754111e-03  6.588226e-03  1.652488e-03
## lysoPC.a.C16.0_T0 -1.888892e-02  7.119173e-02 -1.024627e-01  3.386555e-02
## lysoPC.a.C16.1_T0 -4.645715e-02  2.996157e-02 -5.568621e-02  1.882351e-02
## lysoPC.a.C17.0_T0 -5.085825e-02  1.727304e-02 -1.904967e-02  1.555739e-02
## lysoPC.a.C18.0_T0 -1.950736e-02  5.616350e-02 -1.097307e-01  3.486232e-02
## lysoPC.a.C18.1_T0 -2.845384e-02  5.606669e-02 -7.789242e-02  2.511448e-02
## lysoPC.a.C18.2_T0 -2.672044e-02  1.336755e-02 -8.544482e-02  9.233331e-03
## lysoPC.a.C20.3_T0 -4.491626e-02  2.014642e-03 -5.759618e-02  2.154599e-02
## lysoPC.a.C20.4_T0 -3.186501e-02  3.680045e-02 -4.321636e-02  1.251284e-02
## lysoPC.a.C24.0_T0 -5.259933e-02 -4.423072e-03  1.436010e-03  2.071438e-03
## lysoPC.a.C26.0_T0 -4.635546e-02 -8.375227e-04 -1.651850e-02  2.296390e-02
## lysoPC.a.C26.1_T0 -5.260177e-02 -4.112828e-03  3.559603e-03  7.598541e-04
## lysoPC.a.C28.0_T0 -3.820598e-02 -1.176691e-02 -3.463173e-02  1.541886e-02
## lysoPC.a.C28.1_T0 -5.268437e-02  9.811227e-04 -1.621543e-03  3.456510e-03
## PC.aa.C24.0_T0    -3.450394e-02  1.405759e-02 -1.144507e-02 -1.788179e-02
## PC.aa.C28.1_T0    -5.090210e-02 -5.572934e-03 -8.989270e-03  1.676002e-03
## PC.aa.C30.0_T0    -5.076110e-02 -1.062761e-02 -4.171637e-03  5.860590e-03
## PC.aa.C32.0_T0    -4.047902e-02 -1.049841e-02 -2.208971e-02 -5.985915e-03
## PC.aa.C32.1_T0    -2.933937e-02 -4.226707e-03 -5.208939e-02  1.485087e-02
## PC.aa.C32.3_T0    -5.254305e-02 -8.265740e-03  4.482921e-03  2.645503e-03
## PC.aa.C34.1_T0    -4.679961e-03 -2.399894e-03 -5.577084e-02  2.887736e-02
## PC.aa.C34.2_T0    -5.839240e-03 -5.785504e-02 -3.948971e-02  3.145463e-02
## PC.aa.C34.3_T0    -3.396562e-02 -2.782546e-02 -5.344318e-02  2.916803e-02
## PC.aa.C34.4_T0    -5.193914e-02 -1.288689e-02 -9.518723e-04  7.521392e-03
## PC.aa.C36.0_T0    -5.113143e-02  1.940279e-02 -1.720350e-02  1.913615e-02
## PC.aa.C36.1_T0    -2.407247e-02 -2.667213e-03 -7.121721e-02  5.948390e-02
## PC.aa.C36.2_T0    -8.060236e-03 -4.625739e-02 -6.466646e-02  3.938564e-02
## PC.aa.C36.3_T0    -6.934274e-03 -4.507168e-02 -5.754152e-02  4.404455e-02
## PC.aa.C36.4_T0     7.269459e-03 -1.944508e-02 -3.726738e-03  2.456261e-02
## PC.aa.C36.5_T0    -2.377410e-02  1.315630e-02 -1.724130e-02  5.942156e-02
## PC.aa.C38.0_T0    -4.909721e-02  1.107830e-02 -2.123144e-02  1.879520e-02
## PC.aa.C38.1_T0    -4.226607e-02  3.070666e-02 -3.542939e-04  3.360842e-02
## PC.aa.C38.3_T0    -7.449456e-03 -2.944195e-02 -8.810957e-02  4.505764e-02
## PC.aa.C38.4_T0     4.819443e-03 -1.588009e-02 -3.221724e-02  1.804556e-02
## PC.aa.C38.5_T0    -8.793357e-03 -1.005540e-02 -2.803464e-02  3.987593e-02
```

```
## PC.aa.C38.6_T0     3.682175e-03 -5.666075e-03 -2.843888e-03  1.717111e-02
## PC.aa.C40.1_T0    -3.532944e-02 -7.802958e-03 -5.990080e-02  3.450067e-02
## PC.aa.C40.2_T0    -4.948016e-02  2.974843e-02 -2.947529e-02  2.070395e-02
## PC.aa.C40.3_T0    -5.078606e-02  2.243592e-02 -2.445202e-02  1.596725e-02
## PC.aa.C40.4_T0    -4.967989e-02  1.463772e-02 -4.116538e-02  2.311102e-02
## PC.aa.C40.5_T0    -4.508810e-02 -1.838723e-02 -3.328628e-02  1.524805e-02
## PC.aa.C40.6_T0    -6.193391e-03 -4.759122e-03 -2.173514e-02  3.784442e-02
## PC.aa.C42.0_T0    -5.263114e-02  3.232547e-03 -4.267086e-03  5.772590e-03
## PC.aa.C42.1_T0    -5.270876e-02  1.840187e-03 -2.633491e-03  5.705166e-03
## PC.aa.C42.2_T0    -5.235307e-02  9.994660e-03 -1.034820e-02  9.934093e-03
## PC.aa.C42.4_T0    -5.233352e-02  7.367583e-03 -1.202856e-02  9.351755e-03
## PC.aa.C42.5_T0    -5.269167e-02 -1.287353e-03 -3.441852e-03  4.735824e-03
## PC.aa.C42.6_T0    -4.304304e-02 -4.360666e-03 -1.269944e-02  1.956059e-02
## PC.ae.C30.0_T0    -5.254257e-02 -7.763565e-03  3.942071e-03  1.377721e-03
## PC.ae.C32.1_T0    -5.134015e-02 -1.054594e-02 -4.191666e-03 -3.771758e-03
## PC.ae.C32.2_T0    -5.254671e-02 -7.489553e-03  4.855983e-04  2.477391e-03
## PC.ae.C34.0_T0    -5.252130e-02 -7.244882e-03  4.246375e-03  3.879746e-03
## PC.ae.C34.1_T0    -4.894199e-02 -1.066988e-02 -7.958685e-03  5.977845e-03
## PC.ae.C34.2_T0    -3.815767e-02 -3.850930e-02 -2.519837e-02  1.399627e-02
## PC.ae.C34.3_T0    -4.339116e-02 -3.679685e-02 -1.527740e-02  6.648265e-03
## PC.ae.C36.0_T0    -5.256366e-02  2.609808e-03 -5.873801e-03  8.648078e-03
## PC.ae.C36.1_T0    -3.954970e-02  6.158932e-02 -5.052192e-02  3.778079e-02
## PC.ae.C36.2_T0    -4.722815e-02  1.147999e-02 -1.985515e-02  2.980044e-02
## PC.ae.C36.3_T0    -4.564989e-02 -2.815843e-02 -1.852507e-02  1.615099e-02
## PC.ae.C36.4_T0    -3.142224e-02 -3.911879e-02 -2.895686e-02  1.854032e-02
## PC.ae.C36.5_T0    -3.810759e-02 -3.191308e-02 -1.404708e-02  1.430391e-02
## PC.ae.C38.0_T0    -5.176857e-02 -5.287911e-03 -5.684053e-03  1.273141e-02
## PC.ae.C38.2_T0    -3.518049e-02  5.861955e-02 -6.408075e-02  3.593753e-02
## PC.ae.C38.3_T0    -4.089493e-02  5.243691e-02 -5.363895e-02  5.026145e-02
## PC.ae.C38.4_T0    -4.569485e-02  2.625039e-03 -2.448249e-02  2.056411e-02
## PC.ae.C38.5_T0    -3.192526e-02 -2.723012e-02 -1.504464e-02  2.332015e-03
## PC.ae.C38.6_T0    -4.613645e-02 -2.173110e-02 -5.003642e-03  1.414507e-02
## PC.ae.C40.1_T0    -5.084815e-02  2.284226e-02 -2.106949e-02  1.964880e-02
## PC.ae.C40.2_T0    -4.710051e-02  3.631939e-02 -3.784293e-02  2.692484e-02
## PC.ae.C40.3_T0    -4.020022e-02  5.202573e-02 -5.975111e-02  4.330442e-02
## PC.ae.C40.4_T0    -4.713499e-02  3.459666e-02 -4.848859e-02  2.895913e-02
## PC.ae.C40.5_T0    -4.608866e-02  4.094891e-02 -4.734619e-02  3.106280e-02
## PC.ae.C40.6_T0    -5.133905e-02 -7.497071e-03  4.333479e-05  7.503292e-03
## PC.ae.C42.1_T0    -5.172954e-02  1.700410e-02 -1.623628e-02  1.361134e-02
## PC.ae.C42.2_T0    -5.214331e-02  1.293312e-02 -1.249491e-02  1.188298e-02
## PC.ae.C42.3_T0    -5.140818e-02  1.829603e-02 -1.933481e-02  1.482275e-02
## PC.ae.C42.4_T0    -5.233265e-02  5.381443e-03 -1.570495e-02  9.153817e-03
## PC.ae.C42.5_T0    -5.075374e-02  1.856278e-02 -2.889258e-02  1.420372e-02
## PC.ae.C44.3_T0    -5.262260e-02  2.564179e-03 -4.501264e-03  6.423447e-03
## PC.ae.C44.4_T0    -5.264653e-02 -3.518964e-03 -8.844004e-04  3.910610e-03
## PC.ae.C44.5_T0    -5.078459e-02 -1.029710e-02  2.150019e-03 -1.480206e-03
## PC.ae.C44.6_T0    -5.167694e-02 -1.145469e-02 -1.209864e-03 -4.265336e-04
## SM..OH..C14.1_T0  -4.928159e-02 -1.530486e-02  9.313234e-03 -5.387356e-03
## SM..OH..C16.1_T0  -5.181957e-02 -8.689912e-03  1.020446e-02  1.710025e-03
## SM..OH..C22.1_T0  -3.615453e-02 -4.026414e-02  6.899994e-03  1.878689e-02
## SM..OH..C22.2_T0  -4.666995e-02  3.342502e-03 -1.296481e-02  1.391046e-02
## SM..OH..C24.1_T0  -5.226009e-02 -1.048512e-02  8.167850e-03  2.521347e-03
## SM.C16.0_T0       -7.828120e-03 -8.261959e-03 -3.558635e-03 -1.574621e-02
## SM.C16.1_T0       -3.170526e-02 -2.404167e-02 -1.615961e-02 -1.583856e-02
```

```
## SM.C18.0_T0        -2.901498e-02  1.082833e-02  1.581796e-02  2.361674e-03
## SM.C18.1_T0        -3.905315e-02 -3.637680e-03  8.261195e-03 -6.774493e-03
## SM.C20.2_T0        -5.245883e-02 -9.913957e-03  6.340898e-03  1.655250e-03
## SM.C24.0_T0        -3.504443e-02 -1.304466e-02 -3.389091e-02  1.930640e-02
## SM.C24.1_T0        -1.898667e-02  1.945600e-02  5.773657e-03  3.296389e-03
## MEDDM_T2           -5.245502e-02 -9.564885e-03  6.632371e-03  1.080576e-03
## MEDCOL_T2          -5.245502e-02 -9.564885e-03  6.632371e-03  1.080576e-03
## MEDINF_T2          -5.211366e-02 -1.071897e-02  8.022690e-03  4.077391e-03
## MEDHTA_T2          -5.191703e-02 -6.650130e-03  4.399277e-04  8.783626e-03
## GLU_T2              3.825544e-02  4.252139e-03  8.247871e-03  6.010433e-03
## INS_T2             -4.434399e-04 -1.904866e-02  1.892014e-02 -5.602655e-02
## HOMA_T2            -3.988365e-02 -1.687539e-02  1.363804e-02 -2.900605e-02
## HBA1C_T2           -4.792830e-02 -1.739073e-02  2.556816e-02 -3.765267e-03
## HBA1C.mmol.mol_T2   3.171862e-02 -1.119653e-02  5.174481e-02 -1.201423e-02
## PESO_T2             3.380214e-02 -1.080327e-02 -2.174437e-02 -3.614014e-02
## bmi_T2              3.172695e-02 -1.218128e-02 -2.219040e-02 -1.703664e-02
## CC_T2              -5.216416e-02 -9.795423e-03  8.209750e-03  2.364884e-03
## CINT_T2             4.225017e-02 -1.881668e-03 -4.082745e-03 -4.780645e-03
## CAD_T2              4.179569e-02 -1.463169e-02 -1.059582e-02 -2.620972e-02
## TAD_T2              4.150739e-02  7.100272e-03 -2.455751e-02  2.022245e-02
## TAS_T2              4.103215e-02  8.194423e-03 -1.866632e-02  1.211508e-03
## TG_T2               2.816995e-02 -5.140429e-03  2.624929e-02  8.348114e-03
## COL_T2              3.435067e-02 -1.191990e-02  5.950666e-02  3.593478e-03
## LDL_T2              3.493637e-02  6.368421e-04  5.580306e-02  1.864162e-02
## HDL_T2              1.345175e-02 -2.688160e-02  6.069721e-02 -3.438750e-03
## VLDL_T2             9.435087e-03 -2.811919e-02  3.547613e-02 -1.640362e-02
## PCR_T2             -1.412898e-02 -7.153529e-04  4.185992e-02 -2.697978e-02
## LEP_T2              1.146354e-02 -1.097647e-02  1.987033e-02  2.800129e-03
## ADIPO_T2           -9.355925e-03  1.016337e-03 -1.348928e-02 -1.676050e-02
## GOT_T2              2.360281e-03 -4.586665e-02  1.356174e-02  1.720447e-02
## GPT_T2              7.162270e-03 -2.884920e-02 -5.487717e-03  1.160603e-02
## GGT_T2              7.795501e-03 -2.681696e-02  2.554680e-02  1.852152e-02
## URICO_T2           -3.060967e-02 -2.809284e-02  2.956216e-03 -3.039250e-02
## CREAT_T2           -5.200608e-02 -1.218551e-02  7.035255e-03 -3.953076e-03
## UREA_T2             1.408107e-02  1.329286e-02  7.416350e-02 -2.894384e-02
## HIERRO_T2           1.517783e-02 -1.184607e-02  1.936847e-02  6.521345e-02
## TRANSF_T2           4.199475e-02 -1.246983e-02  2.440201e-02  2.761878e-03
## FERR_T2             7.675456e-03  1.469931e-02 -8.725266e-03 -1.736711e-02
## Ile_T2             -8.621274e-03  5.497743e-02 -2.719281e-03 -1.001336e-01
## Leu_T2             -7.991628e-03  5.891425e-02  4.834765e-04 -1.001346e-01
## Val_T2             -1.093656e-02  6.777186e-02  1.628445e-02 -1.015320e-01
## Ala_T2             -1.674712e-02  7.820676e-02  3.684546e-02 -5.876309e-02
## Pro_T2             -1.612979e-02  6.755335e-02  1.536146e-02 -8.844213e-02
## Gly_T2             -5.530842e-03  3.553334e-02  4.204593e-02 -7.628765e-02
## Ser_T2             -3.100467e-03  4.105752e-02  2.107735e-02 -8.835778e-02
## Trp_T2             -2.766754e-03  6.757765e-02  7.508522e-02 -8.954113e-02
## Phe_T2             -5.692661e-03  6.578627e-02  2.704765e-02 -9.437132e-02
## Met_T2             -1.747946e-02  3.089153e-03  4.662582e-02 -1.477892e-02
## Orn_T2             -3.148348e-03  4.774206e-02  3.655536e-02 -7.391611e-02
## Arg_T2             -8.224978e-03  9.950650e-02  5.251903e-02 -3.832805e-02
## His_T2             -1.376341e-02  9.003043e-02  2.209472e-02 -8.717891e-02
## Asn_T2             -2.652617e-02  6.025220e-02  6.064235e-02 -7.983231e-02
## Asp_T2             -9.997523e-04  3.315654e-02  3.544431e-02 -6.710315e-03
## Glu_T2              7.175733e-03  6.425496e-02 -7.479546e-02  3.022789e-02
```

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## Gln_T2            -1.947893e-02  3.329075e-02  8.014939e-02 -7.136711e-02
## Cit_T2            -2.384681e-02  6.593545e-02  5.819803e-02 -5.666937e-02
## Tyr_T2            -1.792655e-02  7.871440e-02  4.156426e-02 -6.984888e-02
## Thr_T2            -1.433721e-02  3.885229e-02  2.722568e-02 -8.101006e-02
## Lys_T2            -6.023937e-03  7.755619e-02  4.519433e-02 -8.510239e-02
## Creatinine_T2     -7.893733e-03  6.571092e-02  3.904550e-02 -8.615072e-02
## Kynurenine_T2     -5.067728e-02  4.045360e-03  2.062023e-02 -2.315684e-02
## Putrescine_T2      3.431822e-03  2.780524e-02 -1.475363e-02  9.083907e-03
## Sarcosine_T2      -2.618470e-02  3.965990e-02  4.400402e-02 -3.359136e-02
## Serotonin_T2      -1.023620e-02  9.753314e-03 -1.196136e-02 -3.510861e-02
## Taurine_T2        -1.174743e-02  3.227001e-02  9.268447e-03 -3.186203e-03
## SDMA_T2           -5.156382e-02  1.879322e-03  1.290757e-02 -1.335975e-02
## C0_T2             -1.308114e-02  7.750648e-02  4.646985e-02 -1.763247e-02
## C2_T2             -3.077317e-02 -1.270500e-02  1.518447e-04 -2.498897e-02
## C3.OH_T2          -5.246289e-02 -8.721361e-03  7.433831e-03 -4.104634e-04
## C6..C4.1.DC._T2   -8.794011e-05  1.910149e-02 -3.701148e-02  7.061459e-02
## C5.DC..C6.OH._T2  -3.966257e-02 -7.223369e-05  1.488706e-02  2.432386e-02
## C7.DC_T2          -3.945212e-02 -1.980068e-02  2.891609e-02 -2.036034e-02
## C8_T2             -5.247319e-02 -5.835348e-03  9.043459e-03 -1.731781e-03
## C10_T2            -5.246485e-02 -5.304002e-03  8.694260e-03 -3.987744e-03
## C10.1_T2          -5.247819e-02 -8.349250e-03  6.843062e-03 -1.389674e-03
## C10.2_T2          -5.247311e-02 -8.721554e-03  8.374141e-03  8.508383e-05
## C14.1_T2          -5.246604e-02 -8.142849e-03  7.353581e-03  1.867183e-04
## C14.2_T2          -5.244983e-02 -8.934934e-03  7.540346e-03  5.834426e-04
## C16.1_T2          -5.246639e-02 -8.892285e-03  7.453149e-03  9.577030e-04
## C16.2_T2          -5.246115e-02 -9.330009e-03  6.968043e-03  1.079576e-03
## C16.2.OH_T2       -5.246307e-02 -9.292363e-03  6.853309e-03  1.052735e-03
## C18.1_T2          -5.247788e-02 -8.059194e-03  7.173419e-03 -1.154825e-03
## C18.1.OH_T2       -5.245963e-02 -9.360578e-03  7.095768e-03  9.061943e-04
## C18.2_T2          -5.244010e-02 -9.203304e-03  7.137599e-03  3.176886e-04
## lysoPC.a.C14.0_T2 -2.281826e-02 -2.099876e-02  4.869040e-02 -6.003401e-02
## lysoPC.a.C16.0_T2  3.493555e-03  7.448962e-02  1.829048e-02 -6.285216e-02
## lysoPC.a.C16.1_T2 -4.670242e-02  2.390311e-02  2.526679e-02 -3.617971e-02
## lysoPC.a.C17.0_T2 -5.072213e-02  1.749478e-02  1.964668e-02 -1.388903e-02
## lysoPC.a.C18.0_T2 -5.873949e-03  9.003806e-02  4.289960e-02 -3.330789e-02
## lysoPC.a.C18.1_T2 -7.168029e-03  6.562473e-02  5.350483e-02 -8.218161e-02
## lysoPC.a.C18.2_T2 -1.281283e-02  5.720073e-02  6.436753e-02 -6.609383e-02
## lysoPC.a.C20.3_T2 -4.856328e-02  2.413333e-02  2.823329e-02 -1.929142e-02
## lysoPC.a.C20.4_T2 -3.083156e-02  3.708094e-02  2.702869e-02 -6.382399e-02
## lysoPC.a.C24.0_T2 -5.250876e-02 -5.785631e-03  7.645641e-03  8.490239e-04
## lysoPC.a.C26.0_T2 -3.694677e-02 -1.823682e-03  8.127999e-03 -2.731974e-02
## lysoPC.a.C26.1_T2 -5.242601e-02 -8.338033e-03  6.106086e-03 -1.056458e-03
## lysoPC.a.C28.0_T2 -2.936705e-02  1.280549e-02  1.984245e-02 -3.555226e-02
## lysoPC.a.C28.1_T2 -5.241746e-02 -4.941967e-03  6.137211e-03 -8.752074e-04
## PC.aa.C24.0_T2    -1.427309e-02 -3.239714e-02  2.729680e-02 -2.494807e-02
## PC.aa.C28.1_T2    -5.132681e-02  9.440831e-03  3.099406e-02 -1.031314e-02
## PC.aa.C30.0_T2    -5.202883e-02 -1.455034e-03  1.700220e-02 -8.123360e-03
## PC.aa.C32.0_T2    -3.869823e-02  2.687094e-02  5.608107e-02 -5.442115e-02
## PC.aa.C32.1_T2    -3.876949e-02  3.663324e-02  3.865157e-02 -3.515520e-02
## PC.aa.C32.3_T2    -5.253778e-02 -7.642949e-03  6.702165e-03 -4.443329e-04
## PC.aa.C34.1_T2    -6.871676e-03  6.605999e-02  9.276728e-02 -7.690553e-02
## PC.aa.C34.2_T2    -2.109368e-02  6.180995e-02  9.372387e-02 -4.502574e-02
## PC.aa.C34.3_T2    -3.948838e-02  4.385576e-02  5.840761e-02 -2.567210e-02
## PC.aa.C34.4_T2    -5.250617e-02 -4.972344e-03  1.066010e-02 -1.586721e-03
```

```
## PC.aa.C36.0_T2    -5.175205e-02  6.067581e-03  7.172530e-03 -6.574652e-03
## PC.aa.C36.1_T2    -2.360321e-02  8.036915e-02  7.265451e-02 -3.386197e-02
## PC.aa.C36.2_T2    -1.459649e-02  7.477894e-02  9.525628e-02 -1.698838e-02
## PC.aa.C36.3_T2    -1.460511e-02  7.605173e-02  8.745500e-02 -3.869726e-02
## PC.aa.C36.4_T2    -2.018011e-02  3.026035e-02  5.029445e-02 -5.499788e-02
## PC.aa.C36.5_T2    -2.966238e-02  3.767907e-02  4.615501e-02 -4.347601e-02
## PC.aa.C38.0_T2    -5.023731e-02  2.264437e-02  2.931214e-02 -1.909040e-02
## PC.aa.C38.1_T2    -4.530134e-02  2.682410e-02 -2.860314e-04 -2.944812e-02
## PC.aa.C38.3_T2    -2.349020e-02  8.168681e-02  7.646186e-02 -2.231953e-02
## PC.aa.C38.4_T2    -1.962151e-02  6.048277e-02  8.261177e-02 -5.319342e-02
## PC.aa.C38.5_T2    -2.422760e-02  5.609569e-02  8.694400e-02 -6.374079e-02
## PC.aa.C38.6_T2    -1.405830e-02  4.953124e-02  6.309218e-02 -3.780808e-02
## PC.aa.C40.1_T2    -2.775708e-02  3.940623e-02  7.435345e-03 -5.730204e-02
## PC.aa.C40.2_T2    -5.042860e-02  6.634111e-03 -1.657985e-02  7.106866e-03
## PC.aa.C40.3_T2    -5.124411e-02  4.023766e-03 -5.621997e-03 -1.139779e-03
## PC.aa.C40.4_T2    -5.049400e-02  2.448560e-02  1.043348e-02 -7.141418e-03
## PC.aa.C40.5_T2    -4.539757e-02  3.198962e-02  5.355546e-02 -2.436583e-02
## PC.aa.C40.6_T2    -2.352882e-02  6.759138e-02  6.843537e-02 -2.036261e-02
## PC.aa.C42.0_T2    -5.253554e-02 -3.916966e-03  5.710044e-03 -2.719226e-03
## PC.aa.C42.1_T2    -5.252381e-02 -5.140283e-03  5.328132e-03 -1.722765e-03
## PC.aa.C42.2_T2    -5.233557e-02 -2.056739e-03 -3.032401e-03  9.581801e-04
## PC.aa.C42.4_T2    -5.248732e-02 -2.398640e-03 -3.353148e-03  4.047450e-03
## PC.aa.C42.5_T2    -5.260377e-02 -4.255415e-03  4.630732e-03  2.678030e-04
## PC.aa.C42.6_T2    -4.149299e-02 -7.351144e-03  2.247671e-02 -1.793776e-02
## PC.ae.C30.0_T2    -3.866764e-02  3.953976e-03  1.315170e-02 -3.710400e-02
## PC.ae.C32.1_T2    -5.158453e-02  6.287398e-03  2.350916e-02 -1.888153e-02
## PC.ae.C32.2_T2    -5.256756e-02 -3.948619e-03  1.068639e-02 -4.495941e-03
## PC.ae.C34.0_T2    -5.254033e-02 -3.562469e-03  1.103251e-02 -1.613118e-03
## PC.ae.C34.1_T2    -4.645693e-02  3.031314e-02  4.895663e-02 -3.711287e-02
## PC.ae.C34.2_T2    -4.617194e-02  3.563224e-02  5.082014e-02 -3.033178e-02
## PC.ae.C34.3_T2    -4.844283e-02  2.431023e-02  4.018361e-02 -1.851495e-02
## PC.ae.C36.0_T2    -5.258760e-02  3.198468e-03  7.374210e-03 -5.493374e-03
## PC.ae.C36.1_T2    -3.117258e-02  4.693675e-02 -3.237397e-02  2.902262e-03
## PC.ae.C36.2_T2    -4.512749e-02  3.938198e-02  4.617977e-02 -2.463510e-02
## PC.ae.C36.3_T2    -5.032515e-02  1.799711e-02  3.413767e-02 -1.825086e-02
## PC.ae.C36.4_T2    -4.494408e-02  3.281381e-02  5.986622e-02 -4.009632e-02
## PC.ae.C36.5_T2    -4.560357e-02  3.248087e-02  5.238353e-02 -3.456805e-02
## PC.ae.C38.0_T2    -5.193794e-02  8.088385e-03  1.926391e-02 -7.537202e-03
## PC.ae.C38.2_T2    -4.133954e-02  3.041475e-02 -3.077215e-02  1.035068e-02
## PC.ae.C38.3_T2    -3.854888e-02  3.658070e-02 -4.625893e-02  1.518008e-02
## PC.ae.C38.4_T2    -4.751978e-02  3.432325e-02  4.018823e-02 -3.478686e-02
## PC.ae.C38.5_T2    -4.018721e-02  3.833141e-02  6.642050e-02 -5.865205e-02
## PC.ae.C38.6_T2    -4.949739e-02  2.061775e-02  4.156293e-02 -2.144402e-02
## PC.ae.C40.1_T2    -5.075761e-02  1.149852e-02 -2.444890e-03 -3.679518e-03
## PC.ae.C40.2_T2    -5.148151e-02  1.317231e-02  1.771201e-03 -1.264436e-03
## PC.ae.C40.3_T2    -4.283186e-02  2.643150e-02 -4.399597e-02  1.635151e-02
## PC.ae.C40.4_T2    -4.962281e-02  2.384091e-02 -1.290434e-02 -5.914266e-03
## PC.ae.C40.5_T2    -4.731579e-02  3.100539e-02 -3.163526e-02  1.496497e-03
## PC.ae.C40.6_T2    -5.135826e-02  1.550029e-02  2.686007e-02 -1.532806e-02
## PC.ae.C42.1_T2    -5.189516e-02  1.142358e-03 -4.632406e-03  2.442960e-03
## PC.ae.C42.2_T2    -5.205024e-02  3.838136e-04  6.813092e-04 -2.497777e-03
## PC.ae.C42.3_T2    -5.176138e-02  6.231921e-03 -6.203187e-03  5.002887e-04
## PC.ae.C42.4_T2    -5.244812e-02  1.292784e-03  6.611616e-03 -3.296204e-03
## PC.ae.C42.5_T2    -5.161280e-02  1.297242e-02  2.214389e-03 -1.455270e-02
```

```
## PC.ae.C44.3_T2    -4.360104e-02  4.895102e-03  2.906923e-02  1.629672e-02
## PC.ae.C44.4_T2    -5.251256e-02 -5.460471e-03  6.536077e-03 -1.266908e-03
## PC.ae.C44.5_T2    -5.106780e-02  6.549883e-04  2.490225e-02 -2.180662e-02
## PC.ae.C44.6_T2    -5.192468e-02 -4.793165e-03  1.721883e-02 -1.264488e-02
## SM..OH..C14.1_T2  -4.880184e-02  1.826323e-02  4.003102e-02 -1.380669e-02
## SM..OH..C16.1_T2  -5.176974e-02  5.650898e-03  2.461918e-02 -1.938129e-03
## SM..OH..C22.1_T2  -4.818457e-02  3.029608e-02  4.304018e-02 -3.241965e-03
## SM..OH..C22.2_T2  -4.752607e-02  3.605125e-02  3.492114e-02 -9.298929e-03
## SM..OH..C24.1_T2  -5.255810e-02 -3.864650e-03  1.121992e-02  1.417137e-04
## SM.C16.0_T2       -1.835175e-02  6.548734e-02  9.266550e-02 -2.850688e-02
## SM.C16.1_T2       -3.935709e-02  3.352410e-02  4.735321e-02 -2.714570e-02
## SM.C18.0_T2       -2.653149e-02  3.169499e-02  6.981412e-02  3.677801e-03
## SM.C18.1_T2       -3.952526e-02  1.579564e-02  4.474842e-02 -9.096202e-03
## SM.C20.2_T2       -5.248856e-02 -8.014355e-03  8.616849e-03  2.762188e-05
## SM.C24.0_T2       -4.357222e-02  4.788717e-02  4.456885e-02 -4.712070e-03
## SM.C24.1_T2       -2.191563e-02  6.491029e-02  8.093587e-02 -2.428401e-02
## MEDDM_T4          -5.232131e-02 -9.587091e-03  7.383240e-03  9.461683e-04
## MEDCOL_T4         -5.233223e-02 -8.672076e-03  8.971278e-03  1.229109e-03
## MEDINF_T4         -5.225489e-02 -1.098214e-02  6.806316e-03  3.441130e-03
## MEDHTA_T4         -5.196640e-02 -4.783872e-03  3.883764e-03  9.995554e-03
## GLU_T4             3.776587e-02  9.708867e-03 -5.493337e-03  1.930741e-02
## INS_T4            -8.057575e-03 -1.788429e-02  2.241292e-02 -8.746490e-03
## HOMA_T4           -4.833630e-02 -1.115173e-02  9.878051e-03  3.709219e-03
## HBA1C_T4          -4.735918e-02 -1.419374e-02  2.491052e-02 -2.095342e-03
## HBA1C.mmol.mol_T4  3.326820e-02  2.034290e-03  4.200954e-02 -4.767790e-03
## PESO_T4            3.288495e-02 -8.905267e-03 -2.403978e-02 -4.005446e-02
## bmi_T4             3.094881e-02 -1.048555e-02 -2.612189e-02 -2.319682e-02
## CC_T4             -5.215970e-02 -1.286902e-02  5.244644e-03 -1.417420e-03
## CINT_T4            4.279097e-02 -2.547964e-03 -8.785329e-03 -1.697358e-02
## CAD_T4             3.820527e-02  1.988881e-03  6.380656e-03 -1.054119e-02
## TAD_T4             3.888406e-02 -2.264666e-03  4.187906e-02  7.651832e-03
## TAS_T4             4.239813e-02  5.360805e-03  4.200986e-02 -1.453473e-02
## TG_T4              2.759201e-02 -1.336125e-02  3.672854e-03  5.090264e-02
## COL_T4             3.621061e-02 -8.279696e-03  7.623956e-02 -7.160270e-04
## LDL_T4             3.831689e-02  8.833033e-03  5.682105e-02  1.894601e-02
## HDL_T4             1.453665e-02 -1.994897e-02  6.643885e-02 -2.669485e-02
## VLDL_T4            9.870018e-03 -4.033020e-02  1.802995e-02  5.564629e-02
## PCR_T4            -2.166929e-02  1.217278e-03  2.833520e-02 -2.946358e-02
## LEP_T4             1.455570e-02 -1.568257e-02  2.964047e-02 -6.557642e-03
## ADIPO_T4          -4.504446e-03  1.127687e-02 -3.303355e-02 -2.535030e-02
## GOT_T4            -5.335540e-03 -2.610513e-02 -1.364143e-02  2.050602e-02
## GPT_T4             3.887932e-03 -2.283927e-02 -3.072336e-02  4.906735e-02
## GGT_T4             3.954245e-03  4.308871e-03  1.292712e-02  1.266444e-03
## URICO_T4          -4.156842e-02 -2.498072e-02  2.054556e-02  1.741022e-02
## CREAT_T4          -5.211554e-02 -1.083001e-02  5.202121e-03 -1.823182e-03
## UREA_T4            1.689411e-02  1.411641e-02 -1.529797e-02  3.072658e-02
## HIERRO_T4          4.260482e-03 -2.449553e-02  1.241037e-02  4.234017e-02
## TRANSF_T4          3.678199e-02 -7.980235e-03  6.058152e-02  1.680610e-03
## FERR_T4            1.079934e-02  1.832551e-02 -3.696757e-02  8.470054e-03
## Ile_T4             9.262907e-03 -8.640029e-02 -4.576903e-02 -4.984219e-02
## Leu_T4             9.422412e-03 -8.796663e-02 -5.678239e-02 -7.056775e-02
## Val_T4             1.001374e-02 -8.727427e-02 -5.667147e-02 -8.636508e-02
## Ala_T4             1.726081e-02 -9.022227e-02 -5.553461e-02 -3.481182e-02
## Pro_T4             1.140462e-02 -6.154606e-02 -6.179894e-02 -7.409421e-02
```

```
## Gly_T4              5.556011e-03 -7.761374e-02 -3.371671e-03 -7.567848e-02
## Ser_T4              1.463329e-02 -3.311877e-02 -3.267483e-03 -2.855353e-02
## Trp_T4             -1.381675e-03 -9.454777e-02 -3.107555e-02 -8.216936e-02
## Phe_T4              5.788309e-03 -7.585191e-02 -6.444177e-02 -9.993741e-02
## Met_T4             -1.699207e-02 -6.642533e-02  4.589468e-02 -8.793842e-02
## Orn_T4             -2.300997e-03 -8.507586e-02 -3.841970e-02 -7.731444e-02
## Arg_T4              1.727422e-02 -8.035830e-02 -5.833876e-02 -5.863482e-02
## His_T4              7.593302e-03 -8.450097e-02 -7.253802e-02 -7.107802e-02
## Asn_T4             -3.936490e-03 -1.044768e-01 -1.519229e-02 -8.279319e-02
## Asp_T4             -3.282301e-03 -2.824834e-02 -6.080712e-02 -1.565711e-02
## Glu_T4              2.061895e-02  1.539095e-02 -1.058520e-01  5.156662e-02
## Gln_T4              1.303804e-02 -1.050770e-02  5.064413e-02  4.448180e-02
## Cit_T4             -6.288421e-03 -9.584785e-02 -3.430173e-02 -6.596955e-02
## Tyr_T4              1.554605e-03 -8.454654e-02 -4.993335e-02 -8.158786e-02
## Thr_T4              4.983106e-03 -8.900770e-02  2.722186e-03 -7.678495e-02
## Lys_T4              1.420386e-02 -5.292446e-02  1.299039e-02 -2.968199e-02
## Creatinine_T4       7.388703e-03 -8.833154e-02 -5.368672e-02 -5.931448e-02
## Kynurenine_T4      -1.549081e-02 -5.617529e-03 -3.994171e-02 -5.408187e-02
## Putrescine_T4      -4.081949e-03 -1.353443e-02  2.323255e-02  2.264845e-03
## Sarcosine_T4       -2.423799e-02 -3.485859e-02 -8.979389e-03  1.408779e-02
## Serotonin_T4       -1.100873e-02  7.112118e-03 -9.217869e-03 -3.478700e-02
## Taurine_T4          1.222143e-02  3.682083e-02  5.409716e-02  1.939982e-02
## SDMA_T4            -5.133143e-02 -2.778403e-02  3.129862e-04 -1.014630e-02
## C0_T4              -1.577593e-03 -8.812115e-02 -6.511264e-02 -6.111451e-02
## C2_T4              -2.191465e-02 -8.727889e-02  1.952410e-02 -4.392680e-03
## C3.OH_T4           -5.240269e-02 -9.192554e-03  6.613897e-03 -5.927405e-04
## C6..C4.1.DC._T4     3.086552e-03  3.624466e-02  3.066182e-02  3.903211e-02
## C5.DC..C6.OH._T4   -3.776637e-02  2.286194e-03 -2.741385e-02 -3.617937e-02
## C7.DC_T4           -3.918263e-02 -1.357590e-02 -1.525952e-02  1.418613e-03
## C8_T4              -5.231163e-02 -1.044477e-02  6.015725e-03 -2.244538e-03
## C10_T4             -5.228081e-02 -1.251930e-02  4.201056e-03 -2.919374e-03
## C10.1_T4           -5.238288e-02 -1.119416e-02  5.391740e-03  2.614087e-05
## C10.2_T4           -5.242964e-02 -9.721547e-03  7.304200e-03  2.644183e-04
## C14.1_T4           -5.239423e-02 -1.100754e-02  6.403639e-03 -5.615208e-04
## C14.2_T4           -5.242313e-02 -1.008755e-02  6.660763e-03  4.522273e-04
## C16.1_T4           -4.571691e-02 -1.086825e-02  8.704063e-05  6.301137e-03
## C16.2_T4           -5.244948e-02 -9.609665e-03  6.737918e-03  1.135550e-03
## C16.2.OH_T4        -4.582815e-02 -1.059955e-02  1.470787e-05  6.660216e-03
## C18.1_T4           -5.239694e-02 -1.156639e-02  6.212138e-03 -7.840861e-04
## C18.1.OH_T4        -5.244969e-02 -9.540861e-03  6.859970e-03  9.155320e-04
## C18.2_T4           -5.243200e-02 -1.009875e-02  6.686196e-03  5.992285e-04
## lysoPC.a.C16.0_T4  1.713196e-02 -7.138737e-02 -8.209270e-02 -2.393707e-02
## lysoPC.a.C16.1_T4 -4.027309e-02 -4.251549e-02 -5.491598e-02 -2.199679e-02
## lysoPC.a.C17.0_T4 -4.928457e-02 -2.815814e-02 -2.395334e-02 -1.046205e-02
## lysoPC.a.C18.0_T4  1.413599e-02 -5.238706e-02 -8.900639e-02 -1.301527e-02
## lysoPC.a.C18.1_T4  5.038105e-03 -8.082093e-02 -6.025839e-02 -4.679781e-02
## lysoPC.a.C18.2_T4 -1.491493e-03 -9.588730e-02 -5.134648e-02 -4.717478e-02
## lysoPC.a.C20.3_T4 -4.884794e-02 -3.296966e-02 -2.043077e-02 -3.217678e-03
## lysoPC.a.C20.4_T4 -1.417880e-02 -8.274569e-02 -3.412700e-02 -2.518016e-02
## lysoPC.a.C24.0_T4 -5.245141e-02 -8.250638e-03  3.619021e-03  1.066094e-03
## lysoPC.a.C26.0_T4 -3.856085e-02 -1.341375e-02 -1.743713e-02 -1.873343e-02
## lysoPC.a.C26.1_T4 -5.247117e-02 -9.099357e-03  8.160507e-04  8.595927e-04
## lysoPC.a.C28.0_T4 -3.516106e-02 -1.816111e-02 -4.611272e-02 -2.461032e-02
## lysoPC.a.C28.1_T4 -5.246483e-02 -9.063869e-03 -1.807198e-03  3.881281e-03
```

```
## PC.aa.C24.0_T4    -3.164774e-02 -9.324753e-03 -2.704195e-02 -1.317167e-02
## PC.aa.C28.1_T4    -5.022019e-02 -3.179324e-02  5.546469e-03 -1.914702e-02
## PC.aa.C30.0_T4    -5.100076e-02 -2.698390e-02  4.912533e-03 -1.292185e-03
## PC.aa.C32.0_T4    -1.647288e-02 -1.003664e-01 -1.015577e-02 -4.807823e-02
## PC.aa.C32.1_T4    -1.892055e-02 -5.558125e-02 -3.809191e-02 -2.862147e-02
## PC.aa.C32.3_T4    -5.245559e-02 -1.173329e-02  5.160321e-03  3.379655e-04
## PC.aa.C34.1_T4     1.409626e-02 -9.104317e-02 -2.358959e-02 -4.504567e-02
## PC.aa.C34.2_T4     7.209987e-03 -1.071653e-01 -1.273804e-02 -3.102972e-02
## PC.aa.C34.3_T4    -2.922942e-02 -6.244900e-02 -3.840478e-02 -1.984401e-02
## PC.aa.C34.4_T4    -5.232296e-02 -1.387201e-02  4.576163e-03  2.359843e-04
## PC.aa.C36.0_T4    -5.173241e-02 -1.899562e-02 -8.811460e-03  2.920794e-03
## PC.aa.C36.1_T4    -1.032037e-02 -7.115917e-02 -1.124918e-02 -3.905434e-02
## PC.aa.C36.2_T4     7.555204e-03 -9.117929e-02 -9.295991e-03 -4.530973e-02
## PC.aa.C36.3_T4     7.099049e-03 -8.297010e-02 -7.683613e-03 -3.580771e-02
## PC.aa.C36.4_T4     7.730246e-03 -9.962392e-02  1.352704e-02 -2.015163e-02
## PC.aa.C36.5_T4    -2.533029e-02 -6.763111e-02 -2.385244e-02 -3.456147e-02
## PC.aa.C38.0_T4    -4.772214e-02 -4.203239e-02 -5.801135e-03 -2.038307e-02
## PC.aa.C38.1_T4    -5.116917e-02 -1.275558e-02 -1.823866e-02  1.873576e-02
## PC.aa.C38.3_T4     5.259916e-03 -7.584427e-02 -9.527220e-03 -3.220045e-02
## PC.aa.C38.4_T4     9.180663e-03 -9.531148e-02  8.052211e-03 -4.808785e-02
## PC.aa.C38.5_T4    -4.965751e-04 -1.000333e-01 -1.741344e-03 -6.204222e-02
## PC.aa.C38.6_T4     6.768952e-03 -8.764110e-02 -3.264173e-03 -7.004607e-03
## PC.aa.C40.1_T4    -3.389054e-02 -2.801690e-02 -3.432404e-02 -4.573889e-03
## PC.aa.C40.2_T4    -5.092157e-02  2.877257e-03 -2.541736e-02  2.067328e-02
## PC.aa.C40.3_T4    -5.171111e-02 -4.423233e-03 -1.904176e-02  1.375604e-02
## PC.aa.C40.4_T4    -4.928070e-02 -1.818413e-02 -3.194470e-02  2.931356e-03
## PC.aa.C40.5_T4    -4.256444e-02 -5.996266e-02  1.345227e-03 -2.742795e-02
## PC.aa.C40.6_T4    -3.830424e-03 -8.943352e-02 -1.282754e-03 -1.147365e-02
## PC.aa.C42.0_T4    -5.226782e-02 -1.641383e-02 -2.238445e-03 -4.074328e-04
## PC.aa.C42.1_T4    -5.249195e-02 -1.077104e-02 -8.519061e-04  2.153613e-03
## PC.aa.C42.2_T4    -5.244790e-02 -6.414860e-03 -7.032941e-03  6.514160e-03
## PC.aa.C42.4_T4    -5.236424e-02 -4.784123e-03 -9.218458e-03  8.616250e-03
## PC.aa.C42.5_T4    -5.249389e-02 -1.162755e-02 -5.653181e-04  3.276734e-03
## PC.aa.C42.6_T4    -4.000615e-02 -2.481959e-02 -2.258466e-02 -2.354775e-03
## PC.ae.C30.0_T4    -4.191499e-02 -2.207880e-02  1.030657e-02 -1.402873e-02
## PC.ae.C32.1_T4    -4.939342e-02 -4.074802e-02  1.649122e-03 -2.386045e-02
## PC.ae.C32.2_T4    -5.218699e-02 -1.831028e-02  1.679541e-03 -6.964678e-03
## PC.ae.C34.0_T4    -5.235198e-02 -1.424114e-02  5.960183e-03 -3.558667e-03
## PC.ae.C34.1_T4    -4.061515e-02 -6.791533e-02 -7.813853e-04 -5.040955e-02
## PC.ae.C34.2_T4    -3.655876e-02 -7.488340e-02 -6.960182e-03 -5.563544e-02
## PC.ae.C34.3_T4    -3.940780e-02 -6.814094e-02  2.714868e-03 -3.598939e-02
## PC.ae.C36.0_T4    -5.216797e-02 -1.831788e-02  1.998333e-04 -5.647126e-03
## PC.ae.C36.1_T4    -2.673550e-02  1.030129e-02 -9.677020e-02  2.719704e-02
## PC.ae.C36.2_T4    -3.968251e-02 -6.131523e-02 -2.554263e-02 -2.504375e-02
## PC.ae.C36.3_T4    -4.817349e-02 -4.603808e-02 -2.356109e-03 -2.583440e-02
## PC.ae.C36.4_T4    -3.365439e-02 -7.733988e-02  5.319832e-07 -6.532096e-02
## PC.ae.C36.5_T4    -3.441805e-02 -8.167917e-02  4.396732e-03 -5.025253e-02
## PC.ae.C38.0_T4    -5.107980e-02 -2.705217e-02 -4.754182e-03 -4.389361e-03
## PC.ae.C38.2_T4    -3.251521e-02  2.003178e-02 -8.543100e-02  3.823899e-02
## PC.ae.C38.3_T4    -3.276376e-02  2.145729e-02 -8.128558e-02  5.303645e-02
## PC.ae.C38.4_T4    -4.280554e-02 -5.827199e-02 -5.359582e-03 -3.899559e-02
## PC.ae.C38.5_T4    -2.607104e-02 -8.351648e-02  8.783486e-03 -6.756581e-02
## PC.ae.C38.6_T4    -4.580786e-02 -5.588684e-02  5.973495e-03 -3.165104e-02
## PC.ae.C40.1_T4    -5.151444e-02 -1.217382e-02 -2.108544e-02  1.003652e-02
```

```
## PC.ae.C40.2_T4    -5.039888e-02 -7.600454e-03 -2.773796e-02  1.385474e-02
## PC.ae.C40.3_T4    -3.745867e-02  1.955701e-02 -7.309303e-02  4.691796e-02
## PC.ae.C40.4_T4    -4.579107e-02 -5.475700e-03 -5.314290e-02  1.583794e-02
## PC.ae.C40.5_T4    -4.060167e-02 -1.832017e-03 -7.349792e-02  1.551030e-02
## PC.ae.C40.6_T4    -5.003971e-02 -3.815300e-02 -1.241504e-03 -1.623764e-02
## PC.ae.C42.1_T4    -5.215109e-02 -5.778763e-03 -1.236943e-02  1.216057e-02
## PC.ae.C42.2_T4    -5.232789e-02 -7.731871e-03 -9.310328e-03  7.574398e-03
## PC.ae.C42.3_T4    -5.168568e-02 -8.194570e-03 -2.183769e-02  8.086212e-03
## PC.ae.C42.4_T4    -5.212115e-02 -1.336107e-02 -8.742491e-03  6.003850e-04
## PC.ae.C42.5_T4    -4.821479e-02 -3.021840e-02 -3.149769e-02 -1.430080e-02
## PC.ae.C44.3_T4    -5.253276e-02 -6.925788e-03 -2.523453e-03  4.897807e-03
## PC.ae.C44.4_T4    -5.243211e-02 -1.157038e-02  2.064830e-03  8.127635e-04
## PC.ae.C44.5_T4    -4.808206e-02 -4.321302e-02 -2.101124e-03 -2.560206e-02
## PC.ae.C44.6_T4    -5.065478e-02 -3.031522e-02  4.643480e-04 -1.293130e-02
## SM..OH..C14.1_T4  -4.464298e-02 -5.221134e-02  8.126572e-03 -3.452157e-02
## SM..OH..C16.1_T4  -5.067272e-02 -2.882120e-02  9.079541e-03 -8.134989e-03
## SM..OH..C22.1_T4  -4.611094e-02 -4.176324e-02  1.554981e-02 -1.572935e-02
## SM..OH..C22.2_T4  -4.448851e-02 -5.348386e-02 -7.229770e-03 -1.426901e-02
## SM..OH..C24.1_T4  -5.228478e-02 -1.415748e-02  6.742601e-03 -1.739562e-04
## SM.C16.0_T4        1.291397e-02 -1.018592e-01  6.458077e-03 -4.162183e-02
## SM.C16.1_T4       -1.581946e-02 -9.631158e-02 -7.437753e-03 -4.231515e-02
## SM.C18.0_T4       -4.589461e-03 -8.799884e-02  8.513275e-03  1.926188e-02
## SM.C18.1_T4       -2.294203e-02 -8.631346e-02 -3.427442e-03 -1.092838e-02
## SM.C20.2_T4       -5.237725e-02 -1.306412e-02  6.092465e-03  6.583696e-04
## SM.C24.0_T4       -3.860105e-02 -5.500487e-02 -2.059113e-03 -1.005620e-02
## SM.C24.1_T4        1.019777e-02 -9.989735e-02 -7.273003e-03 -6.633316e-03
## MEDDM_T5          -5.150080e-02 -3.237145e-03  3.624385e-03  1.010204e-02
## MEDCOL_T5         -5.150080e-02 -3.237145e-03  3.624385e-03  1.010204e-02
## MEDINF_T5         -5.147425e-02 -4.520952e-03  3.540330e-03  1.237918e-02
## MEDHTA_T5         -5.144547e-02 -3.290546e-03  4.415548e-03  9.942508e-03
## GLU_T5            -4.932356e-03  8.827364e-02  2.392928e-02  2.086465e-02
## INS_T5            -3.656599e-02  3.097675e-02  1.651235e-02  3.040778e-02
## HOMA_T5           -5.044671e-02  5.744586e-03  1.377589e-02  1.311854e-02
## HBA1C_T5          -4.740661e-02 -1.726258e-02  2.114776e-02 -1.609879e-03
## HBA1C.mmol.mol_T5  3.435251e-02 -8.243280e-03  2.939061e-02 -3.154598e-03
## PESO_T5           -2.080252e-02  8.848088e-02 -3.398739e-02 -6.389771e-03
## bmi_T5            -2.237296e-02  9.031373e-02 -3.361876e-02  6.714940e-03
## CC_T5             -5.219480e-02 -1.130866e-02  8.353558e-03  1.394497e-03
## CINT_T5           -2.298862e-02  9.986742e-02 -2.057489e-02  2.146791e-02
## CAD_T5            -1.107638e-02  1.027230e-01 -2.940514e-02  2.045903e-03
## TAD_T5             3.879234e-02  5.728967e-03  2.662768e-02  2.104849e-02
## TAS_T5             4.134639e-02  1.224887e-02  1.332890e-02  2.122096e-03
## TG_T5             -1.296414e-02  5.106110e-02  1.744704e-02  6.915014e-02
## COL_T5             1.030419e-02  3.538364e-02  9.360254e-02  2.797949e-02
## LDL_T5             2.068921e-02  4.370577e-02  6.554792e-02  3.320838e-02
## HDL_T5            -9.409149e-03  7.674480e-03  6.183629e-02  1.216976e-02
## VLDL_T5           -3.165450e-02  3.064036e-02  1.568542e-02  7.712265e-02
## PCR_T5            -3.820377e-02  2.402352e-02  3.775250e-02 -4.537154e-02
## LEP_T5            -2.092006e-02  7.537560e-02  2.583980e-02 -1.449497e-02
## ADIPO_T5          -1.194825e-02  2.956043e-02 -1.952951e-02 -9.008070e-03
## GOT_T5            -2.025115e-02  3.161952e-02 -7.807932e-03  8.795260e-03
## GPT_T5            -1.211157e-02  5.657011e-02 -2.766752e-02  2.687791e-02
## GGT_T5            -3.310877e-02  7.348772e-02 -1.661329e-02  4.534138e-02
## URICO_T5          -4.899134e-02  1.976455e-02  2.344549e-04  2.449636e-02
```

```
## CREAT_T5          -5.247049e-02 -4.689248e-03  6.840599e-03 -1.884846e-04
## UREA_T5           -1.447445e-02  6.533065e-02 -1.604312e-02  4.974307e-03
## HIERRO_T5         -4.796012e-03  6.332439e-04  1.148873e-02  7.421884e-02
## TRANSF_T5         -1.355665e-02  6.965061e-02  3.776407e-02  2.406428e-02
## FERR_T5            1.195663e-03  2.637314e-02 -4.554595e-02  3.085219e-02
## Ile_T5             3.465359e-02 -8.079058e-03 -2.817134e-02  5.863541e-02
## Leu_T5             3.353699e-02 -1.673166e-02 -1.775763e-02  4.551440e-02
## Val_T5             3.577992e-02 -1.360018e-02  3.709242e-03  3.393547e-02
## Ala_T5             4.356605e-02  1.599638e-02 -1.211484e-02  6.537560e-02
## Pro_T5             3.821640e-02 -8.080291e-03 -1.398297e-02  2.737739e-02
## Gly_T5             2.963037e-02 -2.137027e-02  3.944158e-02  3.703296e-02
## Ser_T5             3.513482e-02 -3.761597e-02  3.179049e-02  1.723059e-02
## Trp_T5             2.121909e-02 -1.242593e-02  4.457891e-02  5.339619e-02
## Phe_T5             3.252321e-02 -2.805376e-03 -1.502318e-02  4.471480e-02
## Met_T5             1.384154e-02 -4.230277e-02  8.062186e-02  2.833562e-02
## Orn_T5             2.846596e-02 -2.453541e-02  2.363622e-02  4.225131e-02
## Arg_T5             3.627374e-02 -3.391671e-04 -2.278124e-03  9.847268e-02
## His_T5             3.298185e-02 -8.154079e-03 -4.140090e-02  7.439198e-02
## Asn_T5             3.107940e-02 -3.627083e-02  4.541242e-02  4.194208e-02
## Asp_T5             9.369929e-03  1.453723e-02  2.249959e-02  4.561277e-02
## Glu_T5             2.226301e-02  4.292080e-02 -7.259913e-02  6.952453e-02
## Gln_T5             2.810448e-02 -5.990639e-02  5.466750e-02  4.683917e-02
## Cit_T5             2.621963e-02 -7.355723e-03  2.464372e-02  6.534109e-02
## Tyr_T5             3.543152e-02  9.604781e-04  8.252083e-03  5.498797e-02
## Thr_T5             3.183640e-02 -1.746851e-02  7.279472e-02  1.029979e-02
## Lys_T5             4.125049e-02 -6.561489e-03  1.465518e-02  6.663065e-02
## Creatinine_T5      2.401439e-02 -2.828510e-02  3.363679e-02  3.403529e-02
## Kynurenine_T5     -6.222454e-03 -8.007623e-03  9.589713e-02  1.346008e-02
## Putrescine_T5     -9.185678e-03 -3.704751e-03 -3.769871e-02  4.288502e-02
## Sarcosine_T5      -2.136796e-02  1.621620e-02  3.700191e-03  3.974124e-02
## Serotonin_T5      -1.326178e-02  1.282109e-03 -1.678207e-02 -3.395655e-02
## Taurine_T5         2.022184e-02  1.322162e-02  1.453510e-02 -3.155788e-02
## SDMA_T5           -5.184739e-02 -1.465096e-02  1.452524e-02  1.157047e-02
## C0_T5              2.960476e-02  4.178836e-03  2.111815e-03  8.299266e-02
## C2_T5             -2.218936e-02 -4.967117e-02  6.677972e-02  4.401575e-02
## C3.OH_T5          -5.238162e-02 -1.044421e-02  7.088899e-03  8.029096e-05
## C6..C4.1.DC._T5   -3.334221e-03  1.183923e-03 -4.458075e-02  3.625662e-02
## C5.DC..C6.OH._T5  -4.574370e-02 -6.148936e-03 -3.807181e-03 -1.703594e-02
## C7.DC_T5          -4.120378e-02 -2.786588e-02  2.621333e-02 -2.092243e-02
## C8_T5             -5.225193e-02 -1.067114e-02  8.435645e-03  8.035746e-05
## C10_T5            -5.216199e-02 -1.137957e-02  8.743399e-03  1.517650e-03
## C10.1_T5          -4.000612e-02 -1.197213e-02  1.832955e-02  3.550731e-02
## C10.2_T5          -5.239715e-02 -1.026692e-02  7.127743e-03  6.733821e-04
## C14.1_T5          -5.238913e-02 -1.048270e-02  7.702819e-03  1.966301e-03
## C14.2_T5          -5.241707e-02 -9.923341e-03  6.784646e-03  1.245400e-03
## C16.1_T5          -5.243229e-02 -9.892061e-03  6.872788e-03  1.296953e-03
## C16.2_T5          -5.244414e-02 -9.750169e-03  6.708102e-03  1.064722e-03
## C16.2.OH_T5       -4.268832e-02 -1.280140e-02 -3.344134e-02 -3.205222e-02
## C18.1_T5          -5.242291e-02 -1.004155e-02  7.210269e-03  2.320689e-03
## C18.1.OH_T5       -5.244673e-02 -9.784750e-03  6.796257e-03  1.004841e-03
## C18.2_T5          -5.243392e-02 -9.888021e-03  6.550968e-03  1.622239e-03
## lysoPC.a.C16.0_T5  2.907325e-02 -1.432611e-02 -4.024542e-03  1.049269e-01
## lysoPC.a.C16.1_T5 -3.770926e-02 -3.027073e-03 -3.943408e-02  5.920804e-02
## lysoPC.a.C17.0_T5 -4.988898e-02 -1.018307e-02  1.042120e-02  3.926384e-02
```

```
## lysoPC.a.C18.0_T5  2.688807e-02  6.407240e-03 -4.838211e-03  1.145931e-01
## lysoPC.a.C18.1_T5  2.054466e-02 -2.087372e-02  2.111059e-03  9.161491e-02
## lysoPC.a.C18.2_T5  1.494636e-02 -3.181746e-02  1.523820e-02  1.036081e-01
## lysoPC.a.C20.3_T5 -4.278276e-02 -1.101158e-02 -2.985152e-03  7.534093e-02
## lysoPC.a.C20.4_T5 -1.554890e-03 -2.765956e-02  9.501379e-03  9.692562e-02
## lysoPC.a.C24.0_T5 -5.240408e-02 -8.990629e-03  5.033681e-03  3.108793e-03
## lysoPC.a.C26.0_T5 -5.242397e-02 -6.845549e-03 -4.610364e-04  8.050022e-03
## lysoPC.a.C26.1_T5 -5.240065e-02 -1.021802e-02  5.164947e-03  5.007766e-03
## lysoPC.a.C28.0_T5 -3.058255e-02 -8.995770e-03  4.522804e-03  7.876446e-02
## lysoPC.a.C28.1_T5 -5.243372e-02 -9.282513e-03  4.969562e-03  6.307637e-03
## PC.aa.C24.0_T5    -2.996863e-02 -4.164895e-03  1.888470e-02  1.128320e-02
## PC.aa.C28.1_T5    -5.049343e-02 -2.076042e-02  2.944869e-02  2.385457e-02
## PC.aa.C30.0_T5    -5.022501e-02 -2.013622e-02  1.900439e-02  2.462978e-02
## PC.aa.C32.0_T5    -1.299253e-02 -6.246764e-02  8.907183e-02  9.105592e-02
## PC.aa.C32.1_T5    -1.865142e-02 -2.706506e-02  1.888486e-02  7.939344e-02
## PC.aa.C32.3_T5    -5.244319e-02 -9.996895e-03  7.662508e-03  2.925786e-03
## PC.aa.C34.1_T5     2.626829e-02 -4.407915e-02  7.721770e-02  9.794823e-02
## PC.aa.C34.2_T5     2.013604e-02 -5.430590e-02  8.325655e-02  1.031700e-01
## PC.aa.C34.3_T5    -2.049242e-02 -3.036288e-02  3.911787e-02  1.015323e-01
## PC.aa.C34.4_T5    -5.226023e-02 -1.091114e-02  7.744382e-03  8.979559e-03
## PC.aa.C36.0_T5    -5.154809e-02 -1.118056e-02  3.369201e-03  2.459519e-02
## PC.aa.C36.1_T5     4.658680e-03 -2.188069e-02  7.480618e-02  1.242558e-01
## PC.aa.C36.2_T5     2.214578e-02 -3.266877e-02  8.478424e-02  1.080345e-01
## PC.aa.C36.3_T5     2.272746e-02 -2.994081e-02  7.403130e-02  1.099153e-01
## PC.aa.C36.4_T5     2.079923e-02 -5.039248e-02  7.343672e-02  8.974764e-02
## PC.aa.C36.5_T5    -2.826087e-02 -2.924563e-02  4.671139e-02  7.374215e-02
## PC.aa.C38.0_T5    -4.741980e-02 -2.308284e-02  2.709394e-02  4.359518e-02
## PC.aa.C38.1_T5    -5.146515e-02 -2.945281e-03  5.442573e-03  1.679289e-02
## PC.aa.C38.3_T5     1.901366e-02 -1.173850e-02  4.177319e-02  1.178720e-01
## PC.aa.C38.4_T5     2.479418e-02 -2.775443e-02  8.306046e-02  9.282048e-02
## PC.aa.C38.5_T5     1.667028e-02 -4.590749e-02  9.941096e-02  8.733492e-02
## PC.aa.C38.6_T5     2.250060e-02 -3.984975e-02  6.314098e-02  1.135098e-01
## PC.aa.C40.1_T5    -3.198230e-02 -1.566270e-02  1.408036e-02  5.864319e-02
## PC.aa.C40.2_T5    -5.171586e-02 -8.661961e-04 -1.559623e-02  1.466316e-02
## PC.aa.C40.3_T5    -5.169293e-02 -3.568410e-03 -9.416695e-03  1.329640e-02
## PC.aa.C40.4_T5    -5.019211e-02 -1.275151e-02  8.257900e-03  3.448146e-02
## PC.aa.C40.5_T5    -4.094825e-02 -2.889005e-02  6.263438e-02  6.649408e-02
## PC.aa.C40.6_T5     1.455528e-02 -2.215147e-02  6.117233e-02  1.346530e-01
## PC.aa.C42.0_T5    -5.231470e-02 -1.156519e-02  6.669088e-03  9.862692e-03
## PC.aa.C42.1_T5    -5.246941e-02 -9.591139e-03  4.616755e-03  5.910429e-03
## PC.aa.C42.2_T5    -5.244764e-02 -6.501827e-03 -2.657050e-03  7.038771e-03
## PC.aa.C42.4_T5    -5.238574e-02 -5.151438e-03 -5.477948e-03  8.607143e-03
## PC.aa.C42.5_T5    -5.243442e-02 -9.774281e-03  3.240068e-03  8.726349e-03
## PC.aa.C42.6_T5    -3.883784e-02  4.446195e-03  2.885925e-02  2.931491e-02
## PC.ae.C30.0_T5    -4.338125e-02  1.230531e-03  3.757069e-02  1.544726e-02
## PC.ae.C32.1_T5    -5.000798e-02 -2.463406e-02  3.016962e-02  2.190687e-02
## PC.ae.C32.2_T5    -5.228321e-02 -1.246613e-02  1.090434e-02  6.976597e-03
## PC.ae.C34.0_T5    -5.230840e-02 -1.059795e-02  1.208706e-02  5.232805e-03
## PC.ae.C34.1_T5    -4.048060e-02 -3.767488e-02  7.322595e-02  5.577031e-02
## PC.ae.C34.2_T5    -3.249940e-02 -4.658514e-02  7.552751e-02  6.816774e-02
## PC.ae.C34.3_T5    -3.650278e-02 -4.556629e-02  5.977278e-02  5.952175e-02
## PC.ae.C36.0_T5    -5.214814e-02 -1.253209e-02  1.143110e-02  1.205451e-02
## PC.ae.C36.1_T5    -3.399604e-02  1.780170e-02 -4.805272e-02  6.258786e-02
## PC.ae.C36.2_T5    -3.698098e-02 -2.696795e-02  5.992963e-02  7.752413e-02
```

```
## PC.ae.C36.3_T5    -4.709058e-02 -2.744909e-02  4.481171e-02  3.842484e-02
## PC.ae.C36.4_T5    -3.082485e-02 -5.476116e-02  7.887616e-02  5.879241e-02
## PC.ae.C36.5_T5    -2.920571e-02 -5.754691e-02  7.636102e-02  6.537629e-02
## PC.ae.C38.0_T5    -5.165695e-02 -1.421318e-02  1.269032e-02  2.363743e-02
## PC.ae.C38.2_T5    -3.716498e-02  1.788185e-02 -5.784289e-02  4.351079e-02
## PC.ae.C38.3_T5    -4.081035e-02  2.084427e-02 -4.695902e-02  6.010280e-02
## PC.ae.C38.4_T5    -4.211157e-02 -2.866996e-02  6.975017e-02  5.656605e-02
## PC.ae.C38.5_T5    -1.797039e-02 -5.504358e-02  9.519102e-02  6.293168e-02
## PC.ae.C38.6_T5    -4.478431e-02 -3.529525e-02  5.314466e-02  4.865804e-02
## PC.ae.C40.1_T5    -5.186014e-02 -7.165665e-03 -3.167751e-03  2.450672e-02
## PC.ae.C40.2_T5    -5.085067e-02 -1.608866e-03 -1.202225e-02  3.121778e-02
## PC.ae.C40.3_T5    -4.110564e-02  2.082306e-02 -5.621954e-02  5.430364e-02
## PC.ae.C40.4_T5    -4.847784e-02  2.352035e-03 -1.905042e-02  4.171745e-02
## PC.ae.C40.5_T5    -4.479973e-02  5.846900e-03 -3.249299e-02  5.523492e-02
## PC.ae.C40.6_T5    -5.007202e-02 -2.001413e-02  3.049148e-02  3.288928e-02
## PC.ae.C42.1_T5    -5.233277e-02 -5.129929e-03 -5.171104e-03  1.303470e-02
## PC.ae.C42.2_T5    -5.240686e-02 -5.697950e-03 -1.342449e-03  1.010146e-02
## PC.ae.C42.3_T5    -5.199001e-02 -5.210296e-03 -7.307275e-03  1.701921e-02
## PC.ae.C42.4_T5    -5.214477e-02 -9.002769e-03  2.398404e-03  1.525033e-02
## PC.ae.C42.5_T5    -4.919234e-02 -1.100229e-02 -5.061407e-03  4.178787e-02
## PC.ae.C44.3_T5    -5.249463e-02 -6.823690e-03 -1.539056e-03  6.112403e-03
## PC.ae.C44.4_T5    -5.240055e-02 -9.800095e-03  6.272282e-03  6.023474e-03
## PC.ae.C44.5_T5    -4.952147e-02 -2.504813e-02  2.919331e-02  2.260792e-02
## PC.ae.C44.6_T5    -5.081071e-02 -2.120197e-02  1.897629e-02  1.626352e-02
## SM..OH..C14.1_T5  -4.546559e-02 -2.704088e-02  6.026883e-02  3.922726e-02
## SM..OH..C16.1_T5  -5.123293e-02 -1.555209e-02  2.719542e-02  1.767007e-02
## SM..OH..C22.1_T5  -4.081127e-02 -3.174242e-02  5.776840e-02  4.910790e-02
## SM..OH..C22.2_T5  -4.280667e-02 -2.744135e-02  4.966870e-02  6.182782e-02
## SM..OH..C24.1_T5  -5.225125e-02 -1.193210e-02  1.215401e-02  4.825664e-03
## SM.C16.0_T5        2.511732e-02 -4.307651e-02  9.079469e-02  9.336900e-02
## SM.C16.1_T5       -1.059342e-02 -5.039223e-02  9.530042e-02  7.761307e-02
## SM.C18.0_T5        1.577563e-03 -3.445486e-02  7.254780e-02  1.129948e-01
## SM.C18.1_T5       -2.851238e-02 -3.932298e-02  6.635420e-02  7.695362e-02
## SM.C20.2_T5       -5.234350e-02 -1.104973e-02  8.486420e-03  4.303566e-03
## SM.C24.0_T5       -2.543839e-02 -4.487573e-02  4.899507e-02  9.971798e-02
## SM.C24.1_T5        1.987643e-02 -4.300291e-02  6.967238e-02  1.136367e-01
##                             PC5
## GLU_T0            -1.720638e-02
## INS_T0            -4.680565e-03
## HOMA_T0           -3.741193e-03
## HBA1C_T0          -3.651642e-03
## HBA1C.mmol.mol_T0 -7.868349e-03
## PESO_T0           -1.571945e-02
## bmi_T0            -1.499417e-02
## CC_T0             -8.783815e-04
## CINT_T0           -9.147539e-03
## CAD_T0            -2.201109e-02
## TAD_T0             1.245867e-03
## TAS_T0            -2.246549e-03
## TG_T0              1.198507e-03
## COL_T0            -5.145790e-02
## LDL_T0            -5.645871e-02
## HDL_T0             9.606825e-03
## VLDL_T0            1.300810e-03
```

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## PCR_T0            -6.154786e-02
## LEP_T0            -2.436914e-02
## ADIPO_T0           5.368261e-02
## GOT_T0             6.108101e-02
## GPT_T0             2.777123e-02
## GGT_T0             4.267764e-03
## URICO_T0          -5.514437e-03
## CREAT_T0          -2.221790e-03
## UREA_T0            2.230292e-02
## HIERRO_T0         -1.598124e-02
## TRANSF_T0         -2.004095e-02
## FERR_T0            1.554729e-02
## Ile_T0            -5.137047e-02
## Leu_T0            -4.979559e-02
## Val_T0            -8.624054e-02
## Ala_T0            -5.731886e-02
## Pro_T0            -5.170318e-02
## Gly_T0             2.584130e-02
## Ser_T0             9.667183e-03
## Trp_T0            -7.514744e-02
## Phe_T0            -3.174623e-03
## Met_T0            -4.561577e-02
## Orn_T0             2.626557e-02
## Arg_T0             3.758529e-03
## His_T0            -5.772913e-02
## Asn_T0            -7.118079e-02
## Asp_T0             5.081524e-02
## Glu_T0             2.911597e-02
## Gln_T0            -8.858740e-02
## Cit_T0            -3.348022e-02
## Tyr_T0            -3.299038e-02
## Thr_T0            -3.758933e-02
## Lys_T0            -8.360483e-02
## Creatinine_T0     -7.268980e-02
## Kynurenine_T0     -2.184854e-02
## Putrescine_T0     -1.422032e-03
## Sarcosine_T0      -4.702161e-02
## Serotonin_T0      -3.660384e-02
## Taurine_T0         2.303257e-02
## SDMA_T0           -1.083996e-02
## C0_T0             -5.546852e-02
## C2_T0             -5.357889e-02
## C3.OH_T0          -8.049950e-04
## C6..C4.1.DC._T0    6.345401e-02
## C5.DC..C6.OH._T0   7.444036e-03
## C7.DC_T0           2.030726e-02
## C8_T0              5.195643e-04
## C10_T0            -5.194313e-03
## C10.1_T0          -1.149999e-03
## C10.2_T0          -1.097363e-03
## C14.1_T0          -1.253263e-03
## C14.2_T0          -3.964059e-04
## C16.1_T0          -4.211973e-04
## C16.2_T0           1.722988e-04
```

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## C16.2.OH_T0        3.835253e-04
## C18.1_T0          -5.738362e-04
## C18.1.OH_T0        3.341911e-04
## C18.2_T0          -8.515310e-05
## lysoPC.a.C16.0_T0 -1.995276e-02
## lysoPC.a.C16.1_T0 -9.343917e-03
## lysoPC.a.C17.0_T0  3.778171e-04
## lysoPC.a.C18.0_T0 -4.834820e-03
## lysoPC.a.C18.1_T0 -4.225782e-02
## lysoPC.a.C18.2_T0 -5.465281e-02
## lysoPC.a.C20.3_T0 -3.637022e-02
## lysoPC.a.C20.4_T0 -6.922633e-02
## lysoPC.a.C24.0_T0 -1.561510e-03
## lysoPC.a.C26.0_T0 -1.616745e-02
## lysoPC.a.C26.1_T0 -1.729590e-03
## lysoPC.a.C28.0_T0  4.752687e-03
## lysoPC.a.C28.1_T0 -1.893440e-03
## PC.aa.C24.0_T0    -7.983825e-03
## PC.aa.C28.1_T0    -3.280171e-02
## PC.aa.C30.0_T0    -2.399085e-02
## PC.aa.C32.0_T0    -8.762208e-02
## PC.aa.C32.1_T0    -7.973980e-02
## PC.aa.C32.3_T0    -1.634294e-03
## PC.aa.C34.1_T0    -1.358691e-01
## PC.aa.C34.2_T0    -1.123287e-01
## PC.aa.C34.3_T0    -7.558684e-02
## PC.aa.C34.4_T0    -1.114983e-02
## PC.aa.C36.0_T0    -4.866453e-03
## PC.aa.C36.1_T0    -8.995863e-02
## PC.aa.C36.2_T0    -1.065939e-01
## PC.aa.C36.3_T0    -1.181771e-01
## PC.aa.C36.4_T0    -1.507633e-01
## PC.aa.C36.5_T0    -9.318970e-02
## PC.aa.C38.0_T0    -2.628371e-02
## PC.aa.C38.1_T0     4.536433e-02
## PC.aa.C38.3_T0    -1.040235e-01
## PC.aa.C38.4_T0    -1.375978e-01
## PC.aa.C38.5_T0    -1.360513e-01
## PC.aa.C38.6_T0    -1.293613e-01
## PC.aa.C40.1_T0    -1.007534e-02
## PC.aa.C40.2_T0     2.054555e-02
## PC.aa.C40.3_T0     1.201239e-02
## PC.aa.C40.4_T0    -7.560109e-03
## PC.aa.C40.5_T0    -6.222018e-02
## PC.aa.C40.6_T0    -1.213666e-01
## PC.aa.C42.0_T0    -3.384753e-04
## PC.aa.C42.1_T0     1.536492e-03
## PC.aa.C42.2_T0     6.728100e-03
## PC.aa.C42.4_T0     1.079405e-02
## PC.aa.C42.5_T0     2.582920e-03
## PC.aa.C42.6_T0    -9.373045e-03
## PC.ae.C30.0_T0    -1.338956e-03
## PC.ae.C32.1_T0    -2.601108e-02
## PC.ae.C32.2_T0    -4.140936e-03
```

```
## PC.ae.C34.0_T0    -4.019683e-03
## PC.ae.C34.1_T0    -5.235666e-02
## PC.ae.C34.2_T0    -5.706711e-02
## PC.ae.C34.3_T0    -4.383190e-02
## PC.ae.C36.0_T0    -5.235953e-03
## PC.ae.C36.1_T0     1.866281e-02
## PC.ae.C36.2_T0    -2.318909e-02
## PC.ae.C36.3_T0    -4.434208e-02
## PC.ae.C36.4_T0    -8.882630e-02
## PC.ae.C36.5_T0    -8.686783e-02
## PC.ae.C38.0_T0    -2.114916e-02
## PC.ae.C38.2_T0     4.363949e-02
## PC.ae.C38.3_T0     2.251487e-02
## PC.ae.C38.4_T0    -5.871994e-02
## PC.ae.C38.5_T0    -1.054947e-01
## PC.ae.C38.6_T0    -6.775473e-02
## PC.ae.C40.1_T0     1.541510e-03
## PC.ae.C40.2_T0     2.551408e-02
## PC.ae.C40.3_T0     3.878820e-02
## PC.ae.C40.4_T0     1.397030e-02
## PC.ae.C40.5_T0     1.469480e-02
## PC.ae.C40.6_T0    -2.906283e-02
## PC.ae.C42.1_T0     9.406807e-03
## PC.ae.C42.2_T0     5.632220e-03
## PC.ae.C42.3_T0     9.635461e-03
## PC.ae.C42.4_T0     1.946517e-03
## PC.ae.C42.5_T0     7.755425e-04
## PC.ae.C44.3_T0     7.631846e-03
## PC.ae.C44.4_T0     1.008239e-03
## PC.ae.C44.5_T0    -2.071341e-02
## PC.ae.C44.6_T0    -9.501347e-03
## SM..OH..C14.1_T0  -3.141344e-02
## SM..OH..C16.1_T0  -1.588592e-02
## SM..OH..C22.1_T0  -9.104943e-02
## SM..OH..C22.2_T0  -5.178924e-02
## SM..OH..C24.1_T0  -8.649874e-03
## SM.C16.0_T0       -1.488562e-01
## SM.C16.1_T0       -1.130297e-01
## SM.C18.0_T0       -9.126795e-02
## SM.C18.1_T0       -7.861392e-02
## SM.C20.2_T0       -2.986033e-03
## SM.C24.0_T0       -1.067910e-01
## SM.C24.1_T0       -1.317757e-01
## MEDDM_T2           5.335230e-04
## MEDCOL_T2          5.335230e-04
## MEDINF_T2         -1.621857e-03
## MEDHTA_T2         -2.083133e-04
## GLU_T2            -1.118887e-02
## INS_T2             3.090819e-02
## HOMA_T2            1.440253e-02
## HBA1C_T2          -2.599691e-04
## HBA1C.mmol.mol_T2  2.548893e-03
## PESO_T2           -8.422618e-03
## bmi_T2            -7.553918e-03
```

```
## CC_T2               5.353340e-04
## CINT_T2            -3.628044e-03
## CAD_T2             -1.030943e-02
## TAD_T2              1.854242e-02
## TAS_T2             -8.975238e-03
## TG_T2              -8.086568e-03
## COL_T2             -3.390887e-02
## LDL_T2             -3.529295e-02
## HDL_T2             -1.236031e-02
## VLDL_T2             1.734216e-02
## PCR_T2             -5.048295e-02
## LEP_T2             -2.613449e-02
## ADIPO_T2            5.953662e-02
## GOT_T2              7.922514e-02
## GPT_T2              2.797536e-02
## GGT_T2              4.093309e-03
## URICO_T2            6.449549e-03
## CREAT_T2           -1.101451e-03
## UREA_T2             9.246213e-03
## HIERRO_T2           2.519406e-02
## TRANSF_T2          -5.450842e-03
## FERR_T2             2.388398e-02
## Ile_T2              1.721587e-02
## Leu_T2              4.333834e-02
## Val_T2              3.157222e-02
## Ala_T2              2.511295e-02
## Pro_T2              2.682571e-02
## Gly_T2              6.015683e-02
## Ser_T2              5.469659e-02
## Trp_T2              1.051456e-02
## Phe_T2              5.919895e-02
## Met_T2             -4.358204e-02
## Orn_T2              8.910752e-02
## Arg_T2              6.449982e-02
## His_T2              5.598669e-02
## Asn_T2              2.967764e-02
## Asp_T2              9.303459e-02
## Glu_T2              8.972179e-02
## Gln_T2              1.979045e-03
## Cit_T2              5.104910e-02
## Tyr_T2              3.225136e-02
## Thr_T2              1.405572e-02
## Lys_T2              3.625433e-02
## Creatinine_T2       1.057030e-02
## Kynurenine_T2       4.286078e-03
## Putrescine_T2      -3.517004e-02
## Sarcosine_T2       -1.389908e-02
## Serotonin_T2        2.677595e-03
## Taurine_T2         -1.135532e-02
## SDMA_T2             2.331070e-03
## C0_T2               5.136815e-02
## C2_T2              -2.619780e-02
## C3.OH_T2           -3.210998e-04
## C6..C4.1.DC._T2    -4.799601e-02
```

```
## C5.DC..C6.OH._T2   7.977267e-03
## C7.DC_T2          -6.199207e-03
## C8_T2             -2.038211e-04
## C10_T2             2.576608e-04
## C10.1_T2           8.993374e-04
## C10.2_T2          -1.168486e-03
## C14.1_T2           8.301354e-04
## C14.2_T2           3.900950e-04
## C16.1_T2           3.981869e-04
## C16.2_T2           1.703462e-04
## C16.2.OH_T2        4.198663e-04
## C18.1_T2           9.965502e-04
## C18.1.OH_T2        3.563761e-04
## C18.2_T2           5.830794e-04
## lysoPC.a.C14.0_T2  4.129155e-02
## lysoPC.a.C16.0_T2  4.867326e-02
## lysoPC.a.C16.1_T2  2.112456e-02
## lysoPC.a.C17.0_T2  1.870546e-02
## lysoPC.a.C18.0_T2  4.008249e-02
## lysoPC.a.C18.1_T2  3.494736e-02
## lysoPC.a.C18.2_T2  2.967963e-02
## lysoPC.a.C20.3_T2  1.679209e-02
## lysoPC.a.C20.4_T2  2.233838e-02
## lysoPC.a.C24.0_T2  2.521468e-03
## lysoPC.a.C26.0_T2  2.314272e-02
## lysoPC.a.C26.1_T2  3.669205e-03
## lysoPC.a.C28.0_T2  7.017750e-02
## lysoPC.a.C28.1_T2  7.107187e-03
## PC.aa.C24.0_T2     4.792009e-02
## PC.aa.C28.1_T2    -2.258037e-03
## PC.aa.C30.0_T2    -2.257284e-03
## PC.aa.C32.0_T2    -1.661137e-02
## PC.aa.C32.1_T2    -1.024004e-02
## PC.aa.C32.3_T2     1.468439e-03
## PC.aa.C34.1_T2    -2.412064e-02
## PC.aa.C34.2_T2    -2.398395e-02
## PC.aa.C34.3_T2     4.525159e-03
## PC.aa.C34.4_T2     4.276433e-04
## PC.aa.C36.0_T2     1.408847e-02
## PC.aa.C36.1_T2    -1.320246e-02
## PC.aa.C36.2_T2    -1.097672e-02
## PC.aa.C36.3_T2    -7.883061e-03
## PC.aa.C36.4_T2    -6.117462e-02
## PC.aa.C36.5_T2    -1.562816e-02
## PC.aa.C38.0_T2     3.405639e-04
## PC.aa.C38.1_T2     3.208236e-02
## PC.aa.C38.3_T2    -1.969331e-03
## PC.aa.C38.4_T2    -3.046920e-02
## PC.aa.C38.5_T2    -2.194054e-02
## PC.aa.C38.6_T2    -5.865667e-02
## PC.aa.C40.1_T2     5.417858e-02
## PC.aa.C40.2_T2     3.038402e-02
## PC.aa.C40.3_T2     2.418455e-02
## PC.aa.C40.4_T2     2.630906e-02
```

```
## PC.aa.C40.5_T2     1.564386e-03
## PC.aa.C40.6_T2    -3.603300e-02
## PC.aa.C42.0_T2     7.115705e-03
## PC.aa.C42.1_T2     6.158132e-03
## PC.aa.C42.2_T2     1.484610e-02
## PC.aa.C42.4_T2     1.320982e-02
## PC.aa.C42.5_T2     6.680523e-03
## PC.aa.C42.6_T2     1.996814e-02
## PC.ae.C30.0_T2     3.344044e-02
## PC.ae.C32.1_T2    -1.816300e-03
## PC.ae.C32.2_T2     2.381175e-03
## PC.ae.C34.0_T2     5.582478e-04
## PC.ae.C34.1_T2    -2.985953e-04
## PC.ae.C34.2_T2     5.129224e-04
## PC.ae.C34.3_T2    -1.300180e-03
## PC.ae.C36.0_T2     4.867268e-03
## PC.ae.C36.1_T2     7.044954e-02
## PC.ae.C36.2_T2     1.700228e-02
## PC.ae.C36.3_T2    -1.330701e-03
## PC.ae.C36.4_T2    -1.206516e-02
## PC.ae.C36.5_T2    -1.529576e-02
## PC.ae.C38.0_T2     2.152600e-03
## PC.ae.C38.2_T2     6.184151e-02
## PC.ae.C38.3_T2     6.958746e-02
## PC.ae.C38.4_T2    -4.858352e-03
## PC.ae.C38.5_T2    -2.245176e-02
## PC.ae.C38.6_T2    -1.416607e-02
## PC.ae.C40.1_T2     2.618671e-02
## PC.ae.C40.2_T2     2.075903e-02
## PC.ae.C40.3_T2     6.288411e-02
## PC.ae.C40.4_T2     3.725597e-02
## PC.ae.C40.5_T2     5.241610e-02
## PC.ae.C40.6_T2    -3.626280e-03
## PC.ae.C42.1_T2     1.789557e-02
## PC.ae.C42.2_T2     1.548579e-02
## PC.ae.C42.3_T2     2.289089e-02
## PC.ae.C42.4_T2     9.786602e-03
## PC.ae.C42.5_T2     2.097016e-02
## PC.ae.C44.3_T2     4.489241e-02
## PC.ae.C44.4_T2     4.575337e-03
## PC.ae.C44.5_T2     2.689887e-04
## PC.ae.C44.6_T2     2.281602e-03
## SM..OH..C14.1_T2  -3.128806e-03
## SM..OH..C16.1_T2  -2.337047e-03
## SM..OH..C22.1_T2  -1.460378e-02
## SM..OH..C22.2_T2   1.103739e-02
## SM..OH..C24.1_T2  -1.506632e-03
## SM.C16.0_T2       -2.953034e-02
## SM.C16.1_T2       -2.176239e-02
## SM.C18.0_T2       -3.193996e-02
## SM.C18.1_T2       -2.472939e-02
## SM.C20.2_T2        3.941159e-04
## SM.C24.0_T2       -2.343616e-02
## SM.C24.1_T2       -2.394161e-02
```

```
## MEDDM_T4            6.221103e-04
## MEDCOL_T4          -1.279878e-03
## MEDINF_T4          -1.077442e-03
## MEDHTA_T4          -7.630148e-04
## GLU_T4             -2.335043e-02
## INS_T4              6.262207e-03
## HOMA_T4            -2.913690e-03
## HBA1C_T4           -3.063996e-03
## HBA1C.mmol.mol_T4 -7.227177e-03
## PESO_T4            -5.581687e-03
## bmi_T4             -6.284647e-03
## CC_T4              -4.927665e-04
## CINT_T4            -2.542294e-03
## CAD_T4             -3.078503e-03
## TAD_T4             -1.051695e-02
## TAS_T4             -3.017671e-02
## TG_T4              -4.312488e-02
## COL_T4             -3.381578e-02
## LDL_T4             -2.136601e-02
## HDL_T4             -2.386914e-02
## VLDL_T4            -5.363939e-03
## PCR_T4             -8.291790e-02
## LEP_T4             -5.003410e-02
## ADIPO_T4            8.887434e-02
## GOT_T4              7.457260e-02
## GPT_T4              3.503376e-02
## GGT_T4             -1.079355e-02
## URICO_T4            1.216507e-02
## CREAT_T4            3.378724e-04
## UREA_T4             8.528855e-03
## HIERRO_T4          -1.662116e-02
## TRANSF_T4          -3.037808e-02
## FERR_T4             2.197843e-02
## Ile_T4              6.543982e-02
## Leu_T4              5.069705e-02
## Val_T4              2.705059e-02
## Ala_T4              3.727865e-02
## Pro_T4              3.423316e-02
## Gly_T4              8.064635e-02
## Ser_T4              1.068991e-02
## Trp_T4             -6.711082e-03
## Phe_T4              5.878263e-02
## Met_T4             -1.571397e-02
## Orn_T4              4.259717e-02
## Arg_T4              3.475179e-02
## His_T4              6.035623e-02
## Asn_T4              4.563756e-02
## Asp_T4              4.927814e-02
## Glu_T4              6.366851e-02
## Gln_T4             -3.129997e-02
## Cit_T4              1.129950e-02
## Tyr_T4              3.542797e-02
## Thr_T4              3.370749e-02
## Lys_T4              3.838280e-02
```

```
## Creatinine_T4       3.309588e-02
## Kynurenine_T4       2.106531e-02
## Putrescine_T4      -8.483349e-02
## Sarcosine_T4       -1.177845e-02
## Serotonin_T4       -4.284880e-03
## Taurine_T4         -8.950146e-03
## SDMA_T4             3.687367e-03
## C0_T4              -5.141620e-03
## C2_T4               1.784705e-02
## C3.OH_T4            7.054105e-04
## C6..C4.1.DC._T4    -9.159270e-02
## C5.DC..C6.OH._T4    2.287131e-03
## C7.DC_T4            1.676911e-02
## C8_T4               2.049179e-03
## C10_T4              4.748747e-03
## C10.1_T4            2.951269e-03
## C10.2_T4            2.915242e-04
## C14.1_T4            1.352338e-03
## C14.2_T4            8.748731e-04
## C16.1_T4           -4.802218e-03
## C16.2_T4            4.153818e-04
## C16.2.OH_T4        -4.855219e-03
## C18.1_T4            1.489636e-03
## C18.1.OH_T4         5.327879e-04
## C18.2_T4            7.599482e-04
## lysoPC.a.C16.0_T4  5.121107e-02
## lysoPC.a.C16.1_T4  4.814257e-02
## lysoPC.a.C17.0_T4  2.472207e-02
## lysoPC.a.C18.0_T4  5.044986e-02
## lysoPC.a.C18.1_T4  5.704308e-02
## lysoPC.a.C18.2_T4  5.740298e-02
## lysoPC.a.C20.3_T4  1.076509e-02
## lysoPC.a.C20.4_T4  4.276114e-02
## lysoPC.a.C24.0_T4  3.073208e-03
## lysoPC.a.C26.0_T4  2.253355e-02
## lysoPC.a.C26.1_T4  2.694835e-03
## lysoPC.a.C28.0_T4  2.424564e-02
## lysoPC.a.C28.1_T4  6.121849e-03
## PC.aa.C24.0_T4      2.262053e-02
## PC.aa.C28.1_T4      2.529209e-03
## PC.aa.C30.0_T4      4.828646e-03
## PC.aa.C32.0_T4      2.773994e-02
## PC.aa.C32.1_T4      1.250721e-02
## PC.aa.C32.3_T4      1.584270e-03
## PC.aa.C34.1_T4     -7.739405e-03
## PC.aa.C34.2_T4     -1.050548e-02
## PC.aa.C34.3_T4     -4.237827e-03
## PC.aa.C34.4_T4     -2.372289e-03
## PC.aa.C36.0_T4      1.056348e-02
## PC.aa.C36.1_T4     -4.853587e-02
## PC.aa.C36.2_T4     -3.864676e-02
## PC.aa.C36.3_T4     -4.622997e-02
## PC.aa.C36.4_T4     -1.250613e-02
## PC.aa.C36.5_T4     -3.147602e-02
```

```
## PC.aa.C38.0_T4    -1.271027e-03
## PC.aa.C38.1_T4     1.431643e-02
## PC.aa.C38.3_T4    -4.019321e-02
## PC.aa.C38.4_T4    -2.762959e-02
## PC.aa.C38.5_T4    -2.445981e-02
## PC.aa.C38.6_T4    -3.976156e-02
## PC.aa.C40.1_T4     3.734444e-02
## PC.aa.C40.2_T4     2.635762e-02
## PC.aa.C40.3_T4     2.113096e-02
## PC.aa.C40.4_T4     2.783414e-02
## PC.aa.C40.5_T4    -6.780948e-03
## PC.aa.C40.6_T4    -4.344218e-02
## PC.aa.C42.0_T4     6.296538e-03
## PC.aa.C42.1_T4     6.544875e-03
## PC.aa.C42.2_T4     1.236289e-02
## PC.aa.C42.4_T4     1.430619e-02
## PC.aa.C42.5_T4     6.171514e-03
## PC.aa.C42.6_T4     1.434618e-02
## PC.ae.C30.0_T4     1.476514e-02
## PC.ae.C32.1_T4     1.118356e-02
## PC.ae.C32.2_T4     6.153218e-03
## PC.ae.C34.0_T4     2.179413e-03
## PC.ae.C34.1_T4    -2.327206e-04
## PC.ae.C34.2_T4     7.819869e-03
## PC.ae.C34.3_T4     1.518003e-02
## PC.ae.C36.0_T4     4.670864e-03
## PC.ae.C36.1_T4     7.921180e-02
## PC.ae.C36.2_T4     8.465069e-03
## PC.ae.C36.3_T4    -5.425292e-03
## PC.ae.C36.4_T4    -1.336785e-02
## PC.ae.C36.5_T4    -6.016467e-03
## PC.ae.C38.0_T4    -5.724563e-03
## PC.ae.C38.2_T4     7.826863e-02
## PC.ae.C38.3_T4     6.469396e-02
## PC.ae.C38.4_T4    -2.321748e-03
## PC.ae.C38.5_T4    -1.349853e-02
## PC.ae.C38.6_T4    -1.652951e-02
## PC.ae.C40.1_T4     1.627970e-02
## PC.ae.C40.2_T4     2.914961e-02
## PC.ae.C40.3_T4     6.497773e-02
## PC.ae.C40.4_T4     4.842871e-02
## PC.ae.C40.5_T4     6.573000e-02
## PC.ae.C40.6_T4    -5.342771e-03
## PC.ae.C42.1_T4     1.395189e-02
## PC.ae.C42.2_T4     1.026979e-02
## PC.ae.C42.3_T4     2.066689e-02
## PC.ae.C42.4_T4     1.111241e-02
## PC.ae.C42.5_T4     3.459445e-02
## PC.ae.C44.3_T4     9.049013e-03
## PC.ae.C44.4_T4     4.294970e-03
## PC.ae.C44.5_T4     3.222760e-03
## PC.ae.C44.6_T4     6.820969e-03
## SM..OH..C14.1_T4   6.848421e-03
## SM..OH..C16.1_T4   3.716071e-03
```

```
## SM..OH..C22.1_T4  -4.151286e-02
## SM..OH..C22.2_T4  -1.095592e-03
## SM..OH..C24.1_T4  -2.392746e-03
## SM.C16.0_T4       -1.278910e-02
## SM.C16.1_T4        1.136336e-02
## SM.C18.0_T4        3.942216e-03
## SM.C18.1_T4        1.761095e-02
## SM.C20.2_T4        1.036226e-03
## SM.C24.0_T4       -5.001151e-02
## SM.C24.1_T4       -2.457752e-02
## MEDDM_T5           1.541726e-03
## MEDCOL_T5          1.541726e-03
## MEDINF_T5          3.187324e-04
## MEDHTA_T5          1.631254e-03
## GLU_T5            -5.527453e-02
## INS_T5            -4.071259e-03
## HOMA_T5           -6.458594e-03
## HBA1C_T5          -4.985002e-03
## HBA1C.mmol.mol_T5 -1.389622e-02
## PESO_T5           -2.179200e-02
## bmi_T5            -2.150532e-02
## CC_T5              7.249484e-05
## CINT_T5           -2.481676e-02
## CAD_T5            -2.543660e-02
## TAD_T5            -9.652439e-03
## TAS_T5            -5.731885e-04
## TG_T5             -5.063651e-02
## COL_T5            -6.630115e-02
## LDL_T5            -5.459825e-02
## HDL_T5            -2.128401e-02
## VLDL_T5           -3.610009e-02
## PCR_T5            -5.089461e-02
## LEP_T5            -4.461574e-02
## ADIPO_T5           8.287738e-02
## GOT_T5             7.650004e-02
## GPT_T5             3.879732e-02
## GGT_T5             1.371071e-02
## URICO_T5           1.750099e-03
## CREAT_T5           1.526487e-04
## UREA_T5            1.438625e-03
## HIERRO_T5          2.410971e-03
## TRANSF_T5         -4.065783e-02
## FERR_T5            7.311036e-03
## Ile_T5             6.129890e-02
## Leu_T5             6.038558e-02
## Val_T5             4.059724e-02
## Ala_T5             3.978706e-02
## Pro_T5             6.525215e-02
## Gly_T5             6.657912e-02
## Ser_T5             7.261378e-02
## Trp_T5             8.324534e-03
## Phe_T5             7.489163e-02
## Met_T5            -1.800008e-02
## Orn_T5             1.019141e-01
```

```
## Arg_T5              4.046660e-02
## His_T5              8.287990e-02
## Asn_T5              8.112528e-02
## Asp_T5              9.374592e-02
## Glu_T5              5.834101e-02
## Gln_T5              4.361097e-02
## Cit_T5              1.036185e-01
## Tyr_T5              7.664697e-02
## Thr_T5              9.804510e-03
## Lys_T5              1.077291e-02
## Creatinine_T5       2.462916e-02
## Kynurenine_T5      -8.851449e-03
## Putrescine_T5      -2.499126e-02
## Sarcosine_T5       -1.309200e-03
## Serotonin_T5        5.082634e-03
## Taurine_T5          4.741507e-02
## SDMA_T5             3.343762e-03
## C0_T5               3.783909e-02
## C2_T5               1.585730e-02
## C3.OH_T5            7.224460e-04
## C6..C4.1.DC._T5    -1.463326e-02
## C5.DC..C6.OH._T5    3.343447e-03
## C7.DC_T5           -9.757712e-03
## C8_T5               1.766735e-03
## C10_T5              3.071888e-03
## C10.1_T5            6.500279e-02
## C10.2_T5            6.744042e-04
## C14.1_T5            1.246695e-03
## C14.2_T5            1.016741e-03
## C16.1_T5            5.211026e-04
## C16.2_T5            5.672111e-04
## C16.2.OH_T5         1.053338e-02
## C18.1_T5            1.956490e-03
## C18.1.OH_T5         4.442292e-04
## C18.2_T5            9.834848e-04
## lysoPC.a.C16.0_T5   7.080223e-02
## lysoPC.a.C16.1_T5   6.900519e-02
## lysoPC.a.C17.0_T5   2.487366e-02
## lysoPC.a.C18.0_T5   7.257095e-02
## lysoPC.a.C18.1_T5   8.201356e-02
## lysoPC.a.C18.2_T5   7.123098e-02
## lysoPC.a.C20.3_T5   3.758766e-02
## lysoPC.a.C20.4_T5   5.087245e-02
## lysoPC.a.C24.0_T5   5.029244e-03
## lysoPC.a.C26.0_T5   1.097507e-02
## lysoPC.a.C26.1_T5   3.042777e-03
## lysoPC.a.C28.0_T5   3.751754e-02
## lysoPC.a.C28.1_T5   4.852821e-03
## PC.aa.C24.0_T5      1.902667e-02
## PC.aa.C28.1_T5      3.609349e-03
## PC.aa.C30.0_T5      2.100091e-03
## PC.aa.C32.0_T5      4.776583e-02
## PC.aa.C32.1_T5      3.429672e-02
## PC.aa.C32.3_T5      1.614148e-03
```

```
## PC.aa.C34.1_T5     2.761893e-02
## PC.aa.C34.2_T5     1.135867e-02
## PC.aa.C34.3_T5     1.254723e-02
## PC.aa.C34.4_T5     3.564734e-04
## PC.aa.C36.0_T5     1.467368e-02
## PC.aa.C36.1_T5     4.375246e-02
## PC.aa.C36.2_T5     1.135660e-02
## PC.aa.C36.3_T5     7.302783e-03
## PC.aa.C36.4_T5    -4.911345e-03
## PC.aa.C36.5_T5    -2.951559e-02
## PC.aa.C38.0_T5     7.425625e-03
## PC.aa.C38.1_T5     2.500377e-02
## PC.aa.C38.3_T5     8.581703e-03
## PC.aa.C38.4_T5    -1.115337e-03
## PC.aa.C38.5_T5    -2.278655e-03
## PC.aa.C38.6_T5    -3.171722e-02
## PC.aa.C40.1_T5     5.057646e-02
## PC.aa.C40.2_T5     2.390378e-02
## PC.aa.C40.3_T5     2.484305e-02
## PC.aa.C40.4_T5     2.942227e-02
## PC.aa.C40.5_T5     1.472309e-02
## PC.aa.C40.6_T5    -2.198015e-02
## PC.aa.C42.0_T5     5.582795e-03
## PC.aa.C42.1_T5     4.445405e-03
## PC.aa.C42.2_T5     1.134628e-02
## PC.aa.C42.4_T5     1.436200e-02
## PC.aa.C42.5_T5     8.337280e-03
## PC.aa.C42.6_T5     3.059907e-02
## PC.ae.C30.0_T5     4.123809e-02
## PC.ae.C32.1_T5     1.464275e-02
## PC.ae.C32.2_T5     6.155085e-03
## PC.ae.C34.0_T5     4.204792e-03
## PC.ae.C34.1_T5     1.500118e-02
## PC.ae.C34.2_T5     2.976525e-02
## PC.ae.C34.3_T5     3.032804e-02
## PC.ae.C36.0_T5     1.012804e-02
## PC.ae.C36.1_T5     8.529050e-02
## PC.ae.C36.2_T5     3.312514e-02
## PC.ae.C36.3_T5     1.091268e-02
## PC.ae.C36.4_T5     1.211299e-02
## PC.ae.C36.5_T5     9.455720e-03
## PC.ae.C38.0_T5    -3.775928e-04
## PC.ae.C38.2_T5     7.961989e-02
## PC.ae.C38.3_T5     5.944123e-02
## PC.ae.C38.4_T5     2.808046e-03
## PC.ae.C38.5_T5    -5.304074e-04
## PC.ae.C38.6_T5    -5.422127e-03
## PC.ae.C40.1_T5     1.752307e-02
## PC.ae.C40.2_T5     3.109212e-02
## PC.ae.C40.3_T5     6.101128e-02
## PC.ae.C40.4_T5     4.657417e-02
## PC.ae.C40.5_T5     5.827443e-02
## PC.ae.C40.6_T5    -1.353030e-05
## PC.ae.C42.1_T5     1.483743e-02
```

```
## PC.ae.C42.2_T5     1.296022e-02
## PC.ae.C42.3_T5     2.042069e-02
## PC.ae.C42.4_T5     1.466883e-02
## PC.ae.C42.5_T5     3.683359e-02
## PC.ae.C44.3_T5     9.591082e-03
## PC.ae.C44.4_T5     5.611252e-03
## PC.ae.C44.5_T5     2.671604e-03
## PC.ae.C44.6_T5     6.367906e-03
## SM..OH..C14.1_T5   5.134371e-03
## SM..OH..C16.1_T5   1.238687e-03
## SM..OH..C22.1_T5  -2.926439e-02
## SM..OH..C22.2_T5  -3.646292e-03
## SM..OH..C24.1_T5  -1.647480e-03
## SM.C16.0_T5        9.260947e-03
## SM.C16.1_T5        2.088881e-02
## SM.C18.0_T5        2.338297e-03
## SM.C18.1_T5        1.268094e-02
## SM.C20.2_T5        2.555965e-04
## SM.C24.0_T5       -2.272492e-02
## SM.C24.1_T5        3.476208e-03
```

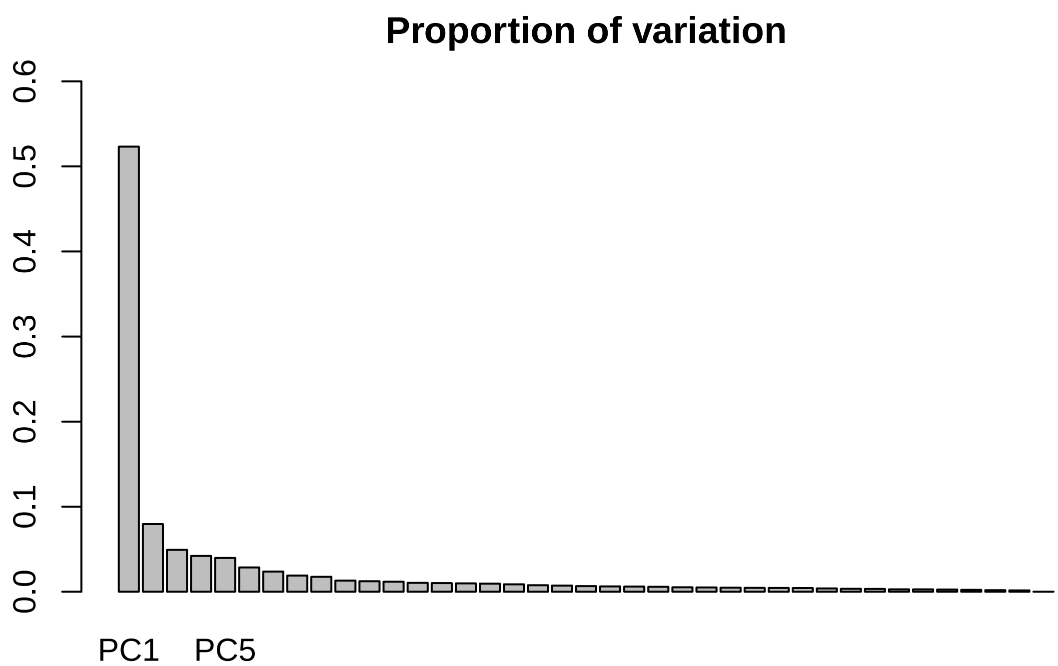
```
summary(PCAseMA)
```

```
## Importance of first k=5 (out of 39) components:
##                            PC1     PC2     PC3     PC4     PC5
## Standard deviation     18.9326 7.37793 5.80678 5.36651 5.21288
## Proportion of Variance  0.5233 0.07947 0.04922 0.04204 0.03967
## Cumulative Proportion   0.5233 0.60274 0.65196 0.69401 0.73368
```

L'anàlisi no és molt concloent. Per obtenir un 70% de la variancia ens calen 5 components, el que és molt. Entenc que hi ha massa variables en joc i segurament caldria reduir abans d'alguna altra manera. Segurament hi ha explicacions biològiques que estaria bé aplicar, però a mi se m'escapen ja que no comprenc totes les variable i per tant, acoto aquí.

Tot i això, per veure com evoluciona l'explicació de la variança faig un gràfic de barres amb una nova PCA que porta tots els components

```
barplot(summary(PCAseMA)$importance[2,],ylim=c(0,0.6))
title(main="Proportion of variation",line=1)
```

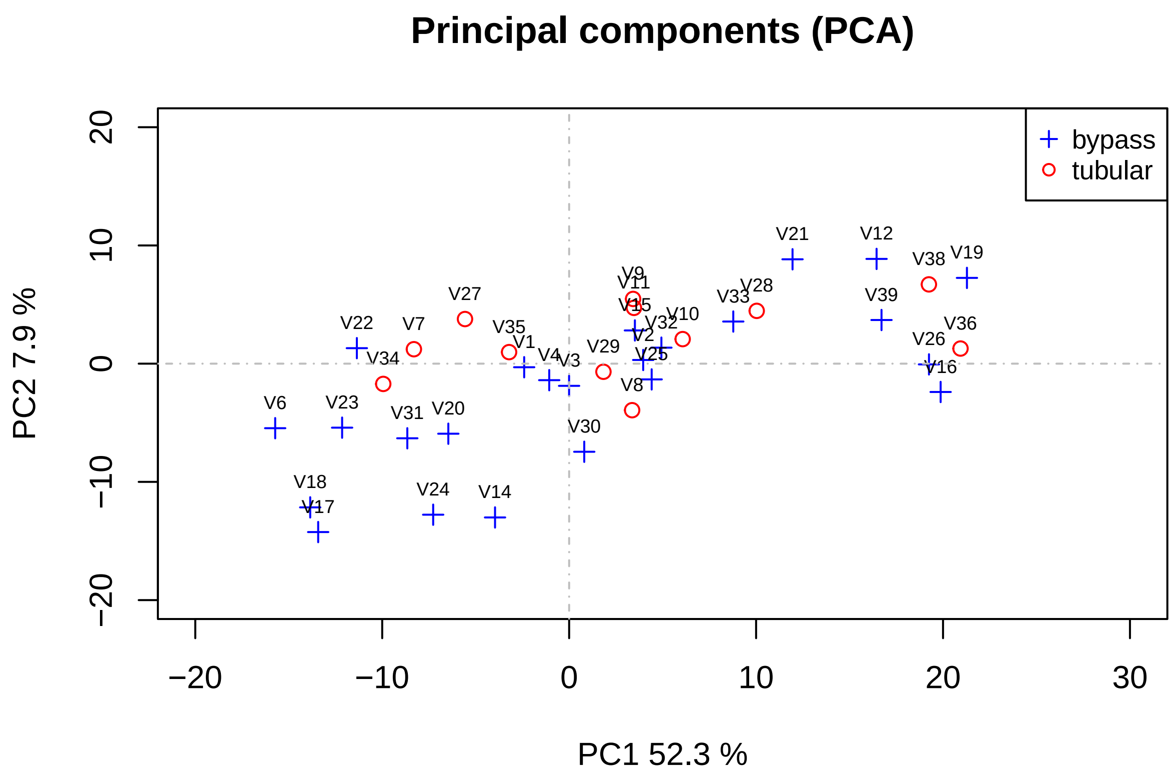


S'parecia força clar com la primera aporta força i les següents van afegint poc. Tot i això, podriem dir que fins a la cinquena sembla interessant agafarles.

Grafiquem ara les primeres components i ho fem en relació al tipus de cirugia que és el que cerca l'article original. De moment, dues

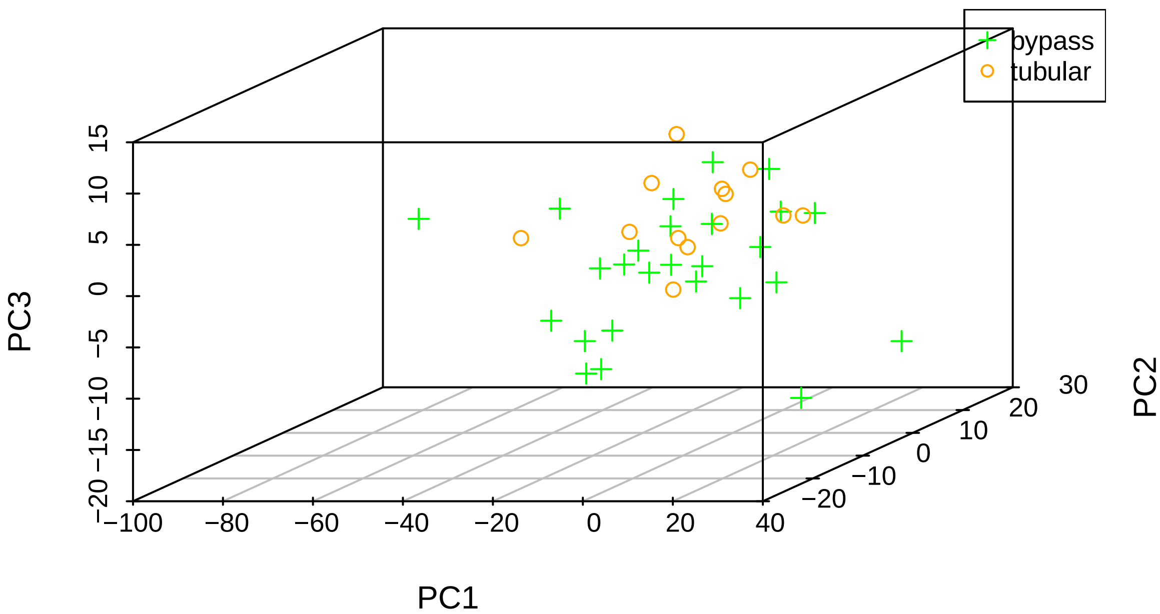
```
# carreguem les dades pel gràfic
cirugia <- as.factor(rowData(seMA)$SURGERY)
cirugian <- as.numeric(cirugia)
# Amb això hem passat les dades de cirugia amb 1: by pass i 2: tubular
loads <- round(PCAseMA$sdev^2/sum(PCAseMA$sdev^2)*100,1)
xlab <- c(paste("PC1",loads[1],"%"))
ylab <- c(paste("PC2",loads[2],"%"))
# grafiquem
plot(PCAseMA$x[,1:2],
     xlab=xlab,
     ylab=ylab,
     pch = c(3,21)[cirugia],
     col=c("blue","red")[cirugia],
     main ="Principal components (PCA)",
     ylim=c(-20,20),
     xlim=c(-20,30))
names <- rownames(PCAseMA$x)
text(PCAseMA$x[,1],PCAseMA$x[,2],names, pos=3, cex=.6)
abline(h=0,v=0,lty=4,col="gray")
legend("topright",
       pch=c(3,21),
       col=c("blue","red"),
```

```
       cex=0.8,
       legend=c("bypass","tubular"))
```



Sembla que les dues components ens separen els dos tipus de cirugia, el que implicaria una resposta diferent dels metabolits a la cirugia. Provem si amb 3 dimensions millora

```
gr3Ddata <- as.data.frame(PCAseMA$x)
gr3Ddata$cirugia <- cirugian
library(scatterplot3d)
# scatterplot3d(PCAseMA$x[,1], PCAseMA$x[,2], PCAseMA$x[,3],
scatterplot3d(gr3Ddata[,1], gr3Ddata[,2], gr3Ddata[,3],
              xlab="PC1", ylab="PC2", zlab="PC3",
              color = ifelse(gr3Ddata$cirugia == "1", "green", "orange"),
              pch = ifelse(gr3Ddata$cirugia == "1", 3, 21))
legend("topright",
       pch=c(3,21),
       col=c("green","orange"),
       cex=0.8,
       legend=c("bypass","tubular"))
```

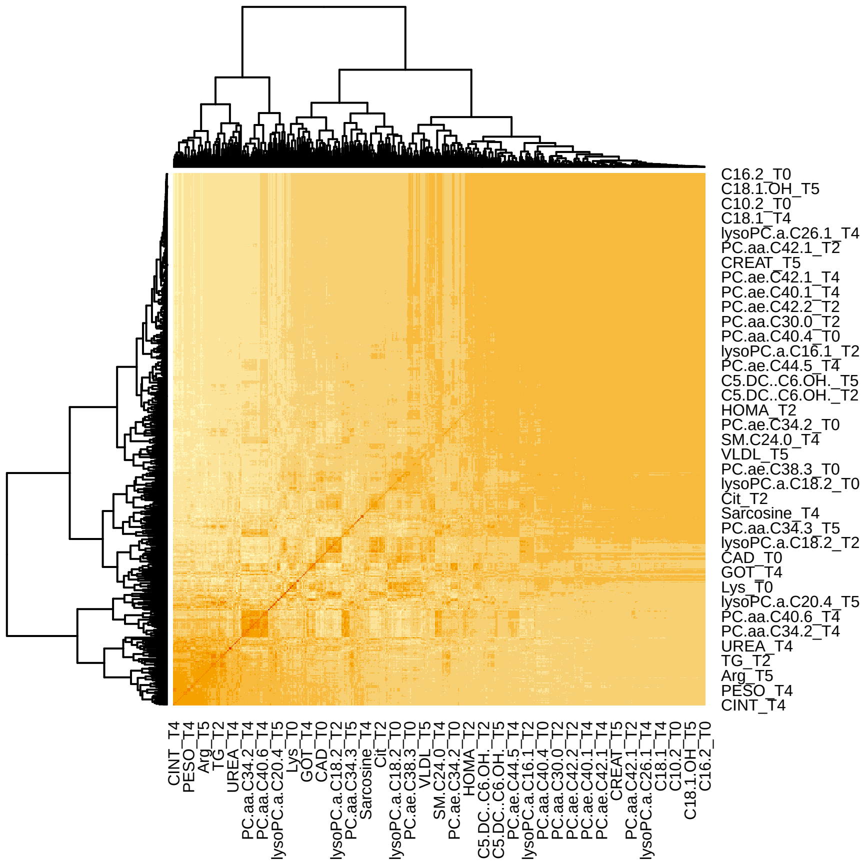


Sembla que en 3 dimensions encara es veu més l'efecte de la cirugia

7. Cerca de patrons

Ho farem amb un heatmap de la matriu de correlacions que permetria veure totes les relacions

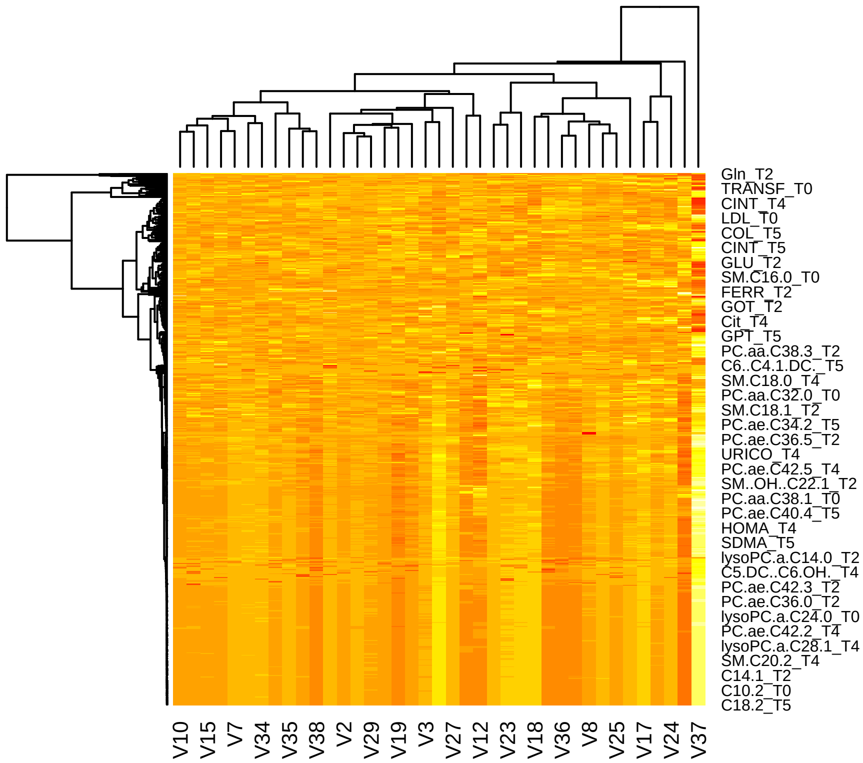
```
cor_matrix <- cor(na.omit(assay(seMAnormalized)))
heatmap(cor_matrix)
```



Es veuen dos grups de variables més correlacionades, però en general les correlacions són debils.

Ara fem un altre heatmap que calcula distancies entre les variables i les mostres

```
dist_matrix <- t(assay(seMAnormalized))
heatmap(dist_matrix, col=heat.colors(16))
```



Hi ha tantes mostres i variables que no es poden veure be totes les relacions, però sembla que hi ha un patró clar de proximitat en les primeres variables del gràfic i algunes que estan a molta distància cap a baix.